United States Patent	12406749
Kind Code	B2
Date of Patent	September 02, 2025
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Systems and methods for predicting repair outcomes in genetic engineering

Abstract

The specification provides methods for introducing a desired genetic change in a nucleotide sequence using a double-strand break (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the desired genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site.

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Appl. No.:	**16/772747**
Filed (or PCT Filed):	**December 15, 2018**
PCT No.:	**PCT/US2018/065886**

PCT Pub. No.: **WO2019/118949**

PCT Pub. Date: June 20, 2019

Prior Publication Data

Document Identifier	Publication Date
US 20220238182 A1	Jul. 28, 2022

Related U.S. Application Data

us-provisional-application US 62669771 20180510
us-provisional-application US 62599623 20171215

Publication Classification

Int. Cl.: **G16B20/30** (20190101); **A61K31/7088** (20060101); **A61K38/46** (20060101); **C12N9/22** (20060101); **C12N15/10** (20060101); **C12N15/11** (20060101); **G16B40/20** (20190101)

U.S. Cl.:

CPC **G16B20/30** (20190201); **A61K31/7088** (20130101); **A61K38/465** (20130101); **C12N9/22** (20130101); **C12N15/1089** (20130101); **C12N15/11** (20130101); **G16B40/20** (20190201); C12N2310/20 (20170501); C12N2800/80 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING
(1) The contents of the electronic sequence listing (B119570051US01-SUBSEQ-ARM.txt; Size: 2,771,180 bytes; and Date of Creation: Nov. 7, 2023) is herein incorporated by reference in its entirety.
RELATED APPLICATIONS
(2) This application is a national stage filing under 35 U.S.C. § 371 of international PCT application, PCT/US2018/065886, filed Dec. 15, 2018, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application, U.S. Ser. No. 62/599,623, filed on Dec. 15, 2017, and to U.S. Provisional Patent Application, U.S. Ser. No. 62/669,771, filed May 10, 2018, each of which is incorporated herein by reference.
BACKGROUND OF THE INVENTION
(3) CRISPR (clustered regularly interspaced short palindromic repeats)-Cas9 has revolutionized genome editing, providing powerful research tools and promising agents for the potential treatment of genetic diseases.sup.1-3. The DNA-targeting capabilities of Cas9 have been improved by the development of gRNA design principles.sup.4, modeling of factors leading to off-target DNA cleavage, enhancement of Cas9 sequence fidelity by modifications to the nuclease.sup.5,6 and gRNA.sup.7, and the evolution or engineering of Cas9 variants with alternative PAM

sequences.sup.8-10. Similarly, control over the product distribution of genome editing has been advanced by the development of base editing to achieve precise and efficient single-nucleotide mutations.sup.7,11,12, and the improvement of template-directed homology-directed repair (HDR) of double strand breaks.sup.13-15.

(4) Non-template directed repair systems, including non-homologous end-joining (NHEJ) and microhomology-mediated end-joining (MMEJ), are major pathways involved in the repair of Cas9-mediated double-strand breaks that can result in highly heterogeneous repair outcomes that generate hundreds of genotypes following DNA cleavage at a single site. While end-joining repair of Cas9-mediated double-stranded DNA breaks has been harnessed to facilitate knock-in of DNA templates.sup.18-21 or deletion of intervening sequence between two cleavage sites.sup.22, NHEJ and MMEJ are not generally considered useful for precision genome editing applications. Recent work has found that the heterogeneous distribution of Cas9-mediated editing products at a given target site is reproducible and dependent on local sequence context.sup.20,21, but no general methods have been described to predict genotypic products following Cas9-induced double-stranded DNA breaks.

(5) The generally accepted view is that DNA double-strand break repair (i.e., template-free, non-homology-dependent repair) following cleavage by genome editing systems produces stochastic and heterogenous repair products and is therefore impractical for applications beyond gene disruption. Further, template-free repair processes (e.g., MMEJ and NHEJ) following DNA double-strand break, despite being more efficient than homology-based repair, are generally not viewed as feasible solutions to precision repair applications, such as restoring the function of a defective gene with a gain-of-function genetic change. Accordingly, methods and solutions enabling the judicious application of template-free genome editing systems, including CRISPR/Cas, TALEN, or Zinc-Finger genome editing systems, would significantly advance the field of genome editing.

SUMMARY OF THE INVENTION

(6) The present inventors have unexpectedly found through computational analyses that template-free DNA/genome editing systems, e.g., CRISPR/Cas9, Cas-based, Cpf1-based, or other DSB (double-strand break)-based genome editing systems, produce a predictable set of repair genotypes thereby enabling the use of such editing systems for applications involving or requiring precise manipulation of DNA, e.g., the correction of a disease-causing genetic mutation or modifying a wildtype sequence to confer a genetic advantage. This finding is contrary to the accepted view that DNA double-strand break repair (i.e., template-free, non-homology-dependent repair) following cleavage by genome editing systems produces stochastic and heterogenous repair products and are therefore impractical for applications beyond gene disruption. Thus, the specification describes and discloses in various aspects and embodiments computational-based methods and systems for practically harnessing the innate efficiencies of template-free DNA repair systems for carrying out precise DNA and/or genomic editing without the reliance upon homology-based repair.

(7) Accordingly, the specification provides in one aspect a method of introducing a desired genetic change in a nucleotide sequence using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the desired genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site.

(8) In another aspect, the specification provides a method of treating a genetic disease by correcting a disease-causing mutation using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence comprising a disease-causing mutation; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for correcting the disease-causing mutation in the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing

genome editing system, thereby correcting the disease-causing mutation and treating the disease.
(9) In yet another aspect, the specification provides a method of altering a genetic trait by introducing a genetic change in a nucleotide sequence using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site and consequently altering the associated genetic trait.
(10) In another aspect, the specification provides a method of selecting a guide RNA for use in a Cas-genome editing system capable of introducing a genetic change into a nucleotide sequence of a target genomic location, the method comprising: identifying in a nucleotide sequence of a target genomic location one or more available cut sites for a Cas-based genome editing system; and analyzing the nucleotide sequence and cut site with a computational model to identify a guide RNA capable of introducing the genetic change into the nucleotide sequence of the target genomic location.
(11) In still another aspect, the specification provides a method of introducing a genetic change in the genome of a cell with a Cas-based genome editing system comprising: selecting a guide RNA for use in the Cas-based genome editing system in accordance with the method of the above aspect; and contacting the genome of the cell with the guide RNA and the Cas-based genome editing system, thereby introducing the genetic change.
(12) In various embodiments, the cut sites available in the nucleotide sequence are a function of the particular DSB-inducing genome editing system in use, e.g., a Cas-based genome editing system.
(13) In certain embodiments, the nucleotide sequence is a genome of a cell.
(14) In certain other embodiments, the method for introducing the desired genetic change is done in vivo within a cell or an organism (e.g., a mammal), or ex vivo within a cell isolated or separated from an organism (e.g., an isolated mammalian cancer cell), or in vitro on an isolated nucleotide sequence outside the context of a cell.
(15) In various embodiments, the DSB-inducing genome editing system can be a Cas-based genoe editing system, e.g., a type II Cas-based genome editing system. In other embodiments, the DSB-inducing genome editing system can be a TALENS-based editing system or a Zinc-Finger-based genome editing system. In still other embodiments, the DSB-inducing genome editing system can be any such endonuclease-based system which catalyzes the formation of a double-strand break at a specific one or more cut sites.
(16) In embodiments involving a Cas-based genome editing system, the method can further comprise selecting a cognate guide RNA capable of directing a double-strand break at the optimal cut site by the Cas-based genome editing system.
(17) In certain embodiments, the guide RNA is selected from the group consisting the guide RNA sequences listed in any of Tables 1-6. In various embodiments, the guide RNA can be known or can be newly designed.
(18) In various embodiments, the double-strand brake (DSB)-inducing genome editing system is capable of editing the genome without homology-directed repair.
(19) In other embodiments, the double-strand brake (DSB)-inducing genome editing system comprises a type I Cas RNA-guided endonuclease, or a variant or orthologue thereof.
(20) In still other embodiments, the double-strand brake (DSB)-inducing genome editing system comprises a type II Cas RNA-guided endonuclease, or a functional variant or orthologue thereof.
(21) The double-strand brake (DSB)-inducing genome editing system may comprise a Cas9 RNA-guided endonuclease, or a variant or orthologue thereof in certain embodiments.
(22) In still other embodiments, the double-strand brake (DSB)-inducing genome editing system can comprise a Cpf1 RNA-guided endonuclease, or a variant or orthologue thereof.

(23) In yet further embodiments, the double-strand brake (DSB)-inducing genome editing system can comprise a *Streptococcus pyogenes* Cas9 (SpCas9), *Staphylococcus pyogenes* Cas9 (SpCas9), *Staphyloccocus aureus* Cas (SaCas9), *Francisella novicida* Cas9 (FnCas9), or a functional variant or orthologue thereof.
(24) In various embodiments, the desired genetic change to be introduced into the nucleotide sequence, e.g., a genome, is to a correction to a genetic mutation. In embodiments, the genetic mutation is a single-nucleotide polymorphism, a deletion mutation, an insertion mutation, or a microduplication error.
(25) In still other embodiments, the genetic change can comprises a 2-60-bp deletion or a 1-bp insertion.
(26) The genetic change in other embodiments can comprise a deletion of between 2-20, or 4-40, or 8-80, or 16-160, or 32-320, 64-640, or up to 1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, or 10,000 or more nucleotides. Preferably, the deletion can restore the function of a defective gene, e.g., a gain-of-function frameshift genetic change.
(27) In other embodiments, the desired genetic change is a desired modification to a wildtype gene that confers and/or alters one or more traits, e.g., conferring increased resistance to a pathogen or altering a monogenic trait (e.g., eye color) or polygenic trait (e.g., height or weight).
(28) In embodiments involving correcting a disease-causing mutation, the disease can be a monogenic disease. Such monogenic diseases can include, for example, sickle cell disease, cystic fibrosis, polycystic kidney disease, Tay-Sachs disease, achondroplasia, beta-thalassemia, Hurler syndrome, severe combined immunodeficiency, hemophilia, glycogen storage disease Ia, and Duchenne muscular dystrophy.
(29) In any of the above aspects and embodiments, the step of identifying the available cut sites can involve identifying one or more PAM sequences in the case of a Cas-based genome editing system.
(30) In various embodiments of the above methods, the computational model used to analyze the nucleotide sequence is a deep learning computational model, or a neural network model having one or more hidden layers. In various embodiments, the computational model is trained with experimental data to predict the probability of distribution of indel lengths for any given nucleotide sequence and cut site. In still other embodiments, the computational model is trained with experimental data to predict the probability of distribution of genotype frequencies for any given nucleotide sequence and cut site.
(31) In various embodiments, the computational model comprises one or more training modules for evaluating experimental data.
(32) In various embodiments, the computational model can comprise: a first training module for computing a microhomology score matrix; a second training module for computing a microhomology independent score matrix; and a third training module for computing a probability distribution over 1-bp insertions, wherein once trained with experimental data the computational model computes a probability distribution over indel genotypes and a probability distribution over indel lengths for any given input nucleotide sequence and cut site.
(33) In other embodiments, the computational model predicts genomic repair outcomes for any given input nucleotide sequence and cut site.
(34) In various embodiments, the genomic repair outcomes can comprise microhomology deletions, microhomology-less deletions, and/or 1-bp insertions.
(35) In still other embodiments, the computational model can comprise one or more modules each comprising one more input features selected from the group consisting of: a target site nucleotide sequence; a cut site; a PAM-sequence; microhomology lengths relative at a cut site, % GC content at a cut site; and microhomology deletion lengths at a cut site, and type of DSB-genome editing system.
(36) In various embodiments, the nucleotide sequence analyzed by the computational model is between about 25-100 nucleotides, 50-200 nucleotides, 100-400 nucleotides, 200-800 nucleotides,

400-1600 nucleotides, 800-3200 nucleotides, and 1600-6400 nucleotide, or even up to 7K, 8K, 9K, 10K, 11K, 12K, 13K, 14K, 15K, 16K, 17K, 18K, 19K, 20K nucleotides, or more in length.
(37) In another aspect, the specification relates to guide RNAs which are identified by various methods described herein. In certain embodiments, the guide RNAs can be any of those presented in Tables 1-6, the contents of which form part of this specification.
(38) According to various embodiments, the RNA can be purely ribonucleic acid molecules. However, in other embodiments, the RNA guides can comprise one or more naturally-occurring or non-naturally occurring modifications. In various embodiments, the modifications can including, but are not limited to, nucleoside analogs, chemically modified bases, intercalated bases, modified sugars, and modified phosphate group linkers. In certain embodiments, the guide RNAs can comprise one or more phosphorothioate and/or 5′-N-phosphporamidite linkages.
(39) In still other aspects, the specification discloses vectors comprising one or more nucleotide sequences disclosed herein, e.g., vectors encoding one or more guide RNAs, one or more target nucleotide sequences which are being edited, or a combination thereof. The vectors may comprise naturally occurring sequences, or non-naturally occurring sequences, or a combination thereof.
(40) In still other aspects, the specification discloses host cells comprising the herein disclosed vectors encoding one more more nucleotide sequences embodied herein, e.g., one or more guide RNAs, one or more target nucleotide sequences which are being edited, or a combination thereof.
(41) In other aspects, the specification discloses a Cas-based genome editing system comprising a Cas protein (or homolog, variant, or orthologue thereof) complexed with at least one guide RNA. In certain embodiments, the guide RNA can be any of those disclosed in Tables 1-6, or a functional variant thereof.
(42) In still other aspects, the specification provides a Cas-based genome editing system comprising an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA, wherein the guide RNA can be identified by the methods disclosed herein for selecting a guide RNA.
(43) In yet another aspect, the specification provides a Cas-based genome editing system comprising an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA, wherein the guide RNA can be identified by the methods disclosed herein for selecting a guide RNA.
(44) In still a further aspect, the specification provides a library for training a computational model for selecting a guide RNA sequence for use with a Cas-based genome editing system capable of introducing a genetic change into a genome without homology-directed repair, wherein the library comprises a plurality of vectors each comprising a first nucleotide sequence of a target genomic location having a cut site and a second nucleotide sequence encoding a cognate guide RNA capable of directing a Cas-based genome editing system to carry out a double-strand break at the cut site of the first nucleotide sequence.
(45) In another aspect, the specification provides a library and its use for training a computational model for selecting an optimized cut site for use with a DSB-based genome editing system (e.g., Cas-based system, TALAN-based system, or a Zinc-Finger-based system) that is capable of introducing a desired genetic change into a nucleotide sequence (e.g., a genome) at the selected cut site without homology-directed repair, wherein the library comprises a plurality of vectors each comprising a nucleotide sequence having a cut site, and optionally a second nucleotide sequence encoding a cognate guide RNA (in embodiments involving a Cas-based genome editing system).
(46) In a still further aspect, the specification discloses a computational model.
(47) In certain embodiments, the computational model can predict and/or compute an optimized or preferred cut site for a DSB-based genome editing system for introducing a genetic change into a nucleotide sequence. In preferred embodiments, the repair does not require homology-based repair

mechanisms.
(48) In certain other embodiments, the computational model can predict and/or compute an optimized or preferred cut site for a Cas-based genome editing system for introducing a genetic change into a nucleotide sequence. In preferred embodiments, the repair does not require homology-based repair mechanisms.
(49) In still other embodiments, the computation model provides for the selection of a optimized or preferred guide RNA for use with a Cas-based genome editing system for introducing a genetic change in a genome. In preferred embodiments, the repair does not require homology-based repair mechanisms.
(50) In various embodiments, the computational model is a neural network model having one or more hidden layers.
(51) In other embodiments, the computational model is a deep learning computational model.
(52) In various embodiments, that the DSB-based genome editing system (e.g., a Cas-based genome editing system) edits the genome without relying on homology-based repair.
(53) In various embodiments, that computational model is trained with experimental data to predict the probability of distribution of indel lengths for any given nucleotide sequence and cut site. In other embodiments, computational model is trained with experimental data to predict the probability of distribution of genotype frequencies for any given nucleotide sequence and cut site.
(54) In embodiments, the computational model comprises one or more training modules for evaluating experimental data.
(55) In an embodiment, the computational model comprises: a first training module (**305**) for computing a microhomology score matrix (**305**); a second training module (**310**) for computing a microhomology independent score matrix; and a third training module (**315**) for computing a probability distribution over 1-bp insertions, wherein once trained with experimental data the computational model computes a probability distribution over indel genotypes and a probability distribution over indel lengths for any given input nucleotide sequence and cut site.
(56) In certain embodiments, the computational model predicts genomic repair outcomes for any given input nucleotide sequence (i.e., context sequence) and cut site.
(57) In certain embodiments, the genomic repair outcomes comprise microhomology deletions, microhomology-less deletions, and 1-bp insertions.
(58) In various embodiments, the one or more modules each comprising one more input features selected from the group consisting of: a target site nucleotide sequence; a cut site; a PAM-sequence; microhomology lengths relative at a cut site, % GC content at a cut site; and microhomology deletion lengths at a cut site.
(59) In certain embodiments, the nucleotide sequence analyzed by the computational model is between about 25-100 nucleotides, 50-200 nucleotides, 100-400 nucleotides, 200-800 nucleotides, 400-1600 nucleotides, 800-3200 nucleotides, and 1600-6400 nucleotide, or more.
(60) In yet another aspect, the specification discloses a method for training a computational model, comprising: (i) preparing a library comprising a plurality of nucleic acid molecules each encoding a nucleotide target sequence and a cognate guide RNA, wherein each nucleotide target sequence comprises a cut site; (ii) introducing the library into a plurality of host cells; (iii) contacting the library in the host cells with a Cas-based genome editing system to produce a plurality of genomic repair products; (iv) determining the sequences of the genomic repair products; and (iv) training the computational model with input data that comprises at least the sequences of the nucleotide target sequence and/or the genomic repair products and the cut sites.
(61) In still another aspect, the specification discloses a method for training a computational model, comprising: (i) preparing a library comprising a plurality of nucleic acid molecules each encoding a nucleotide target sequence and a cut site; (ii) introducing the library into a plurality of host cells; (iii) contacting the library in the host cells with a DSB-based genome editing system to produce a plurality of genomic repair products; (iv) determining the sequences of the genomic repair

products; and (iv) training the computational model with input data that comprises at least the sequences of the nucleotide target sequence and/or the genomic repair products and the cut sites.
(62) In certain embodiments, the trained computational models disclosed herein are capable of computing a probability of distribution of indel lengths for any given input nucleotide sequence and input cut site, and/or a probability of distribution of genotype frequencies for any given input nucleotide sequence and input cut site.
(63) In embodiments relating to Cas-based genomic editing systems, the trained computational model is capable of selecting a guide RNA for use with a Cas-based genome editing system for introducing a genetic change into a genome.
(64) The computational model provides a means to produce precision genetic change with a DSB-based genomic editing system. The genetic changes can include microhomology deletion, microhomology-less deletion, and 1-bp insertion. In certain embodiments, the genetic change corrects a disease-causing mutation. In other embodiments, the genetic change modifies a wildtype sequence, which may confer a change in a genetic trait (e.g., a monogenic or polygenic trait). The disease-causing mutation that can be corrected using the computational model with a DSB-based genomic editing sytem can include, but is not limited to, sickle cell disease, cystic fibrosis, polycystic kidney disease, Tay-Sachs disease, achondroplasia, beta-thalassemia, Hurler syndrome, severe combined immunodeficiency, hemophilia, glycogen storage disease Ia, or Duchenne muscular dystrophy.
(65) In another aspect, the disclosure provides a method for selecting one or more guide RNAs (gRNAs) from a plurality of gRNAs for CRISPR, comprising acts of: for at least one gRNA of the plurality of gRNAs, using a local DNA sequence and a cut site targeted by the at least one gRNA to predict a frequency of one or more repair genotypes resulting from template-free repair following application of CRISPR with the at least one gRNA; and determining whether to select the at least one gRNA based at least in part on the predicted frequency of the one or more repaired genotypes.
(66) In embodiments, the one or more repair genotypes correspond to one or more healthy alleles of a gene related to a disease. In other embodiments, the predicted frequency of the one or more repair genotypes is at least about 30%, or at least about 40%, or at least about 50%, or more.
(67) In certain embodiments, the step of predicting the frequency of the one or more repair genotypes comprises: for each deletion length of a plurality of deletion lengths, aligning subsequences of that deletion length on 5′ and 3′ sides of the cut site to identify one or more longest microhomologies; featurizing the identified microhomologies; applying a machine learning model to compute a frequency distribution over the plurality of deletion lengths; and using frequency distribution over the plurality of deletion lengths to determine the frequency of the one or more repair genotypes.
(68) In certain embodiments, the plurality of gRNAs comprise gRNAs for CRISPR/Cas9, and the application of CRISPR comprises application of CRISPR/Cas9.
(69) In yet another aspect, the system comprises: at least one processor; and at least one computer-readable storage medium having encoded thereon instructions which, when executed, cause the at least one processor to perform a herein disclosed computational method.
(70) A method for editing a nucleotide sequence using a DSB-based genomic editing system that introduces a genetic change at a cut site in a nucleotide sequence, wherein the cut site location is informed by a computational model that computes a frequency distribution over the plurality of deletion lengths and/or a frequency distribution of one or more repaired genotypes over the deletion lengths.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** shows an illustrative DNA segment **100**, in accordance with some embodiments.

(2) FIGS. **2**A-D show an illustrative matching of 3′ ends of top and bottom strands of a DNA segment at a cut site and an illustrative repair product, in accordance with some embodiments.

(3) FIG. **3**A shows an illustrative machine learning model **300**, in accordance with some embodiments.

(4) FIG. **3**B shows an illustrative process **350** for building one or more machine learning models for predicting frequencies of deletion genotypes and/or deletion lengths, in accordance with some embodiments.

(5) FIG. **4**A shows an illustrative neural network **400**A for computing microhomology (MH) scores, in accordance with some embodiments.

(6) FIG. **4**B shows an illustrative neural network **400**B for computing MH-less scores, in accordance with some embodiments.

(7) FIG. **4**C shows an illustrative process **400**C for training two neural networks jointly, in accordance with some embodiments.

(8) FIG. **4**D shows an illustrative implementation of the insertion module **315** shown in FIG. **3**A, in accordance with some embodiments.

(9) FIG. **5** shows an illustrative process **500** for processing data collected from CRISPR/Cas9 experiments, in accordance with some embodiments.

(10) FIG. **6** shows an illustrative process **600** for using a machine learning model to predict frequencies of indel genotypes and/or indel lengths, in accordance with some embodiments.

(11) FIG. **7** shows illustrative examples of a blunt-end cut and a staggered cut, in accordance with some embodiments.

(12) FIG. **8**A shows an illustrative plot **800**A of predicted repair genotypes, in accordance with some embodiments.

(13) FIG. **8**B shows another illustrative plot **800**B of predicted repair genotypes, in accordance with some embodiments.

(14) FIG. **8**C shows another illustrative plot **800**C of predicted repair genotypes, in accordance with some embodiments.

(15) FIG. **8**D shows a microhomology identified in the example of FIG. **8**C, in accordance with some embodiments.

(16) FIG. **9** shows another illustrative neural network **900** for computing a frequency distribution over deletion lengths, in accordance with some embodiments.

(17) FIG. **10** shows, schematically, an illustrative computer **1000** on which any aspect of the present disclosure may be implemented.

(18) FIGS. **11**A-**11**C show a high-throughput assessment of Cas9-mediated DNA repair products. FIG. **11**A, A genome-integrated screening library approach for monitoring Cas9 editing products at thousands of target sequences. FIG. **11**B, Frequency of Cas9-mediated repair products by class from 1,996 Lib-A target sequences in mouse embryonic stem cells (mESCs).

(19) FIG. **11**C, Distribution of Cas9-mediated repair products by class in 88 VO target sequences in K562 cells.

(20) FIGS. **12**A-**12**E show modeling of Cas9-mediated indels by inDelphi. FIG. **12**A, Schematic of computational flow for inDelphi modeling. inDelphi separates Cas9-mediated editing products by indel type and uses machine learning tools trained on experimental Lib-A editing products to predict relative frequencies of editing products for any target site. Major editing products include 1- to 60-bp MH deletions, 1-to 60-bp MH-less deletions, and 1-bp insertions. FIG. **12**B, Mechanism depicting microhomology-mediated end-joining repair, which yields distinct repair outcomes that reflect which microhomologous bases are used during repair.

(21) FIG. **12**C, Observed mean frequency of 1-bp insertion genotypes among 1,981 Lib-A target

sequences with varying −4 nucleotides. Error bars show 95% C.I. on sample mean with 1000-fold bootstrapping. Data distributions are shown in FIGS. **18**A-**18**H. FIG. **12**D, Comparison of observed 1-bp insertion frequencies among all Cas9-edited products from 1,996 Lib-A target sequences. The box denotes the 25th, 50th, and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. *P=5.4×10-36; **P=8.6×10-70, two-sided two-sample T-test, test statistic=−13.0 and -18.4, degrees of freedom=777 and 1,994; Hedges' g effect size=0.94 and 0.85, for*and**respectively. FIG. **12**E, Motif representation of base identities that impact the frequency of 1-bp insertions in Lib-A data. Only bases with non-zero linear regression weights in 10,000-fold iterative cross-validation are shown. Median held-out Pearson correlation 0.62, total N=1996.

(22) FIGS. **13**A-**13**F show that Cas9-mediated editing outcomes are accurately predicted by inDelphi. FIG. **13**A, Histogram of the observed fraction of Cas9-mediated editing products whose indel length is included in inDelphi predictions in endogenous VO target sequences in HEK293 (N=86), HCT116 (N=91), and K562 (N=82) cells. FIG. **13**B, Distribution of Pearson correlation values comparing inDelphi predictions to observed product frequencies in VO sequence contexts in HEK293T (N=86), HCT116 (N=91), and K562 (N=82) cells. The box denotes the 25th, 50th, and 75th percentiles, and whiskers show 1.5 times the interquartile range.

(23) FIG. **13**C, Distribution of Pearson correlation values comparing inDelphi predictions to observed indel length frequencies in VO sequence contexts in HEK293 (N=86), HCT116 (N=91), and K562 (N=82) cells. Box plot as in (FIG. **13**B). FIG. **13**D, Comparison of inDelphi and Microhomology-Predictor frameshift predictions to observed frameshift frequencies among 86 VO target sequences in HEK293 cells. The error band represents the 95% C.I. around the regression estimate with 1,000-fold bootstrapping. FIG. **13**E shows 1-bp insertion frequencies among edited outcomes in U2OS and HEK293T cells (n=27 and 26 observations, baseline n=1,958 and 89 target sites, P=4.2×10-8 and 8.1×10-12, respectively), two-sided Welch's t-test. FIG. **13**F shows smoothed predicted distribution of the highest frequency indel among major editing outcomes (+1 to −60 indels) for SpCas9 gRNAs targeting the human genome.

(24) FIGS. **14**A-**14**F show high-precision, template-free Cas9 nuclease-mediated deletion and insertion. FIG. **14**A, Schematic of deletion repair at a designed Lib-B target sequence with a 9-bp microduplication and strong sequence microhomology. FIG. **14**B, Observed frequency of microduplication collapse among all edited products at 56 Lib-B target sequences designed with 7- to 25-bp microduplications. The error band represents the 95% C.I. around the regression estimate with 1,000-fold bootstrapping. FIG. **14**C, Observed frequencies of 1-bp insertions among 205 sequence contexts designed to vary base identity at positions −5 to −2 (relative to the PAM at positions 0-2) in three surrounding low-microhomology sequence contexts. The X-axis is sorted by median 1-bp insertion frequency; see FIGS. **20**A-**20**E for the complete axis. FIG. **14**D, Comparison of the observed 1-bp insertion frequency at 205 Lib-B designed sequences as in (FIG. **14**C) with varying positions −4 and −3. The box denotes the 25th, 50th, and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. *P=0.03; **P=2.98×10-7, two-sided two-sample T-test, test statistic=−2.2 and -6.5, degrees of freedom=185 and 32, Hedges' g effect size=0.58 and 2.3, for*and**respectively. FIG. **14**E, Comparison of predicted precision scores to observed precision scores for microhomology deletions in 86 VO target sites in HEK293 cells. FIG. **14**F, Distribution of the predicted frequency of the most frequent deletion and insertion outcomes among major editing outcomes (1-bp insertions, 1- to 60-bp MH deletions, and 1- to 60-bp MH-less deletions) at 1,063,802 Cas9 gRNAs targeting human exons and introns.

(25) FIGS. **15**A-**15**F show precise template-free Cas9-mediated editing of pathogenic alleles to wild-type genotypes. FIG. **15**A, Using Cas9-nuclease to correct a pathogenic LDLR allele to wild-type. FIG. **15**B, Comparison among ClinVar/HGMD pathogenic alleles of observed and predicted frequencies of repair to wild-type alleles, accompanied by a histogram of observed frequencies. Major editing products include 1-bp insertions, 1- to 60-bp MH deletions, and 1- to 60-bp MH-less

deletions. FIG. **15**C, Comparison of observed and predicted frequencies of frameshift repair to the wild-type frame among ClinVar/HGMD pathogenic alleles, accompanied by a histogram of observed frequencies. Major editing products are defined as in (FIB. 15B).
(26) FIG. **15**D, Histograms of observed frequencies of repair to the wild-type genotype for wild-type mESCs and Prdkc−/−Lig4−/− mESCs at Lib-B pathogenic microduplication alleles with predicted repair frequency ≥50% among all major editing products, defined as in (FIG. **15**B). Dashed lines indicate sample means which differ significantly. P=7.8×10-12; two-sided two-sample T-test, test statistic=−6.9, degrees of freedom=1,297, Hedges' g effect size=0.39. FIG. **15**E, Flow cytometry contour plots showing GFP fluorescence and LDL-DyLight550 uptake in mESCs containing the LDLRdup1662_1669dupGCTGGTGA-P2A-GFP allele (LDLRdup-P2A-GFP) and treated with SpCas9 and gRNA when denoted. FIG. **15**F, Fluorescence microscopy of mESCs containing the LDLRdup1662_1669dupGCTGGTGA-P2A-GFP allele treated with SpCas9 and gRNA, or untreated.
(27) FIGS. **16**A-**16**F show design and cloning of a high-throughput library to assess CRISPR-Cas9-mediated editing products. FIG. **16**A, From left to right, distributions of predicted Cas9 on-target efficiency (Azimuth score), number of nucleotides participating in microhomology in 3-30-bp deletions, GC content, and estimated precision of deletion outcomes derived from 169,279 potential SpCas9 gRNA target sites in the human genome with quintiles marked as used to design Lib-A. FIG. **16**B, Schematic of the cloning process used to clone Lib-A and Lib-B. The cloning process involves ordering a library of oligonucleotides pairing a gRNA protospacer with its 55-bp target sequence, centered on an NGG PAM. To insert the gRNA hairpin between the gRNA protospacer and the target site, the library undergoes an intermediate Gibson Assembly circularization step, restriction enzyme linearization, and Gibson Assembly into a plasmid backbone containing a U6-promoter to facilitate gRNA expression, a hygromycin resistance cassette, and flanking Tol2 transposon sites to facilitate integration into the genome. FIG. **16**C, Analysis of cumulative percentage of all CRISPR-Cas9-mediated deletions from VO target sequences in HEK293 (N=89), HCT116 (N=92), and K562 (N=86) that delete up to the reported number of nucleotides (X-axis). 94% of deletions are 30-bp or shorter. FIG. **16**D shows the number of unique high-confidence editing outcomes called by simulating data subsampling in data in lib-A (n=2,000 target sites) in mESCs (combined data from n=3 independent biological replicates) and U2OS cells (combined data from n=2 independent biological replicates). For ‘all’, the original non-subsampled data are presented. Each box depicts data for 2,000 target sites. Outliers are not depicted. FIG. **16**E shows Pearson's r of genotype frequencies comparing lib-A in mESCs and U2OS cells with endogenous data in HEK293 (n=87 target sites), HCT116 (n=88), and K562 (n=86) cells. Outliers are depicted as diamonds. 1-bp insertion frequency adjustment was performed at each target site by proportionally scaling them to be equal between two cell types. FIG. **16**F shows Pearson's r of genotype frequencies at lib-A target sites, comparing two independent biological replicate experiments in mESCs (n=1,861 target sites, median r=0.89) and U2OS cells (n=1,921, median r=0.77). Outliers are depicted as diamonds. Box plots denote the 25th, 50th and 75th percentiles and whiskers show 1.5 times the interquartile range.
(28) FIGS. **17**A-**17**D show that high-throughput CRISPR-Cas9 editing outcome screening yields replicate-consistent data that is concordant with the repair spectrum at endogenous human genomic loci. FIG. **17**A, Box and swarm plot of the Pearson correlation of the genotypic product frequency spectra at VO target sequences comparing Lib-A in mESCs with endogenous data inHEK293 (N=87), HCT116 (N=88), and K562 (N=86). Each dot represents a target sequence, the box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range. FIG. **17**B, Pearson correlation of the genotypic product frequency spectra at 1,861 Lib-A target sequences comparing two biological replicate experiments in mESCs. Median r=0.89. The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. FIG. **17**C, Distribution of Cas9-mediated genotypic products by

repair category in endogenous data at VO target sequences in K562 (N=88), HCT116 (N=92), and HEK293 (N=89). FIG. **17**D, Frequencies of deletions occurring beyond the Cas9 cutsite by distance as measured by the number of bases between the deleted base nearest to the cutsite and the two bases immediately surrounding the cutsite. Cutsite and distances are oriented with the NGG PAM on the positive side. *P<1×105 for the Pearson correlation between a specific deletion frequency distribution and Cas9 editing rates across target sequences (VO-HEK293T N=96, Lib-A mESC N=2000). Box plot as in (FIG. **17**A), with outliers beyond whiskers not depicted.
(29) FIGS. **18**A-**18**I show that sequence features correlated with higher and lower inDelphi phi scores. FIG. **18**A, Diagram of all unique alignment outcomes at an example 7-bp deletion accompanied with a table of their MH-less end-joining type, MH length, deletion length, and delta-position. FIG. **18**B, Plot of function learned by the neural network modeling MH deletions (MH-NN) mapping MH length and % GC to a numeric score (psi). FIG. **18**C, Plot of function learned by the neural network modeling MH-independent deletions (MHless-NN) mapping deletion length to a numeric score (psi). FIG. **18**D, Histogram of MHless-NN phi scores by deletion length, normalized to sum to 1. FIG. **18**E, Observed frequency of 1-bp insertion genotypes in 1,981 Lib-A target sequences with varying −4 nucleotides. The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. FIG. **18**F, Plot showing 1-bp insertion frequency in 1,996 Lib-A target sequences compared to their total phi score. Pearson correlation=−0.084 (P=1.7×10-4). FIG. **18**G, Relationship between 1-bp insertion frequency in 1,996 Lib-A target sequences compared to the predicted deletion length precision score. Pearson correlation=0.069 (P=2.1×10-3). FIG. **18**H, Diagram of hypothesized repair mechanisms that give rise to the outcome categories used by inDelphi, based on known mechanisms of MMEJ, microhomology-mediated alt-NHEJ and c-NHEJ repair pathways. Microhomology-mediated repair begins with 5′-end resection, allowing overlap of 3′-overhangs. Microhomologous basepairing of the 3′-overhangs temporarily stabilizes the ssDNA ends. In microhomology deletion, non-paired 3′-overhangs are removed and polymerase and ligase fill in and connect the gaps to reconstitute a dsDNA strand. In microhomology-less deletions, one 3′-overhang is ligated to the dsDNA backbone and the opposing strand is removed entirely, giving rise to a unilateral deletion with loss of bases on one side of the cutsite only. DNA polymerase and ligation bridge the ssDNA to create a contiguous dsDNA strand. Microhomology-independent mutations occur as a combined result of exonuclease, polymerase, and ligase activity that results in the joining of modified ends at the double strand break cutsite, giving rise to microhomology-less deletions, insertions, and mixtures thereof. FIG. **18**I shows the categories of Cas9-mediated genotypic outcomes in data from U2OS cells (n=1,958 lib-A target sites), which can be compared to the categories of Cas9-mediated genotypic outcomes shown in FIG. **17**C with regard to data from endogenous contexts at VO target sites in K562 (n=88 target sites), HCT116 (n=92), HEK293 (n=89) cells.
(30) FIGS. **19**A-**19**F show performance of inDelphi at predicting Cas9-mediated indel length and repair genotypes. FIG. **19**A, Box and swarm plot of the Pearson correlation at 189 held-out Lib-A target sequences comparing inDelphi predictions with observed mESC Lib-A genotype product frequencies. The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range. FIG. **19**B, Box and swarm plot of the Pearson correlation at 189 held-out Lib-A target sequences comparing inDelphi predictions with observed mESC Lib-A indel length frequencies for 1-bp insertions to 60-bp deletions. Box plot as in (FIG. **19**A). FIG. **19**C, Distribution of predicted frameshift frequencies among 1-60-bp deletions for SpCas9 gRNAs targeting exons, shuffled exons, and introns in the human genome. Dashed lines indicate means. ***P<10-100. FIG. **19**D, Pie chart depicting the output of Delphi for specific outcome classes. MH deletions (58% of all products) and single-base insertions (9% of all products) are predicted at single-base resolution, and deletion length is predicted for MH-less deletions (25% of all products). FIGS. **19**E and **19**F show a comparison of two methods for frameshift predictions to observed

values with Pearson's r in HCT116 cells (FIG. **19**E, n=91 target sites) and K562 cells (FIG. **19**F, n=82 target sites). The error band represents the 95% confidence intervals around the regression estimate with 1,000-fold bootstrapping.

(31) FIGS. **20**A-**20**K show target sequences with extremely high or low microhomology phi scores skew toward a single predictable Cas9-mediated edited product. FIG. **20**A, Scatter plot of the frequency of microduplication repair in Lib-B target sequences with designed 7-25-bp regions of microduplication as a function of microduplication length in human U2OS (N=32) and HEK293T cells (N=39). The Error band represents the 95% C.I. around the regression estimate with 1,000-fold bootstrapping. FIG. **20**B, Box plots displaying total deletion phi score, total precision scores, and 1-bp insertion frequencies for (blue) 312 Lib-B sequences in the low-microhomology cohort with four randomized bases flanking the cutsite (fourbp), (green) 89 VO sequences (VO), and (red) 71 Lib-B sequences in the high-microhomology cohort with microduplications ranging from 7-25 bp (longdup). Box displays median and first and third quartiles. Whiskers are at 1.5 times interquartile range (IQR). Either swarm plot or outlier fliers depicted for each box plot. *P=6.1×10-9; two-sided two-sample T-test, test statistic=−5.94, degrees of freedom=399, Hedges' g effect size=0.49. FIG. **20**C, Scatterplot of 1-bp insertion frequency among all non-wild-type products when varying four bases surrounding the cutsite (positions −5 to −2 counted from the NGG-PAM at positions 0-2) with all x-tick labels depicted, contained within three target sequences (red, blue, green) from the low-microhomology cohort of Lib-B in mESCs (N=205). FIG. **20**D, Distribution of the total frequency of all non-wild-type Cas9 editing products in the subset of target sequences from the low-microhomology cohort containing four randomized bases flanking the cutsite (fourbp) with >50% overall frequencies of 1-bp insertion (N=50), VO sequences (N=89), and the high-microhomology cohort with microduplications ranging from 7-25 bp (longdup) in Lib-B editing in mESCs (N=56). FIG. **20**E, Scatterplot displaying 1-bp insertion frequencies and Cas9 editing rate in 205 “fourbp” contexts with Pearson correlation of −0.35 (P=3.3e-07). FIG. **20**F shows the frequency of 1-bp insertions in mESCs (n=1,981 lib-A target sites) and U2OS cells (n=1,918) with varying −4 nucleotides. FIGS. **20**G and **20**H show plots of 1-bp insertion frequency in mESCs (n=1,996 lib-A target sites) and U2OS cells (n=1,966) compared to their total phi score (FIG. **20**G) and predicted deletion length precision score (FIG. **20**H) with Pearson's r. FIG. **20**I shows a comparison of 1-bp insertion frequencies among all edited products from 1,966 lib-A target sites in U2OS cells (combined data from n=2 independent biological replicates). FIG. **20**J shows nucleotides and their effect on the frequency of 1-bp insertions in U2OS cells. Only bases with non-zero linear regression weights in 10,000-fold iterative cross-validation are shown. Total n=1,966 lib-A target sites. FIG. **20**K shows the insertion frequency in mESCs (n=205) and U2OS cells (n=217) when varying four bases by the cleavage site (positions −5 to −2 counted from the NGG-PAM at positions 0-2) contained within three target sites designed with weak microhomology.

(32) FIGS. **21**A-**21**D show the precise repair of pathogenic microduplications. FIG. **21**A, Observed frequencies of repair to wild-type genotype at 194 ClinVar pathogenic alleles vs. predicted frequencies in Lib-B in human HEK293T cells. FIG. **21**B, Observed frequencies of repair to wild-type frame at 140 ClinVar pathogenic alleles vs. predicted frequencies in Lib-B in human HEK293T cells. FIG. **21**C, Observed frequencies of repair to wild-type genotype at 49 Clinvar pathogenic alleles vs. predicted frequencies in Lib-B in human U2OS cells. FIG. **21**D, Observed frequencies of repair to wild-type frame at 37 ClinVar pathogenic alleles vs. predicted frequencies in Lib-B in human U2OS cells.

(33) FIGS. **22**A-**22**E show altered distribution of Cas9-mediated genotypic products in Prkdc−/−Lig4−/− mESCs as compared to wild-type mESCs. FIG. **22**A, Distribution of Cas9-mediated genotypic products by repair outcome class in Prkdc−/−Lig4−/− mESC for 1,446 target sequences.

(34) FIG. **22**B, Comparison of observed mean frequency of deletion products contributed by microhomology-less unilateral joining and medial joining deletions among all deletions comparing

1,995 Lib-A target sequences in wildtype mESC to 1,850 Lib-A target sequences in Prkdc−/−Lig4−/− mESC. *P<10-66; two-sided two-sample T-test, test statistic >17.7, degrees of freedom=3,843; Hedges' g effect size >0.58. FIG. **22**C, Comparison of observed frequency of deletion products contributed by microhomology-less unilateral joining and medial joining deletions among all deletions, between 1,995 Lib-A target sequences in wildtype mESC to 1,850 Lib-A target sequences in Prkdc−/−Lig4−/− mESC.*and**as in (FIG. **22**B). The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as fliers. FIG. **22**D, Observed mean frequency of 1-bp insertion genotypes at 1,055 target sequences with varying −4 nucleotides in Lib-A in Prkdc−/−Lig4−/− mESCs. The error bars show the 95% C.I. on the sample mean with 1,000-fold bootstrapping. FIG. **22**E, Observed frequency of 1-bp insertion genotypes at 1,055 target sequences with varying −4 nucleotides in Lib-A in Prkdc−/−Lig4−/− mESCs. Box plot as in (b).

(35) FIGS. **23**A-**23**H show that template-free Cas9-nuclease editing of human cells containing pathogenic LDLR microduplication alleles restores LDL uptake. FIG. **23**A, Flow cytometric contour plots showing GFP fluorescence and LDL-Dylight550 uptake in HCT116 cells containing the denoted LDLR alleles and treated with SaCas9 and gRNA when denoted. FIG. **23**B, Fluorescence microscopy of HCT116 cells containing the denoted LDLR alleles and treated with SaCas9 and gRNA when denoted. GFP fluorescence is shown in green, LDL-Dylight550 uptake in red, and Hoechst staining nuclei in blue. FIG. **23**C, Fluorescence microscopy of U2OS cells containing the denoted LDLR alleles and treated with SaCas9 and gRNA when denoted. GFP fluorescence is shown in green, LDL-Dylight550 uptake in red, and Hoechst staining nuclei in blue. FIG. **23**D, Flow cytometry gating strategy used for mESC+LDLRdup-P2A-GFP untreated. FIG. **23**E, Flow cytometry gating strategy used for mESC+LDLRdup-P2A-GFP+SpCas9+gRNA. FIGS. **23**F and **23**G show the results of 12 pathogenic 1-bp deletion alleles selected by inDelphi for high 1-bp insertion frequency (combined data from n=2 independent biological replicates) compared to lib-A (f) and presented in a table (FIG. **23**G). The box denotes the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as diamonds. *P=1.6×10-4, two-sided Welch's t-test. For detailed statistics, see Methods. In the table, the most frequent 1-bp insertion genotype predicted by inDelphi that does not correspond to the wild-type genotype is indicated by an asterisk. In fluorescence microscopy plots, GFP fluorescence is shown in green, LDL-Dylight550 uptake in red, and Hoechst staining nuclei in blue. FIG. **23**H shows mESC-trained inDelphi genotype prediction accuracy as 40 library sites.

(36) FIG. **24**A is a schematic depicting an exemplary method of using a trained computational model (e.g., “inDelphi”) in conjunction with a Cas-based genome editing system to edit a nucleotide sequence (e.g., a genome) to achieve a desired genetic outcome (e.g., a correction to a disease-causing mutation to treat a disease, or modification of a wildtype type gene to confer an improved trait or phenotype). For any given set of inputs (a context sequence and a selected cut site), the trained computational model computes the probability distribution of indel lengths and the probability distribution of genotype frequencies, enabling the user to select the optimal input (e.g., cut site) for conducting editing by a Cas-based genome editing system to achieve the highest frequency of desired genetic output. The computational method may be used to predict, for a given local sequence context, template-free repair genotypes and frequencies of occurrence thereof.

(37) FIG. **24**B is a schematic depicting an exemplary method of using a trained computational model (e.g., “inDelphi”) in conjunction with a double-strand break (DSB)-inducing genome editing system to edit a nucleotide sequence (e.g., a genome) to achieve a desired genetic outcome (e.g., a correction to a disease-causing mutation to treat a disease, or modification of a wildtype type gene to confer an improved trait or phenotype). For any given set of inputs (a context sequence and a selected cut site), the trained computational model computes the probability distribution of indel lengths and the probability distribution of genotype frequencies, enabling the user to select the optimal input (e.g., cut site) for conducting editing by a DSB-inducing genome editing system to

achieve the highest frequency of desired genetic output. The computational method may be used to predict, for a given local sequence context, template-free repair genotypes and frequencies of occurrence thereof.

(38) FIG. **25**A-**25**D provides a characterization of lib-B data including pathogenic microduplication repair in wild-type mESCs, wild-type U2OS cells and mESCs treated with DPKi3, NU7026 and MLN4924. FIG. **25**A shows box plots of the number of unique high-confidence editing outcomes called by simulating data subsampling in data at 2,000 lib-B target sites in mESCs (combined data from n=2 independent technical replicates) and U2OS cells (combined data from n=2 independent biological replicates). In ‘all’, the full non-subsampled data are presented (see Table 8 herein for read counts). Each box depicts data for 2,000 target sites. The box denotes the 25th, 50th, and 75th percentiles and whiskers show 1.5 times the interquartile range. Outliers are not depicted. FIG. **25**B shows the frequencies of repair to wild-type genotype at 567 ClinVar pathogenic alleles versus predicted frequencies in lib-B in human U2OS cells with Pearson's r. FIG. **25**C shows the frequencies of repair to wild-type frame at 437 ClinVar pathogenic alleles versus predicted frequencies in lib-B in human U2OS cells with Pearson's r. FIG. **25**D shows the frequency of pathogenic microduplication repair in wild-type mESCs (n=1,480 target sites) compared to mESCs treated with MLN4924 (n=1,569), NU7041 (n=1,561) and DPKi3 (n=1,563).

(39) FIG. **26**A-**26**G shows the altered distributions of Cas9-mediated genotypic products in Prkdc−/−Lig4−/− mESCs and mESCs treated with DPKi3, NU7026, and MLN4924 compared to wild-type mESCs. FIG. **26**A shows a comparison of MH deletions among all deletions at lib-B target sites in wild-type cells (n=1,909 target sites), cells treated with DPKi3 (n=1,999), MLN4924 (n=1,995) or NU7026 (n=1,999) and Prkdc−/−Lig4−/− cells (n=1,446). Statistical tests performed against wild-type population. *P=5.6×10-5, **P=3.5×10-13, ***P=5.0×10-41, two-sided Welch's t-test. FIG. **26**B shows a comparison of the frequency of each class of MH-less deletions among all deletion products in wild-type (lib-A and lib-B target sites, n=3,829 target sites), DPKi3 (lib-B, n=1,990), MLN4924 (lib-B, n=1,980), NU7026 (lib-B, n=1,992) and Prkdc−/−Lig4−/− (lib-A and lib-B target sites, n=3,344). P values are compared to wild-type, two-sided Welch's t-test. FIG. **26**C shows frequency of 1-bp insertions at 1,055 target sites in lib-A in Prkdc−/−Lig4−/− mESCs. FIG. **26**D Frequencies of deletion repair to wild-type genotype in lib-B in wild-type mESCs (n=1,480 target sites, combined data from two technical replicates) compared to conditions, with combined data from two independent biological replicates for each of Prkdc−/−Lig4−/− (n=1,041 target sites), MLN4924 (n=1,569), NU7026 (n=1,561) and DPKi3 (n=1,563). FIG. **26**E provides a table of Pearson's r of the change in disease correction frequency compared to wild-type at n=791 target sites for each pair of conditions. f, g, Annexin V-568 staining flow cytometry contour plots (FIG. **26**F) and mean±standard deviation values (FIG. **26**G) in wild-type and Prkdc−/−Lig4−/− lib-A mESCs following transfection with SpCas9-P2A-GFP (representative data for n=2 experiments). Box plots denote the 25th, 50th and 75th percentiles, whiskers show 1.5 times the interquartile range, and outliers are depicted as diamonds.

DEFINITIONS

(40) As used herein and in the claims, the singular forms “a,” “an,” and “the” include the singular and the plural reference unless the context clearly indicates otherwise. Thus, for example, a reference to “an agent” includes a single agent and a plurality of such agents. The term “Cas9” or “Cas9 nuclease” refers to an RNA-guided nuclease comprising a Cas9 protein, or a fragment thereof (e.g., a protein comprising an active or inactive DNA cleavage domain of Cas9, and/or the gRNA binding domain of Cas9). A Cas9 nuclease is also referred to sometimes as a casn1 nuclease or a CRISPR (clustered regularly interspaced short palindromic repeat)-associated nuclease. CRISPR is an adaptive immune system that provides protection against mobile genetic elements (viruses, transposable elements and conjugative plasmids). CRISPR clusters contain spacers, sequences complementary to antecedent mobile elements, and target invading nucleic acids. CRISPR clusters are transcribed and processed into CRISPR RNA (crRNA). In type II CRISPR

systems correct processing of pre-crRNA requires a trans-encoded small RNA (tracrRNA), endogenous ribonuclease 3 (rnc) and a Cas9 protein. The tracrRNA serves as a guide for ribonuclease 3-aided processing of pre-crRNA. Subsequently, Cas9/crRNA/tracrRNA endonucleolytically cleaves linear or circular dsDNA target complementary to the spacer. The target strand not complementary to crRNA is first cut endonucleolytically, then trimmed 3′-5′ exonucleolytically. In nature, DNA-binding and cleavage typically requires protein and both RNAs. However, single guide RNAs ("sgRNA", or simply "gNRA") can be engineered so as to incorporate aspects of both the crRNA and tracrRNA into a single RNA species. See, e.g., Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. *Science* 337:816-821(2012), the entire contents of which is hereby incorporated by reference. Cas9 recognizes a short motif in the CRISPR repeat sequences (the PAM or protospacer adjacent motif) to help distinguish self versus non-self. Cas9 nuclease sequences and structures are well known to those of skill in the art (see, e.g., "Complete genome sequence of an M1 strain of *Streptococcus pyogenes*." Ferretti et al., J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najar F. Z., Ren Q., Zhu H., Song L., White J., Yuan X., Clifton S. W., Roe B. A., McLaughlin R. E., Proc. Natl. Acad. Sci. U.S.A. 98:4658-4663(2001); "CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III." Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., Nature 471:602-607(2011); and "A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity." Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. *Science* 337:816-821(2012), the entire contents of each of which are incorporated herein by reference). Cas9 orthologs have been described in various species, including, but not limited to, *S. pyogenes* and *S. thermophilus*. Additional suitable Cas9 nucleases and sequences will be apparent to those of skill in the art based on this disclosure, and such Cas9 nucleases and sequences include Cas9 sequences from the organisms and loci disclosed in Chylinski, Rhun, and Charpentier, "The tracrRNA and Cas9 families of type II CRISPR-Cas immunity systems" (2013) RNA Biology 10:5, 726-737; the entire contents of which are incorporated herein by reference. In some embodiments, a Cas9 nuclease has an inactive (e.g., an inactivated) DNA cleavage domain.

(41) In some embodiments, Cas9 refers to Cas9 from: *Corynebacterium ulcerans* (NCBI Refs: NC_015683.1, NC_017317.1); *Corynebacterium diphtheria* (NCBI Refs: NC_016782.1, NC_016786.1); *Spiroplasma syrphidicola* (NCBI Ref: NC_021284.1); *Prevotella intermedia* (NCBI Ref: NC_017861.1); *Spiroplasma taiwanense* (NCBI Ref: NC_021846.1); *Streptococcus* iniae (NCBI Ref: NC_021314.1); *Belliella baltica* (NCBI Ref: NC_018010.1); *Psychroflexus torques* I (NCBI Ref: NC_018721.1); *Streptococcus thermophilus* (NCBI Ref: YP_820832.1); *Listeria innocua* (NCBI Ref: NP_472073.1); *Campylobacter jejuni* (NCBI Ref: YP_002344900.1); or *Neisseria meningitidis* (NCBI Ref: YP_002342100.1).

(42) The term "Cas-based genome editing system" refers to a system comprising any naturally occurring or variant Cas endonuclease (e.g., Cas9), or functional variant, homolog, or orthologue thereof, and a cognate guide RNA. The term "Cas-based genome editing system" may also refer to an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA.

(43) The term "DSB-based genome editing system" refers to a system comprising any naturally occurring or variant endonuclease which catalyzes the formation of a double strand break at a cut site (e.g., Cas9, Crf1, TALEN, or Zinc Finger), or functional variant, homolog, or orthologue thereof, and a cognate guide RNA if required (e.g., TALENs and Zinc Fingers do not require a guide RNA for targeting to a cut site). The term "DSB-based genome editing system" may also refer to an expression vector having at least one expressible nucleotide sequence encoding a DSB endonuclease protein (or homolog, variant, or orthologue thereof) and at least one other expressible

nucleotide sequence encoding a guide RNA, if required (e.g., as required for Cas9 or Crf1).
(44) The term "effective amount," as used herein, refers to an amount of a biologically active agent that is sufficient to elicit a desired biological response. For example, in some embodiments, an effective amount of a nuclease may refer to the amount of the nuclease that is sufficient to induce cleavage of a target site specifically bound and cleaved by the nuclease. In some embodiments, an effective amount of a recombinase may refer to the amount of the recombinase that is sufficient to induce recombination at a target site specifically bound and recombined by the recombinase. As will be appreciated by the skilled artisan, the effective amount of an agent, e.g., a nuclease, a recombinase, a hybrid protein, a fusion protein, a protein dimer, a complex of a protein (or protein dimer) and a polynucleotide, or a polynucleotide, may vary depending on various factors as, for example, on the desired biological response, the specific allele, genome, target site, cell, or tissue being targeted, and the agent being used.
(45) The term "linker," as used herein, refers to a chemical group or a molecule linking two molecules or moieties, e.g., a binding domain and a cleavage domain of a nuclease. In some embodiments, a linker joins a gRNA binding domain of an RNA-programmable nuclease and the catalytic domain of a recombinase. In some embodiments, a linker joins a dCas9 and a recombinase. Typically, the linker is positioned between, or flanked by, two groups, molecules, or other moieties and connected to each one via a covalent bond, thus connecting the two. In some embodiments, the linker is an amino acid or a plurality of amino acids (e.g., a peptide or protein). In some embodiments, the linker is an organic molecule, group, polymer, or chemical moiety. In some embodiments, the linker is 5-100 amino acids in length, for example, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 30-35, 35-40, 40-45, 45-50, 50-60, 60-70, 70-80, 80-90, 90-100, 100-150, or 150-200 amino acids in length. Longer or shorter linkers are also contemplated.
(46) The term "mutation," as used herein, refers to a substitution of a residue within a sequence, e.g., a nucleic acid or amino acid sequence, with another residue, or a deletion or insertion of one or more residues within a sequence. Mutations are typically described herein by identifying the original residue followed by the position of the residue within the sequence and by the identity of the newly substituted residue. Various methods for making the amino acid substitutions (mutations) provided herein are well known in the art, and are provided by, for example, Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4.sup.th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)). Mutations can include a variety of categories, such as single base polymorphisms, microduplication regions, indel, and inversions, and is not meant to be limiting in any way.
(47) The terms "nucleic acid" and "nucleic acid molecule," as used herein, refer to a compound comprising a nucleobase and an acidic moiety, e.g., a nucleoside, a nucleotide, or a polymer of nucleotides. Typically, polymeric nucleic acids, e.g., nucleic acid molecules comprising three or more nucleotides are linear molecules, in which adjacent nucleotides are linked to each other via a phosphodiester linkage. In some embodiments, "nucleic acid" refers to individual nucleic acid residues (e.g. nucleotides and/or nucleosides). In some embodiments, "nucleic acid" refers to an oligonucleotide chain comprising three or more individual nucleotide residues. As used herein, the terms "oligonucleotide" and "polynucleotide" can be used interchangeably to refer to a polymer of nucleotides (e.g., a string of at least three nucleotides). In some embodiments, "nucleic acid" encompasses RNA as well as single and/or double-stranded DNA. Nucleic acids may be naturally occurring, for example, in the context of a genome, a transcript, an mRNA, tRNA, rRNA, siRNA, snRNA, a plasmid, cosmid, chromosome, chromatid, or other naturally occurring nucleic acid molecule. On the other hand, a nucleic acid molecule may be a non-naturally occurring molecule, e.g., a recombinant DNA or RNA, an artificial chromosome, an engineered genome, or fragment thereof, or a synthetic DNA, RNA, DNA/RNA hybrid, or including non-naturally occurring nucleotides or nucleosides. Furthermore, the terms "nucleic acid," "DNA," "RNA," and/or similar

terms include nucleic acid analogs, e.g., analogs having other than a phosphodiester backbone. Nucleic acids can be purified from natural sources, produced using recombinant expression systems and optionally purified, chemically synthesized, etc. Where appropriate, e.g., in the case of chemically synthesized molecules, nucleic acids can comprise nucleoside analogs such as analogs having chemically modified bases or sugars, and backbone modifications. A nucleic acid sequence is presented in the 5′ to 3′ direction unless otherwise indicated. In some embodiments, a nucleic acid is or comprises natural nucleosides (e.g. adenosine, thymidine, guanosine, cytidine, uridine, deoxyadenosine, deoxythymidine, deoxyguanosine, and deoxycytidine); nucleoside analogs (e.g., 2-aminoadenosine, 2-thiothymidine, inosine, pyrrolo-pyrimidine, 3-methyl adenosine, 5-methylcytidine, 2-aminoadenosine, C5-bromouridine, C5-fluorouridine, C5-iodouridine, C5-propynyl-uridine, C5-propynyl-cytidine, C5-methylcytidine, 2-aminoadenosine, 7-deazaadenosine, 7-deazaguanosine, 8-oxoadenosine, 8-oxoguanosine, O(6)-methylguanine, and 2-thiocytidine); chemically modified bases; biologically modified bases (e.g., methylated bases); intercalated bases; modified sugars (e.g., 2′-fluororibose, ribose, 2′-deoxyribose, arabinose, and hexose); and/or modified phosphate groups (e.g., phosphorothioates and 5′-N-phosphoramidite linkages).
(48) The terms “protein,” “peptide,” and “polypeptide” are used interchangeably herein, and refer to a polymer of amino acid residues linked together by peptide (amide) bonds. The terms refer to a protein, peptide, or polypeptide of any size, structure, or function. Typically, a protein, peptide, or polypeptide will be at least three amino acids long. A protein, peptide, or polypeptide may refer to an individual protein or a collection of proteins. One or more of the amino acids in a protein, peptide, or polypeptide may be modified, for example, by the addition of a chemical entity such as a carbohydrate group, a hydroxyl group, a phosphate group, a farnesyl group, an isofarnesyl group, a fatty acid group, a linker for conjugation, functionalization, or other modification, etc. A protein, peptide, or polypeptide may also be a single molecule or may be a multi-molecular complex. A protein, peptide, or polypeptide may be just a fragment of a naturally occurring protein or peptide. A protein, peptide, or polypeptide may be naturally occurring, recombinant, or synthetic, or any combination thereof. The term “fusion protein” as used herein refers to a hybrid polypeptide which comprises protein domains from at least two different proteins. One protein may be located at the amino-terminal (N-terminal) portion of the fusion protein or at the carboxy-terminal (C-terminal) protein thus forming an “amino-terminal fusion protein” or a “carboxy-terminal fusion protein,” respectively. A protein may comprise different domains, for example, a nucleic acid binding domain (e.g., the gRNA binding domain of Cas9 that directs the binding of the protein to a target site) and a nucleic acid cleavage domain or a catalytic domain of a recombinase. In some embodiments, a protein comprises a proteinaceous part, e.g., an amino acid sequence constituting a nucleic acid binding domain, and an organic compound, e.g., a compound that can act as a nucleic acid cleavage agent. In some embodiments, a protein is in a complex with, or is in association with, a nucleic acid, e.g., RNA. Any of the proteins provided herein may be produced by any method known in the art. For example, the proteins provided herein may be produced via recombinant protein expression and purification, which is especially suited for fusion proteins comprising a peptide linker. Methods for recombinant protein expression and purification are well known, and include those described by Green and Sambrook, Molecular Cloning: A Laboratory Manual (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)), the entire contents of which are incorporated herein by reference.
(49) The term “RNA-programmable nuclease,” and “RNA-guided nuclease” are used interchangeably herein and refer to a nuclease that forms a complex with (e.g., binds or associates with) one or more RNA that is not a target for cleavage. In some embodiments, an RNA-programmable nuclease, when in a complex with an RNA, may be referred to as a nuclease:RNA complex. Typically, the bound RNA(s) is referred to as a guide RNA (gRNA). gRNAs can exist as a complex of two or more RNAs, or as a single RNA molecule. gRNAs that exist as a single RNA molecule may be referred to as single-guide RNAs (sgRNAs), though “gRNA” is used

interchangeabley to refer to guide RNAs that exist as either single molecules or as a complex of two or more molecules. Typically, gRNAs that exist as single RNA species comprise two domains: (1) a domain that shares homology to a target nucleic acid (e.g., and directs binding of a Cas9 complex to the target); and (2) a domain that binds a Cas9 protein. In some embodiments, domain (2) corresponds to a sequence known as a tracrRNA, and comprises a stem-loop structure. For example, in some embodiments, domain (2) is homologous to a tracrRNA as depicted in FIG. **1**E of Jinek et al., *Science* 337:816-821(2012), the entire contents of which is incorporated herein by reference. Other examples of gRNAs (e.g., those including domain 2) can be found in U.S. Provisional Patent Application, U.S. Ser. No. 61/874,682, filed Sep. 6, 2013, entitled “Switchable Cas9 Nucleases And Uses Thereof,” and U.S. Provisional Patent Application, U.S. Ser. No. 61/874,746, filed Sep. 6, 2013, entitled “Delivery System For Functional Nucleases,” the entire contents of each are hereby incorporated by reference in their entirety. In some embodiments, a gRNA comprises two or more of domains (1) and (2), and may be referred to as an “extended gRNA.” For example, an extended gRNA will, e.g., bind two or more Cas9 proteins and bind a target nucleic acid at two or more distinct regions, as described herein. The gRNA comprises a nucleotide sequence that complements a target site, which mediates binding of the nuclease/RNA complex to said target site, providing the sequence specificity of the nuclease:RNA complex. In some embodiments, the RNA-programmable nuclease is the (CRISPR-associated system) Cas9 endonuclease, for example Cas9 (Csn1) from *Streptococcus pyogenes* (see, e.g., “Complete genome sequence of an M1 strain of *Streptococcus pyogenes*.” Ferretti J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najar F. Z., Ren Q., Zhu H., Song L., White J., Yuan X., Clifton S. W., Roe B. A., McLaughlin R. E., Proc. Natl. Acad. Sci. U.S.A. 98:4658-4663(2001); “CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III.” Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., Nature 471:602-607(2011); and “A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity.” Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. Science 337:816-821(2012), the entire contents of each of which are incorporated herein by reference.

(50) Because RNA-programmable nucleases (e.g., Cas9) use RNA:DNA hybridization to target DNA cleavage sites, these proteins are able to be targeted, in principle, to any sequence specified by the guide RNA. Methods of using RNA-programmable nucleases, such as Cas9, for site-specific cleavage (e.g., to modify a genome) are known in the art (see e.g., Cong, L. et al. Multiplex genome engineering using CRISPR/Cas systems. *Science* 339, 819-823 (2013); *Mali*, P. et al. RNA-guided human genome engineering via Cas9. *Science* 339, 823-826 (2013); Hwang, W. Y. et al. Efficient genome editing in zebrafish using a CRISPR-Cas system. *Nature biotechnology* 31, 227-229 (2013); Jinek, M. et al. RNA-programmed genome editing in human cells. *eLife* 2, e00471 (2013); Dicarlo, J. E. et al. Genome engineering in *Saccharomyces cerevisiae* using CRISPR-Cas systems. *Nucleic acids research* (2013); Jiang, W. et al. RNA-guided editing of bacterial genomes using CRISPR-Cas systems. *Nature biotechnology* 31, 233-239 (2013); the entire contents of each of which are incorporated herein by reference).

(51) The term “subject,” as used herein, refers to an individual organism, for example, an individual mammal. In some embodiments, the subject is a human. In some embodiments, the subject is a non-human mammal. In some embodiments, the subject is a non-human primate. In some embodiments, the subject is a rodent. In some embodiments, the subject is a sheep, a goat, a cattle, a cat, or a dog. In some embodiments, the subject is a vertebrate, an amphibian, a reptile, a fish, an insect, a fly, or a nematode. In some embodiments, the subject is a research animal. In some embodiments, the subject is genetically engineered, e.g., a genetically engineered non-human subject. The subject may be of either sex and at any stage of development.

(52) The terms “treatment,” “treat,” and “treating,” refer to a clinical intervention aimed to reverse,

alleviate, delay the onset of, or inhibit the progress of a disease or disorder, or one or more symptoms thereof, as described herein. As used herein, the terms "treatment," "treat," and "treating" refer to a clinical intervention aimed to reverse, alleviate, delay the onset of, or inhibit the progress of a disease or disorder, or one or more symptoms thereof, as described herein. In some embodiments, treatment may be administered after one or more symptoms have developed and/or after a disease has been diagnosed. In other embodiments, treatment may be administered in the absence of symptoms, e.g., to prevent or delay onset of a symptom or inhibit onset or progression of a disease. For example, treatment may be administered to a susceptible individual prior to the onset of symptoms (e.g., in light of a history of symptoms and/or in light of genetic or other susceptibility factors). Treatment may also be continued after symptoms have resolved, for example, to prevent or delay their recurrence.

(53) Use of ordinal terms such as "first," "second," "third," etc. in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

(54) Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of "including," "comprising," "having," "containing," "involving," and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

Detailed Description of Certain Embodiments

(55) Major research efforts focus on improving efficiency and specificity of genome editing systems, such as, CRISPR/Cas9, other Cas-based, TALEN-based, and Zinc Finger-based genome editing systems. For instance, with regard to CRISPR/Cas9 systems, efficiency may be improved by predicting optimal Cas9 guide RNA (gRNA) sequences, while specificity may be improved by modeling factors leading to off-target cutting, and by manipulating Cas9 enzymes. Variant Cas9 enzymes and fusion proteins may be developed to alter the protospacer adjacent motif (PAM) sequences acted on by Cas9, and to produce base-editing Cas9 constructs with high efficiency and specificity. For example, Cpf1 (also known as Cas12a) and other alternatives may be used in CRISPR genome editing in addition to, or instead of, Cas9.

(56) The inventors have recognized and appreciated that less attention has been devoted to understanding and modulating repair outcomes. In that respect, nucleotide insertions and/or deletions resulting from template-free repair mechanisms (e.g., NHEJ, MMEJ, etc. and excluding homology-based repair (HDR)) are commonly thought to be random and therefore only suitable for gene knock-out applications. For gene knock-in or gain-of-function applications, a template-based repair mechanism such as HDR is typically used.

(57) CRISPR/Cas with HDR allows arbitrarily designed DNA sequences to be incorporated at precise genomic locations. However, this technique suffers from low efficiency—HDR occurs rarely in typical biological conditions (e.g., around 10% frequency), because cells only permit HDR to occur after sister chromatids are synthesized in S phase but before M phase when mitosis splits the sister chromatids into daughter cells. For many cell-types, the fraction of time spent in S-G2-M phases of a cell cycle is low. In sum, while outcomes are predictable when HDR does occur, HDR occurs infrequently, and therefore a desired DNA sequence will be incorporated into only a small percentage of cells. In addition, in post-mitotic cell-types of interest such as neurons, the HDR repair pathway is no longer used, further limiting HDR's utility for genetic engineering.

(58) Some research has been done to improve efficiency of HDR, for example, through improved homology templates and small molecule modulation. Despite these efforts, template-based repair efficiency remains low, and proposed CRISPR/Cas gene knock-in or gain of function applications have thus far been limited to ex vivo applications where screening may be performed for cells with a desired repair genotype.

(59) Unlike HDR, NHEJ is capable of occurring during any phase of a cell cycle and in post-mitotic cells. However, NHEJ, as discussed above, has been perceived as a random process that produces a large variety of repair genotypes with insertions and/or deletions, and has been used mainly to knock out genes. In short, NHEJ is efficient but unpredictable.
(60) Recent work suggests that outcomes of some template-free repair mechanisms are actually non-random. For instance, it has been observed that MMEJ is involved in repair outcomes. Furthermore, repair outcomes have been analyzed to predict gRNAs that are more likely to produce frameshifts. However, there is still a need for accurate prediction of genotypic outcomes of CRISPR/Cas cutting and ensuing cellular DNA repair.
(61) The present inventors have unexpectedly found through computational analyses that template-free DNA/genome editing systems, e.g., CRISPR/Cas9, Cas-based, Cpf1-based, or other DSB (double-strand break)-based genome editing systems, produce a predictable set of repair genotypes thereby enabling the use of such editing systems for applications involving or requiring precise manipulation of DNA, e.g., the correction of a disease-causing genetic mutation or modifying a wildtype sequence to confer a genetic advantage. This finding is contrary to the accepted view that DNA double-strand break repair (i.e., template-free, non-homology-dependent repair) following cleavage by genome editing systems produces stochastic and heterogenous repair products and are therefore impractical for applications beyond gene disruption. Thus, the specification describes and discloses in various aspects and embodiments computational-based methods and systems for practically harnessing the innate efficiencies of template-free DNA repair systems for carrying out precise DNA and/or genomic editing without the reliance upon homology-based repair.
(62) In accordance with some embodiments, techniques are provided for predicting genotypes of CRISPR/Cas editing outcomes. For instance, a high-throughput approach may be used for monitoring CRISPR/Cas cutting outcomes, and/or a computer-implemented method may be used to predict genotypic repair outcomes for NHEJ and/or MMEJ. The inventors have recognized and appreciated that accurate prediction of repair genotypes may allow development of CRISPR/Cas gene knock-in or gain-of-function applications based on one or more template-free repair mechanisms. This approach may simplify a genome editing process, by reducing or eliminating a need to introduce exogenous DNA into a cell as a template.
(63) Additionally, or alternatively, using one or more template-free repair mechanisms for gene knock-in may provide improved efficiency. For instance, the inventors have recognized and appreciated that NHEJ and MMEJ may account for a large portion of CRISPR/Cas repair products. While template-free repair mechanisms may not always produce desired repair genotypes with sufficiently high frequencies, one or more desired repair genotypes may occur with sufficiently high frequencies in some specific local sequence contexts. For such a local sequence context, template-free repair mechanisms may outperform HDR with respect to simplicity and efficiency.
(64) In some embodiments, one or more of the techniques provided herein may be used to predict, for a given local sequence context, template-free repair genotypes and frequencies of occurrence thereof, which may facilitate designs of gene knock-in or gain-of-function applications. For example, the inventors have recognized and appreciated that some disease-causing alleles, when cut at a selected location by CRISPR/Cas, may exhibit one or just a few repair outcomes that occur at a high frequency and transform the disease-causing allele into one or more healthy alleles. Disease-causing alleles may occur in genomic sequences that code for proteins or regulatory RNAs, or genomic sequences that regulate transcription or other genomic functions.
(65) In some embodiments, one or more of the techniques provided herein may be used to predict, for a given local sequence context, template-free repair genotypes and frequencies of occurrence thereof, which may be used to select desirable one or more guide RNAs when HDR is employed to edit DNA. Since HDR does not occur 100% of the time, the template-free repair genotypes predicted by this method will be a natural byproduct of sites where HDR failed to occur. The one or more techniques provided herein allow these failed HDR byproducts to be predicted and one or

more guide RNAs chosen that will produce the most desirable byproducts for HDR failures. For example, a disease-causing allele may be targeted for HDR repair, but if HDR does not occur at a specific site the template-free repair products can be chosen to transform a disease-causing allele into one or more healthy alleles or to not have deleterious effects. Deleterious effects could result from template-free repair that changed a weakly functional allele into a non-functional allele or into a dominant allele that negatively impacted health. In some embodiments, guide RNA selection consists of considering all guide RNAs that are compatible with HDR repair of a disease-causing allele, and for each guide RNA using one or more of the techniques provided herein to predict its template-free repair genotypes. One or more guide RNAs are then selected for use with the HDR template that have the template-free repair genotypes that are most advantageous for health. Alternatively in some embodiments, one or more guide RNAs are then selected for use with the HDR template that have the template-free repair genotypes that are most likely to disrupt gene function.

(66) It should be appreciated that the techniques disclosed herein may be implemented in any of numerous ways, as the disclosed techniques are not limited to any particular manner of implementation. Examples of details of implementation are provided solely for illustrative purposes. For instance, while examples are given where CRISPR/Cas9 is used to perform genome editing, it should be appreciated that aspects of the present application are not so limited. In some embodiments, another genome editing technique, such as CRISPR/Cpf1, may be used. Furthermore, the disclosed techniques may be used individually or in any suitable combination, as aspects of the present disclosure are not limited to the use of any particular technique or combination of techniques.

(67) FIG. **1** shows an illustrative DNA segment **100**, in accordance with some embodiments. For instance, the DNA segment **100** may be exon 43 of a dystrophin gene. About 4% of Duchenne's muscular dystrophy cases are caused by mutations in this exon. Therapeutic solutions showing success in clinical trials use antisense oligonucleotides to cause this exon to be skipped during translation, thereby restoring normal dystrophin function.

(68) The inventors have recognized and appreciated that another therapeutic approach may be possible, using genome editing to make permanent changes to dystrophin exon 43. For instance, in some embodiments, CRISPR/Cas9 (or another suitable technique for cutting a DNA sequence, such as CRISPR/Cpf1) may be used to disrupt a donor splice site motif of dystrophin exon 43, and one or more template-free repair mechanisms may restore normal dystrophin function.

(69) In one aspect, the specification discloses a computational model.

(70) In certain embodiments, the computational model can predict and/or compute an optimized or preferred cut site for a DSB-based genome editing system for introducing a genetic change into a nucleotide sequence. In preferred embodiments, the repair does not require homology-based repair mechanisms.

(71) In certain other embodiments, the computational model can predict and/or compute an optimized or preferred cut site for a Cas-based genome editing system for introducing a genetic change into a nucleotide sequence. In preferred embodiments, the repair does not require homology-based repair mechanisms.

(72) In still other embodiments, the computation model provides for the selection of a optimized or preferred guide RNA for use with a Cas-based genome editing system for introducing a genetic change in a genome. In preferred embodiments, the repair does not require homology-based repair mechanisms.

(73) In various embodiments, the computational model is a neural network model having one or more hidden layers.

(74) In other embodiments, the computational model is a deep learning computational model.

(75) In various embodiments, that the DSB-based genome editing system (e.g., a Cas-based genome editing system) edits the genome without relying on homology-based repair.

(76) In various embodiments, that computational model is trained with experimental data to predict the probability of distribution of indel lengths for any given nucleotide sequence and cut site. In other embodiments, computational model is trained with experimental data to predict the probability of distribution of genotype frequencies for any given nucleotide sequence and cut site.
(77) In embodiments, the computational model comprises one or more training modules for evaluating experimental data.
(78) In an embodiment, the computational model comprises: a first training module (**305**) for computing a microhomology score matrix (**305**); a second training module (**310**) for computing a microhomology independent score matrix; and a third training module (**315**) for computing a probability distribution over 1-bp insertions, wherein once trained with experimental data the computational model computes a probability distribution over indel genotypes and a probability distribution over indel lengths for any given input nucleotide sequence and cut site.
(79) In certain embodiments, the computational model predicts genomic repair outcomes for any given input nucleotide sequence (i.e., context sequence) and cut site.
(80) In certain embodiments, the genomic repair outcomes comprise microhomology deletions, microhomology-less deletions, and 1-bp insertions.
(81) In various embodiments, the one or more modules each comprising one more input features selected from the group consisting of: a target site nucleotide sequence; a cut site; a PAM-sequence; microhomology lengths relative at a cut site, % GC content at a cut site; and microhomology deletion lengths at a cut site.
(82) In certain embodiments, the nucleotide sequence analyzed by the computational model is between about 25-100 nucleotides, 50-200 nucleotides, 100-400 nucleotides, 200-800 nucleotides, 400-1600 nucleotides, 800-3200 nucleotides, and 1600-6400 nucleotide, or more. In various embodiments, the computation model concerns predicting genetic repair outcomes at double-strand breaks cleaves induced by any DSB-based genomic editing system (e.g., CRISPR/Cas9, Cas-base, Cfr1-based, or others). FIG. **1** depicts the anatomy of a double strand break. In the example shown in FIG. **1**, the DNA segment **100** includes a top strand **105**A and a bottom strand **105**B. These two strands are complementary and therefore encode the same information. In some embodiments, CRISPR/Cas9 may be used to create a double strand cut at a selected donor splice site motif, which may be a specific sequence of 6-10 nucleotides. In the example of FIG. **1**, an NGG PAM may be used, as underlined and shown at **115**, so that a cut site **110** would occur within the selected donor splice site motif. Any suitable algorithm may be used to detect presence or absence of the splice site motif in repair products, thereby verifying if the splice site motif has been successfully eliminated.
(83) FIGS. **2**A-D show an illustrative matching of 3′ ends of top and bottom strands of a DNA segment at a cut site and an illustrative repair product, in accordance with some embodiments. For instance, the strands may be the illustrative top strand **105**A and the illustrative bottom strand **105**B of FIG. **1**, and the cut site may be the illustrative cut site **110** of FIG. **1**. (To avoid clutter, the surrounding sequence context is omitted in FIGS. **2**B-D.)
(84) In some embodiments, a segment of double-stranded DNA may be represented such that the top strand runs 5′ on the left to 3′ on the right. Given a cut in this double stranded DNA, nucleotides and their complementary base-paired nucleotides that lie between the 5′ end of the top strand and the cut site may be said to be located at the 5′ side of the cut site. Likewise, nucleotides and their complementary base-paired nucleotides that lie between the cut site and the 3′ end of the top strand may be said to be located at the 3′ side of the cut site.
(85) In the example shown in FIG. **2**A, a deletion length of 5 base pairs is considered, for example, as a result of 5′ end resection, where the top strand **105**A has an overhang **200**A of length 5 at the 5′ side of the cut site **110**, and the bottom strand **105**B has an overhang **200**B of length 5 at the 3′ side of the cut site **110**. As shown in FIG. **2**B, there is no match between the overhangs **200**A and **200**B in the first three bases, but there is a match in each of the last two bases. Thus, in this example, a

microhomology **205** is present, with a 2 base pair match.
(86) FIG. **2**C shows an illustrative result of flap removal, where the three mismatched bases in the overhang **200**B are removed. For instance, in some embodiments, given a microhomology, some or all nucleotides on the 3′ side of the microhomology on the top strand, and/or some or all nucleotides on the 3′ side of the microhomology on the bottom strand, may be resected. Pictorially, with the top strand running 5′ to 3′, nucleotides to the right of the microhomology on the top strand may be resected, and nucleotides to the left of the microhomology on the bottom strand may be resected.
(87) FIG. **2**D shows an illustrative repair product resulting from polymerase fill-in and ligation, where three matching bases are added to the overhang **200**B.
(88) FIG. **3**A shows an illustrative machine learning model **300**, in accordance with some embodiments. The machine learning model **300** may be trained using experimental data to compute, given an input DNA sequence seq and a cut site location, a probability distribution over any suitable set of deletion and/or insertion genotypes, and/or a probability distribution over any suitable set of deletion and/or insertion lengths. For instance, in some embodiments, 1 base pair insertions and 1-60 base pair deletions may be considered. (These repair outcomes may also be referred to herein as +1 to −60 indels.) The inventors have observed empirically that indels outside of this range occur infrequently. However, it should be appreciated that aspects of the present disclosure are not limited to any particular set of repair outcomes. In some embodiments, only insertions (e.g., 1-2 base pair insertions), or only deletions (e.g., 1-28 base pair deletions), may be considered, for example, based on availability of training data.
(89) The inventors have recognized and appreciated that accurate predictions of repair outcomes may be facilitated by separating the repair outcomes into three classes: microhomology (MH) deletions, microhomology-less (MH-less) deletions, and insertions. The inventors have further recognized and appreciated that different machine learning techniques may be applied to the different classes of repair outcomes. For instance, in the example of FIG. **3**, the machine learning model **300** includes three modules: the MH deletion module **305**, the MH-less deletion module **310**, and the insertion module **315**. As discussed below, these modules may compute scores for various indel genotypes and/or indel lengths, which may in turn be used to compute a probability distribution over indel genotypes and/or a probability distribution over indel lengths. In some embodiments, one or more modules (e.g., the MH deletion module **305** and the MH-less deletion module **310**) may be trained jointly. In some embodiments, a module may be dependent upon one or more other modules. For instance, as discussed below, an input feature used in the insertion module **315** may be derived based on outputs of the MH deletion module **305** and/or the MH-less deletion module **310**.
(90) In some embodiments, MH deletions may include deletions that are derivable analytically by simulating MMEJ. For instance, all microhomologies may be identified for deletion lengths of interest (e.g., deletion lengths 1-60). A genotypic outcome may be derived for each such microhomology by simulating polymerase fill-in, for example, as discussed in connection with FIGS. **2**A-**2**D. (The inventors have recognized and appreciated that there is a one-to-one correspondence between the microhomologies and the genotypic outcomes.) A deletion that is derivable in this manner may be classified as a MH deletion, whereas a deletion that is not derivable in this matter may be classified as a MH-less deletion.
(91) Techniques for identifying microhomologies for a given a deletion length L of interest (e.g., each deletion length between 1 and 60) are described below. However, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular technique for identifying microhomologies.
(92) In some embodiments, an input DNA sequence seq may be represented as a vector with integer indices, where each element of the vector is a nucleotide from the set, {A, C, G, T}, and the cut site is between seq[−1] and seq[0], and seq is oriented 5′ on the left to 3′ on the right. A

subsequence seq[i:j], i<j, may be a vector of length j−i, including elements seq[i] to seq[j−1]. For each deletion length L of interest (e.g., L between 1 and 60), left[L] may be used to denote seq[−L: 0], and right[L] may be used to denote seq[0, L]. Thus, with reference to the example shown in FIGS. **1**, **2**A, left[5] may be ACAAG, and right[5] may be GGTAG. Because the top strand **105**A and the bottom strand **105**B are complementary, a microhomology (e.g., the microhomology **205**) may be identified by looking for exact matches between left[5] and right[5] (which may be equivalent to complementary matches between the overhang **200**A and the overhang **200**B). For instance, a match vector may be constructed for each deletion length L of interest (e.g., L between 1 and 60) as follows: match[L][i]=‘I’ if left[L][i]=right[L][i], otherwise match[L][i]‘.’ Such matching between left[5] and right[5] is illustrated below.
(93) TABLE-US-00001 ACAAG ...|| GGTAG
(94) In some embodiments, a microhomology may be identified by looking for match[L][i:j] such that match[L][k]=‘|’ for all i≤k≤j and match[L][i]!=‘|’ and match[L][j]!=‘|’. For instance, with reference to the example shown in FIG. **1**, there may be no microhomology for deletion length 3, no microhomology for deletion length 4, one microhomology for deletion length 5, three microhomologies for deletion length 6, etc., as illustrated below.
(95) TABLE-US-00002 AAG ... GGT ... CAAG .... GGTA .... ACAAG ...|| GGTAG GACAAG |..|.| GGTAGG
(96) In some embodiments, microhomologies identified for a suitable set of deletion lengths (e.g., 1-60) may be enumerated using indices n=1 . . . N, where N is the number of identified microhomologies. For each n, let G[n] denote the genotypic outcome corresponding to the microhomology n, let ML [n] denote the microhomology length of the microhomology n, let C[n] denote the GC fraction of the microhomology n, and let DL [n] denote the deletion length of the microhomology n.
(97) Although examples of representations of DNA sequences and subsequences are discussed herein, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular representation.
(98) FIG. **3**B shows an illustrative process **350** for building one or more machine learning models for predicting frequencies of deletion genotypes and/or deletion lengths, in accordance with some embodiments. For instance, the process **350** may be used to build the illustrative MH deletion module **305** and/or the illustrative MH-less deletion module **310** in the example of FIG. **3**A. These modules may be used to compute, given an input DNA sequence seq and a cut site location, a probability distribution over any suitable set of deletion genotypes and/or a probability distribution over any suitable set of deletion lengths.
(99) In some embodiments, a probability distribution over deletion lengths from 1-60 may be computed. However, it should be appreciated that aspects of the present disclosure are not limited to any particular set of deletion lengths. In some embodiments, an upper limit of deletion lengths may be determined based on availability of training data and/or any other one or more suitable considerations.
(100) Referring to FIG. **3**B, act **355** of the process **350** may include, for each deletion length L of interest (e.g., each deletion length between 1-60), aligning subsequences of length L on the 5′ and 3′ sides of a cut site in an input DNA sequence to identify one or more microhomologies, as discussed in connection with FIG. **3**A. This may be performed for an input DNA sequence and a cut site for which repair genotype data from an CRISPR/Cas9 experiment is available.
(101) At act **360**, one or more microhomologies identified at act **355** may be featurized. Any suitable one or more features may be used, as aspects of the present disclosure are not so limited. As one example, the inventors have recognized and appreciated that energetic stability of a microhomology may increase proportionately with a length of the microhomology. Accordingly, in some embodiments, a microhomology length j−i may be used as a feature for a microhomology match[L][i:j].

(102) As another example, the inventors have recognized and appreciated that thermodynamic stability of a microhomology may depend on specific base pairings, and that G-C pairings have three hydrogen bonds and therefore have higher thermodynamic stability than A-T pairings, which have two hydrogen bonds. Accordingly, in some embodiments, a GC fraction, as shown below, may be used as a feature for a microhomology match[L][i:j], where indicator(boolean) equals 1 if boolean is true, and 0 otherwise.

(103) $$\frac{\text{.Math.}_{k=i}^{j-1}\ \text{indicator}(\text{top}[L][k] = \text{'}G\text{'or'}C\text{'})}{j-i}$$

(104) In some embodiments, a length N vector may be constructed for each feature (e.g., microhomology length, GC fraction, etc.), where N is the number of microhomologies identified at act **355** for a set of deletion lengths of interest (e.g., 1-60), as discussed in connection with FIG. **3**A. As discussed above, the inventors have recognized and appreciated that there is a one-to-one correspondence between microhomologies and genotypic outcomes that are classified as MH deletions. Therefore, feature vectors for microhomologies may be viewed as feature vectors for MH deletions.

(105) In some embodiments, acts **355** and **360** may be repeated for different input DNA sequences and/or cut sites for which repair genotype data from CRISPR/Cas9 experiments is available.

(106) It should be appreciated that aspects of the present disclosure are not limited to any particular featurization technique. For instance, in some embodiments, two features may be used, such as microhomology length and GC fraction. However, that is not required, as in some embodiments one feature may be used (e.g., microhomology length, GC fraction, or some other suitable feature), or more than two features may be used (e.g., three, four, five, etc.). Examples of features that may be used for a microhomology match[L][i:j] within a deletion of length L include, but are not limited to, a position of the microhomology within the deletion (e.g., as represented by

(107) $\frac{\text{.Math.}_{k=i}^{j-1}\ k}{L*(j-i)}$),

and a ratio between a length of the microhomology (i.e., j−i) and the L*(j−i) deletion length L. As another example, the inventors have recognized and appreciated that deoxyribonuclease (DNase) hypersensitivity may be used to classify genomic sequences into open or closed chromatin, which may impact DNA repair outcomes. Accordingly, in some embodiments, open vs. closed chromatin may be used as a feature. Any one or more of these features, and/or other features, may be used in addition to, or instead of, microhomology length and GC fraction. Furthermore, in some embodiments, explicit featurization may be reduced or eliminated by automatically learning data representations (e.g., using one or more deep learning techniques). Returning to FIG. **3**B, one or more machine learning models may be trained at act **365** to compute one or more target probability distributions. For instance, a neural network model may be built for the illustrative MH deletion module **305** in the example of FIG. **3**A. This model may take as input a length N vector for each of one or more features, as constructed at act **360**, and output a length N vector of MH scores, where N is the number of microhomologies identified at act **355** for a set of deletion lengths of interest (e.g., 1-60). Additionally, or alternatively, a neural network model may be built for the illustrative MH-less deletion module **310** in the example of FIG. **3**A. This model may take as input a vector for each of one or more features, and output a vector of MH-less scores. Both of the input vector and the output vector may be indexed by the set of deletion lengths of interest (e.g., 1-60) These neural network models may then be trained jointly using repair genotype data collected from CRISPR/Cas9 experiments.

(108) FIG. **4**A shows an illustrative neural network **400**A for computing MH scores, in accordance with some embodiments. For instance, the neural network **400**A may be used in the illustrative MH deletion module **305** in the example of FIG. **3**A, and may be trained at act **365** of the illustrative process **350** shown in FIG. **3**B.

(109) In some embodiments, the neural network **400**A may have one input node for each microhomology feature being used. For instance, in the example shown in FIG. **4**A, there are two

input nodes, which are associated with microhomology length and GC fraction, respectively. Each input node may receive a length N vector, where N is the number of microhomologies identified for a set of deletion lengths of interest (e.g., 1-60), for example, as discussed in connection with act **355** in the example of FIG. **3**B.

(110) In some embodiments, the neural network **400**A may include one or more hidden layers, each having one or more nodes. In the example shown in FIG. **4**A, there are two hidden layers, each having 16 nodes. However, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular number of hidden layers or any particular number of nodes in a hidden layer. Furthermore, different hidden layers may have different numbers of nodes.

(111) In some embodiments, the neural network **400**A may be fully connected. (To avoid clutter, the connections are not illustrated in FIG. **4**A.) However, that is not required. For instance, in some embodiments, a dropout technique may be used, where a parameter p may be selected, and during training each node's value is independently set to 0 with probability p. This may result in a neural network that is not fully connected.

(112) In some embodiments, a leaky rectified linear unit (ReLU) nonlinearity sigma may be used in the neural network **400**A. For instance, at hidden layer h and node i, an activation function may be provided as follows:

unit[*h][i*]=sigma(*w[h][i*]*unit[*h*−1]+*b[h][i*]), where sigma(*x*)=max(0, *x*)+0.001*min(0, *x*).

(113) Thus, the neural network **400**A may be parameterized by w[h] and b[h] for each hidden layer h. In some embodiments, these parameters may be initialized randomly, for example, from a spherical Gaussian distribution with some suitable center (e.g., 0) and some suitable variance (e.g., 0.1). These parameters may then be trained using repair genotype data collected from CRISPR/Cas9 experiments, for instance, as discussed below.

(114) In some embodiments, the neural network **400**A may have one output node, producing a length N vector ψ.sub.MH of scores, where N is the number of microhomologies identified for the set of deletion lengths of interest (e.g., 1-60). Thus, there may be one score for each identified microhomology.

(115) In some embodiments, the neural network **400**A may operate independently for each microhomology, taking as input the length of that microhomology (from the first input node) and the GC fraction of that microhomology (from the second input node), transforming those two values into 16 values (at the first hidden layer), then transforming those 16 values into 16 other values (at the second hidden layer), and finally outputting a single value (at the output node). In such an embodiment, parameters for the first hidden layer, w[1][i] and b[1][i], are vectors of length 2 for each node i from 1 to 16, whereas parameters for the second hidden layer, w[2][i] and b[2][i], are vectors of length 16 for each node i from 1 to 16, and parameters for the output layer, w[3][1] and b[3][1], are also vectors of length 16.

(116) In some embodiments, the vector ψ.sub.MH of raw scores may be converted into a vector ϕ.sub.MH of MH scores. The inventors have recognized and appreciated (e.g., from experimental data) that the strength of a microhomology decreases exponentially with deletion length. Accordingly, in some embodiments, an exponential linear model may be used to convert the raw scores into the MH scores. For instance, the following formula may be used:

ϕ.sub.MH[*n*]=exp(ψ.sub.MH*[n]−DL[n]**0.25),

where n is an index for a microhomology (and thus a number between 1 and N), and DL [n] is the deletion length of the microhomology n.

(117) In some embodiments, 0.25 may be a hyperparameter value chosen to improve training speed by appropriate scaling. However, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular hyperparameter value for exponential conversion, or any conversion at all. In some embodiments, the vector P of raw scores may be used directly as MH scores.

(118) FIG. **4**B shows an illustrative neural network **400**B for computing MH-less scores, in

accordance with some embodiments. For instance, the neural network **400**B may be used in the illustrative MH-less deletion module **310** in the example of FIG. **3**A, and may be trained at act **365** of the illustrative process **350** shown in FIG. **3**B.
(119) In some embodiments, deletion length may be modeled explicitly as an input to the neural network **400**B. Thus, in an example where the set of deletion lengths of interest is 1-60, an input node of the neural network **400**B may receive a deletion length vector, [1, 2, . . . , 60].
(120) In some embodiments, the neural network **400**B may include one or more hidden layers, each having one or more nodes. In the example shown in FIG. **4**B, the neural network **400**B has two hidden layers that are similarly constructed as the illustrative neural network **400**A in the example of FIG. **4**A. However, it should be appreciated that aspects of the present disclosure are not limited to the use of a similar construction between the neural network **400**A and the neural network **400**B.
(121) In some embodiments, the neural network **400**B may have an output node producing a vector ψ.sub.MH-less of scores. There may be one score for each deletion length L of interest. Thus, in an example where the set of deletion lengths of interest is 1-60, the length of the vector ψ.sub.MH-less may be 60.
(122) In some embodiments, an exponential linear model may be used to convert the vector ψ.sub.MH-less into a vector ϕ.sub.MH-less of MH-less scores. For instance, the following formula may be used:
ϕ.sub.MH-less$[L]$=exp(ψ.sub.MH-less$[L]-L*$0.25),
where L is a deletion length of interest. However, it should be appreciated that aspects of the present disclosure are not limited to the use of any particular hyperparameter value for exponential conversion, or any conversion at all.
(123) FIG. **4**C shows an illustrative process **400**C for training two neural networks jointly, in accordance with some embodiments. For instance, the process **400**C may be used to jointly train the illustrative neural networks **400**A and **400**B of FIGS. **4**A-**4**B.
(124) In some embodiments, the MH score vector ϕ.sub.MH and the MH-less score vector may be used to predict a probability distribution over MH deletion genotypes and/or a probability distribution over deletion lengths. For instance, given a microhomology n, a frequency may be predicted for the corresponding MH deletion genotype, out of all MH deletion genotypes. As discussed above, the inventors have recognized and appreciated that there is a one-to-one correspondence between microhomologies and genotypic outcomes that are classified as MH deletions. Thus, n=1 . . . N may be used as an index both for microhomologies and for MH deletions.
(125) In some embodiments, a frequency prediction for a microhomology n may depend on whether the microhomology n is full. A microhomology n is said to be full if the length of the microhomology n is the same as the deletion length associated with the microhomology n. For a microhomology n that is not full, a frequency may be predicted as follows, out of all MH deletion genotypes.
(126) $$V_{MHG}[n] = \frac{{}_{MH}[n]}{\text{.Math.}_{m=1}^{N}\ {}_{MH}[m] + \text{.Math.}_{m=1}^{N}\ {}_{\text{MH - less}}[L[m]] * \text{indicator}(\text{microhomology}\,m\,\text{isfull})}$$
Here DL [m] denotes the deletion length of the microhomology m, and indicator(boolean) equals 1 if boolean is true, and 0 otherwise.
(127) The inventors have recognized and appreciated that, for a full microhomology, only a single deletion genotype is possible for the entire deletion length. Moreover, the single genotype may be generated via different pathways, such as MMEJ and MH-less end-joining. Therefore, full microhomologies may be modeled as receiving contributions from MH-dependent and an MH-less mechanisms. Thus, for a microhomology n that is full, a frequency may be predicted as follows, out of all MH deletion genotypes.

(128) $$V_{MHG}[n] = \frac{{}_{MH}[n] + {}_{MH\text{-}less}[DL[n]]}{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m] + .\text{Math.}_{m=1}^{N}\ {}_{\text{MH - less}}[\text{DL}[m]] * \text{indicator(microhomology}\,m\,\text{isfull)}}$$

(129) Because the predicted frequencies are normalized. V.sub.MHG is a probability distribution over all microhomologies identified for the set of deletion lengths of interest, and hence also a probability distribution over all MH deletions.

(130) In some embodiments, given a deletion length L, a frequency may be predicted as follows for the set of all deletions having the deletion length L, out of all deletions, taking into account contributions from MH-dependent and MH-less mechanisms.

(131) $$V_{DL}[L] = \frac{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m] * \text{indicator}(\text{DL}[m] == L) + {}_{\text{MH - less}}[L]}{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m] + .\text{Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]}$$

Here DL [m] denotes the deletion length of the microhomology m, and indicator(boolean) equals 1 if boolean is true, and 0 otherwise.

(132) In some embodiments, the parameters w[h] and b[h] for each hidden layer h of the neural networks **400**A and **400**B may be trained using a gradient descent method with L2-loss:

Loss=Σ.sub.m=1.sup.N(*V*.sub.MHG*[m]*−*V*.sub.MHG**[m]*).sup.2+Σ.sub.l=1.sup.60(*V*.sub.DL*[l]* −*V*.sub.DL**[l]*).sup.2,

where V.sub.MHG*is an observed probability distribution on MH deletion genotypes, and V.sub.DL*is an observed probability distribution on deletion lengths (e.g., based on repair genotype data collected from CRISPR/Cas9 experiments).

(133) In some embodiments, multiple instantiations of the neural networks **400**A and **400**B may be trained with different loss functions. For instance, in addition to, or instead of L2-loss, a squared Pearson correlation function may be used.

Loss=−(pearsonr(*V*.sub.MHG*[m]*, *V*.sub.MHG**[m]*)).sup.2−(pearsonr(*V*.sub.MHG*[m]*,*V*.sub.MHG**[m]*)).sup.2

The function pearsonr(x, y) may be defined as follows for length N vectors x and y, where x and y denote the averages of x and y, respectively.

(134) $$\text{pearsonr}(x, y) = \frac{.\text{Math.}_{m=1}^{N}\ (x[m] - \bar{x})(y[m] - \bar{y})}{\sqrt{.\text{Math.}_{m=1}^{N}\ (x[m] - \bar{x})^2}\sqrt{.\text{Math.}_{m=1}^{N}\ (y[m] - \bar{y})^2}}$$

(135) Although neural networks are used in the examples shown in FIGS. **4**A-**4**C, it should be appreciated that aspects of the present disclosure are not so limited. For instance, in some embodiments, one or more other types of machine learning techniques, such as linear regression, non-linear regression, random-forest regression, etc., may be used additionally or alternatively.

(136) Furthermore, in some embodiments, one or more neural networks that are different from the neural networks **400**A and **400**B may be used additionally or alternatively. As one example, a different activation function may be used for one or more nodes, such as sigma(x)=max(0, x) (rectified linear unit, or ReLU), sigma(x)=max(0.001x, x) (another example of leaky ReLU), sigma(x)=0.5*(tanh (x)+1.0) or

(137) $\text{sigma}(x) = \frac{1}{1 + e^{-x}}$(Sigmoid),

sigma(x)=max(0, x)+min(0, x)*0.5*(tanh(x)+1) (Swish), etc. As another example, batch normalization may be performed at one or more hidden layers. It should be appreciated that aspects of the disclosure are not limited to training the neural networks **400**A and **400**B jointly. For instance, given a microhomology n, a frequency may be predicted as follows for the corresponding MH deletion genotype, out of all MH deletion genotypes.

(138) $$V'_{MHG}[n] = \frac{{}_{MH}[n]}{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m]}$$

Since this prediction does not depend on the ϕ.sub.MH-less scores, the neural network **400**A may be trained independently.

(139) In some embodiments, one or more other probability distributions may be predicted in addition to, or instead of V.sub.MHG and V.sub.DL. As one example, given a microhomology n, a

frequency may be predicted as follows for the corresponding MH deletion genotype, out of all deletion genotypes (both MH and MH-less).

(140) $$V''_{MHG}[n] = \frac{{}_{MH}[n]}{\text{.Math.}_{m=1}^{N}\ {}_{MH}[m] + {}_{l=1}^{60}\ {}_{\text{MH - less}}[l]}$$

(141) As another example, for a microhomology n that is not full, a frequency may be predicted as follows, out of all deletion genotypes (both MH and MH-less).

(142) 0 $$V'''_{MHG}[n] = \frac{{}_{MH}[n]}{\text{.Math.}_{m=1}^{N}\ {}_{MH}[m] + \text{.Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]}$$

(143) For a microhomology n that is full, a frequency may be predicted as follows, out of all deletion genotypes (both MH and MH-less).

(144) $$V'''_{MHG}[n] = \frac{{}_{MH}[n] + {}_{\text{MH - less}}[DL(n)]}{\text{.Math.}_{m=1}^{N}\ {}_{MH}[m] + \text{.Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]}$$

Here DL [n] denotes the deletion length of the microhomology n.

(145) As another example, given a deletion length L, a frequency may be predicted as follows for the set of MH-less deletions having the deletion length L, out of all MH-less deletion genotypes.

(146) $$V'_{DL}[L] = \frac{{}_{\text{MH - less}}[L]}{\text{.Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]}$$

(147) As another example, given a deletion length L, a frequency may be predicted as follows for the set of MH-less deletions having the deletion length L, out of all deletion genotypes (both MH and MH-less).

(148) $$V''_{DL}[L] = \frac{{}_{\text{MH - less}}[L]}{\text{.Math.}_{m=1}^{N}\ {}_{MH}[m] + \text{.Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]}$$

(149) Any one or more of the above predicted probability distributions may be used to train the neural networks **400**A and **400**B, with some suitable loss function.

(150) FIG. **4**D shows an illustrative implementation of the insertion module **315** shown in FIG. **3**A, in accordance with some embodiments. In this example, the insertion module **315** includes two models. First, an insertion rate model **405** may be constructed to predict, given an input DNA sequence and a cut site, a frequency of 1 base pair insertions out of all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions). Second, an insertion base pair model **410** may be constructed to predict frequencies of 1 base pair insertion genotypes (i.e., A, C, G, T), again out of all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions). However, it should be appreciated that aspects of the present disclosure are not limited to any particular set of indels. In some embodiments, a small set of indels (e.g., 1 base pair insertions and 1-28 base pair deletions) may be considered, for instance, when less training data is available.

(151) In some embodiments, the insertion rate model **405** may have one or more input features, which may be encoded as an M-dimensional vector of values for some suitable M. The insertion rate model **405** may have at least one output value. A set of training data for the insertion rate model **405** may include a plurality of M-dimensional training vectors and respective output values. Given an M-dimensional query vector, a k-nearest neighbor (k-NN) algorithm with weighting by inverse distance may be used to compute a predicted output value for the query vector. For instance, k=5 may be used, and five training vectors that are closest to the query vector may be identified, and a predicted output value for the query vector may be computed as a sum of the outputs values corresponding to the five closest training vectors, weighted by inverse distance, as follows.

(152) $$y = \text{.Math.}_{i=1}^{5}\ \hat{y}[i] * \left(\frac{\text{.Math.}_{j=1}^{5}\ d(x,\hat{x}[j]) - d(x,\hat{x}[i])}{\text{.Math.}_{j=1}^{5}\ d(x,\hat{x}[j])}\right)$$

Here x is the query vector, d is a distance function for the M-dimensional vector space, {circumflex over (x)}} [1], . . . , {circumflex over (x)}[**5**] are the five closest training vectors, ŷ[1], . . . , ŷ[5] are the output values corresponding respectively to {circumflex over (x)}[1], . . . , {circumflex over

(x)}[5], and y is the predicted output value for the query vector x.
(153) It should be appreciated that aspects of the present disclosure are not limited to the use of any particular k, or to the use of any k-NN algorithm. For instance, any one or more of the following techniques, and/or Bayesian variants thereof, may be used in addition to, or instead of k-NN: gradient-boosted regression, linear regression, nonlinear regression, multilayer perceptron, deep neural network, etc. Also, any suitable distance metric d may be used, such as Euclidean distance.
(154) In some embodiments, the insertion rate model **405** may have three input features: overall deletion score, precision score, and one or more cut site nucleotides. The overall deletion score may be computed based on outputs of the MH deletion module **305** and the MH-less deletion module **310** in the example of FIG. **3**A, for instance, as follows.

(155) $$= \sum_{m=1}^{N} {}_{\text{MH}}[m] + \sum_{l=1}^{60} {}_{\text{MH - less}}[l]$$

Alternatively, log (ϕ) may be used as the overall deletion score.
(156) In some embodiments, the precision score may be indicative of an amount of entropy in predicted frequencies of a suitable set of deletion lengths. The inventors have recognized and appreciated that it may be desirable to calculate precision based on a large set of deletion lengths, but in some instances a smaller set (e.g., 1-28) may be used due to one or more constraints associated with available data. As discussed above, given a deletion length L, a frequency may be predicted as follows for the set of all deletions having the deletion length L, out of all deletions, taking into account contributions from MH-dependent and MH-less mechanisms.

(157) $$V_{DL}[L] = \frac{\sum_{m=1}^{N} {}_{MH}[m] * \text{indicator}(DL[m] == L) + {}_{MH\text{-}less}[L]}{\sum_{m=1}^{N} {}_{MH}[m] + \sum_{l=1}^{60} {}_{MH\text{-}less}[l]}$$

Here DL [m] denotes the deletion length of the microhomology m, and indicator(boolean) equals 1 if boolean is true, and 0 otherwise. The precision score may be computed as follows.

(158) $$\text{precision} = 1 - \frac{-\sum_{L=1}^{28} V_{DL}[L] * \log(V_{DL}[L])}{\log(28)}$$

(159) In some embodiments, the one or more cut site nucleotides may include nucleotides on either side of the cut site (i.e., seq[−1] and seq[0]). In the example shown in FIG. **1**, the cut site nucleotides are G and G, which are the third and fourth nucleotides to the left of the PAM sequence **115**. However, it should be appreciated that aspects of the present disclosure are not limited to the use of two cut side nucleotides as input features to the insertion rate model **405**. For instance, only one cut side nucleotide (e.g., seq[−1], which may be the fourth nucleotide to the left of the PAM sequence) may be used when less training data is available, whereas more than two cut side nucleotides (e.g., seq[−2], seq[−1], and seq[0], which may be the third, fourth, and fifth nucleotides to the left of the PAM sequence) may be used when more training data is available.
(160) In some embodiments, one or more input features to the insertion rate model **405** may be encoded in some suitable manner. For instance, the one or more cut site nucleotides may be one-hot encoded, for example, as follows.
A=1000,C=0100,G=0010,T=0001
(161) In some embodiments, encoded input features may be concatenated to form an input vector. In an example in which two cut side nucleotides are used, an input vector may have a length of 10: four for each of the two cut side nucleotides, one for the precision score, and one for the overall deletion score.
(162) In some embodiments, training data for a certain input DNA sequence may be organized into a matrix X. Each row in the matrix (X[i, −]) may correspond to a possible cut site, and may store a length M training vector for that cut site (e.g., M=10). In some embodiments, each column in the matrix (X[−,j]) may be normalized to mean 0 and variance 1, as follows.

(163) $$X[i,j] = \frac{X[i,j] - \text{mean}(X[-,j])}{\text{var}(X[-,j])}$$

(164) In some embodiments, values in a query vector may be normalized in a like fashion. For

instance, a jth value in a query vector x may be normalized as follows.

(165) $x[i]=\frac{x[j]\text{ - mean}(X[\text{-},j])}{\text{var}(X[\text{-},j])}$

(166) In some embodiments, an output value may be computed for each row in the training matrix X. For instance, an output value Y[i], i corresponding to a possible cut site, may be a frequency of observed 1 base pair insertions, relative to all observed +1 to −60 indels, at that cut site.

(167) In some embodiments, the insertion base pair model **410** may be constructed to predict frequencies of 1 base pair insertion genotypes (i.e., A, C, G, T). For instance, the insertion base pair model **410** may predict that the probability of a certain insertion genotype given one or more cut site nucleotides is the same as the frequency of that insertion genotype as observed in a subset of training data in which those one or more cut site nucleotides are observed. Thus, given an input DNA sequence seq and a cut site, the insertion base pair model **410** may determine one or more cut site nucleotides (e.g., seq[−1]="C"). The insertion base pair model **410** may then score the insertion genotypes A as follows.

$\phi.\text{sub.ins}[\text{"}A\text{"}]=y*\text{ObsFreq}(\text{"}A\text{"}|\text{seq}[-1]=\text{"}C\text{"})$

Here y is the frequency of 1 base pair insertions as predicted by the insertion rate model **405**, and ObsFreq("A" I seq[−1]="C") is the observed frequency of insertion genotype A given that the nucleotide to the left of the cut site C. The other three insertion genotypes may be scored similarly.

(168) In some embodiments, more than one cut site nucleotides may be considered. For instance, the insertion base pair model **410** may determine that seq[−2]="A", seq[−1]="C", and seq[0]="G". The insertion base pair model **410** may then score the insertion genotypes A as follows, and the other three insertion genotypes may be scored similarly.

$\phi.\text{sub.ins}[\text{"}A\text{"}]=y*\text{ObsFreq}(\text{"}A\text{"}|\text{seq}[-2]=\text{"}A\text{"}\&\ \text{seq}[-1]=\text{"}C\text{"}\&\ \text{seq}[0]=\text{"}G\text{"})$

In some embodiments, a frequency of 1 base pair insertion genotype A, out of all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions), may be predicted as follows. Frequencies for the other three insertion genotypes may be predicted similarly.

(169) $0V_{ins}[\text{"}A\text{"}]=\frac{{}_{ins}[\text{"}A\text{"}]}{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m]+.\text{Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]+{}_{ins}[\text{"}A\text{"}]+{}_{ins}[\text{"}C\text{"}]+{}_{ins}[\text{"}G\text{"}]+{}_{ins}[\text{"}T\text{"}]}$

(170) In some embodiments, a frequency of 1 base pair insertions, out of all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions), may be predicted as follows.

(171) $V_{ins}=\frac{{}_{ins}[\text{"}A\text{"}]+{}_{ins}[\text{"}C\text{"}]+{}_{ins}[\text{"}G\text{"}]+{}_{ins}[\text{"}T\text{"}]}{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m]+.\text{Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]+{}_{ins}[\text{"}A\text{"}]+{}_{ins}[\text{"}C\text{"}]+{}_{ins}[\text{"}G\text{"}]+{}_{ins}[\text{"}T\text{"}]}$

(172) In some embodiments, given a deletion length L, a frequency may be predicted as follows for the set of all deletions having the deletion length L, out of all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions), taking into account contributions from MH-dependent and MH-less mechanisms.

(173) $V_{\text{DL + ins}}[L]=\frac{.\text{Math.}_{m=1}^{N}\ {}_{\text{MH}}[m]*\text{indicator}(\text{DL}[m]==L)+{}_{\text{MH - less}}[L]}{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m]+.\text{Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]+{}_{ins}[\text{"}A\text{"}]+{}_{ins}[\text{"}C\text{"}]+{}_{ins}[\text{"}G\text{"}]+{}_{ins}[\text{"}T\text{"}]}$

(174) In some embodiments, given a deletion length L, a frequency may be predicted as follows for the set of MH-less deletions having the deletion length L, out of all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions).

(175) $V''_{\text{DL + ins}}[L]=\frac{{}_{\text{MH - less}}[L]}{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m]+.\text{Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]+{}_{ins}[\text{"}A\text{"}]+{}_{ins}[\text{"}C\text{"}]+{}_{ins}[\text{"}G\text{"}]+{}_{ins}[\text{"}T\text{"}]}$

(176) In some embodiments, given a microhomology n, a frequency may be predicted as follows for the corresponding MH deletion genotype, out of all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions).

(177) $V''_{\text{MHG + ins}}[n]=\frac{{}_{\text{MH}}[n]}{.\text{Math.}_{m=1}^{N}\ {}_{MH}[m]+.\text{Math.}_{l=1}^{60}\ {}_{\text{MH - less}}[l]+{}_{ins}[\text{"}A\text{"}]+{}_{ins}[\text{"}C\text{"}]+{}_{ins}[\text{"}G\text{"}]+{}_{ins}[\text{"}T\text{"}]}$

(178) In some embodiments, for a microhomology n that is not full, a frequency may be predicted as follows, out of all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions).

$$(179)\ V'''_{\mathrm{MHG+ins}}[n] = \frac{{}_{\mathrm{MH}}[n]}{\sum_{m=1}^{N} {}_{MH}[m] + \sum_{l=1}^{60} {}_{\mathrm{MH\text{-}less}}[l] + {}_{ins}[\text{“}A\text{”}] + {}_{ins}[\text{“}C\text{”}] + {}_{ins}[\text{“}G\text{”}] + {}_{ins}[\text{“}T\text{”}]}$$

For a microhomology n that is full, a frequency may be predicted as follows, out of all all +1 to −60 indels (i.e., 1 base pair insertions and 1-60 base pair deletions).

$$(180)\ V'''_{\mathrm{MHG+ins}}[n] = \frac{{}_{\mathrm{MH}}[n] + {}_{\mathrm{MH\text{-}less}}[\mathrm{DL}(n)]}{\sum_{m=1}^{N} {}_{MH}[m] + \sum_{l=1}^{60} {}_{\mathrm{MH\text{-}less}}[l] + {}_{ins}[\text{“}A\text{”}] + {}_{ins}[\text{“}C\text{”}] + {}_{ins}[\text{“}G\text{”}] + {}_{ins}[\text{“}T\text{”}]}$$

(181) Here DL [n] denotes the deletion length of the microhomology n.

(182) FIG. **5** shows an illustrative process **500** for processing data collected from CRISPR/Cas9 experiments, in accordance with some embodiments. For instance, the process **500** may be performed for each input DNA sequence and CRISPR/Cas9 cut site, and a resulting dataset may be used to train the illustrative computational models described in connection with FIGS. **4**A-**4**D.

(183) At act **505**, repair genotypes observed from CRISPR/Cas 9 experiments may be aligned with an original DNA sequence. Any suitable technique may be used to observe the repair genotypes, such as Illumina DNA sequencing. Any suitable alignment algorithm may be used for alignment, such as a Needleman-Wunsch algorithm with some suitable scoring parameters (e.g., +1 for match, −2 for mismatch, −4 for gap open, and −1 for gap extend, or +1 for match, −1 for mismatch, −5 for gap open, and −0 for gap extend).

(184) At act **510**, one or more filter criteria may be applied to alignment reads from act **505**. For instance, in some embodiments, only those reads in which a deletion includes at least one base directly 5′ or 3′ of the CRISPR/Cas9 cut site are considered. This may filter out deletions that are unlikely to have resulted from CRISPR/Cas9.

(185) At act **515**, frequencies of indels of interest (e.g., from +1 to −60) may be normalized into a probability distribution.

(186) FIG. **6** shows an illustrative process **600** for using a machine learning model to predict frequencies of indel genotypes and/or indel lengths, in accordance with some embodiments. Acts **605** and **610** may be similar to, respectively, acts **355** and **360** of the illustrative process **350** of FIG. **3**B, except that acts **605** and **610** may be performed for an input DNA sequence seq and a cut site location for which repair genotype data from an CRISPR/Cas9 experiment may not be available. At act **615**, one or more machine learning models, such as the machine learning models trained at act **365** of the illustrative process **350** of FIG. **3**B, may be applied to an output of act **610** to compute a frequency distribution over deletion lengths of interest.

(187) The inventors have recognized and appreciated that, while Cas9 is typically understood to induce a blunt-end double-strand break, some evidence suggests that Cas9 may generate a 1 base pair staggered end cut instead. FIG. **7** shows illustrative examples of a blunt-end cut and a staggered cut, in accordance with some embodiments.

(188) FIG. **8**A shows an illustrative plot **800**A of predicted repair genotypes, in accordance with some embodiments. For instance, the plot **800**A may be generated by applying one or more of the illustrative techniques described in connection with FIGS. **2**A-**2**D, **3**A-**3**B, **4**A-**4**D, **5-6** to the example shown in FIG. **1**. Each vertical bar may correspond to a deletion length, and a height of the bar may correspond to a predicted frequency of that deletion length. The lighter color may indicate repair genotypes that successfully eliminate the donor splice site motif, whereas the darker color may indicate failure. In this example, about 90% of repair products in the 3−26 base pair deletion class are predicted to be successful for the illustrative local sequence context and cut site shown in FIG. **1**.

(189) The inventors have recognized and appreciated that the 3−26 base pair deletion class may occur as frequently as 50%, for example, when assaying selected sequences (e.g., patient genotypes underlying certain diseases) integrated into the genome of mouse embryonic stem cells, with a 14-day exposure to CRISPR/Cas9. Thus, in view of the 90% success rate predicted above for the 3−26 base pair deletion class, a genetic editing approach using CRISPR-Cas9 may be provided that achieves a desired result with a 45% rate. In contrast, genetic editing using HDR may achieve a

success rate of 10% or lower, and may require a more complex experimental protocol.
(190) FIG. **8**B shows another illustrative plot **800**B of predicted repair genotypes, in accordance with some embodiments. For instance, the plot **800**B may be generated by applying one or more of the illustrative techniques described in connection with FIGS. **2**A-**2**D, **3**A-**3**B, **4**A-**4**D, **5-6** to an illustrative DNA sequence **805**B, which may be associated with spinal muscular atrophy (SMA).
(191) In some patients, a specific single nucleotide polymorphism (SNP) in exon 7 of the SMA2 gene may induce exon skipping of exon 7, erroneously including exon 8 instead. Exon 8 includes a protein degradation signal (namely, EMLA-STOP, as shown in FIG. **8**B), which causes degradation in the SMA2 gene product, thereby inducing spinal muscular atrophy. In this region, a disease genotype must have precisely EMLA-STOP. Nearly any other genotype is considered healthy.
(192) In the example of FIG. **8**B, each vertical bar corresponds to a deletion length, and a height of the bar corresponds to a predicted frequency of that deletion length. The lighter color may indicate repair genotypes that successfully disrupt the EMLA-STOP signal, whereas the darker color may indicate failure. In this example, over 90% of repair products in the 3−26 base pair deletion class are predicted to be healthy.
(193) FIG. **8**C shows another illustrative plot **800**C of predicted repair genotypes, in accordance with some embodiments. For instance, the plot **800**C may be generated by applying one or more of the illustrative techniques described in connection with FIGS. **2**A-**2**D, **3**A-**3**B, **4**A-**4**D, **5-6** to an illustrative DNA sequence associated with breast-ovarian cancer.
(194) In the example of FIG. **8**C, a clinical observed patient genotype includes an abnormal duplication of 14 base pairs that a wild type sequence from a normal/health individual lacks. The patient genotype is incorporated into the genome of mouse embryonic stem cells, and then CRISPR/Cas9 is applied. It is observed that the 3−26 base pair deletion class occurs 65% out of all repair classes at this local sequence context. Moreover, as shown in FIG. **8**C, repair to wild type is observed to occur at 89% rate among all 3−26 base pair deletions. Thus, an overall wild type repair rate is about 57%.
(195) FIG. **8**D shows a microhomology identified in the example of FIG. **8**C.
(196) As discussed above, the inventors have recognized and appreciated at least two tasks of interest: predicting frequencies of deletion lengths, as well as predicting frequencies of repair genotypes. In some embodiments, a single machine learning model may be provided that performs both tasks.
(197) In some embodiments, repair genotypes corresponding to a deletion of length L may be labeled as follows: for every integer K ranging from 0 to L, a K-genotype associated with deletion length L may be obtained by concatenating left[L][−inf: K] with right[L][K: +inf]. A vector COLLECTION of length Q where each element is a tuple (K, L) may be constructed by enumerating each K-genotype for each deletion length L of interest and removing tuples that have the same repair genotype, e.g., (k′, L) and (k, L) such that left[L][−inf: k′] concatenated with right[L][k′: +inf] is equivalent to left[L][−inf: k] concatenated with right[L][k: +inf], for example, by retaining only the tuple with the larger K. A training data set may be constructed using observational data by constructing a vector X of length Q where X sums to 1 and X[q] represents an observed frequency of a repair genotype generated by COLLECTION[q].
(198) In some embodiments, the vector COLLECTION may be featurized. This may be performed for a given tuple (k, l) by determining whether there is an index i such that match[l][i: k] is a microhomology. If no such i exists, then the tuple (k, l) may be considered to not partake in microhomology.
(199) The inventors have recognized and appreciated that frequencies of repair products may be influenced by certain features of microhomologies such as microhomology length, fraction of G-C pairings, and/or deletion length. The inventors have also recognized and appreciated that some default values may be useful for repair genotypes that are considered to not partake in microhomology.

(200) For example, the inventors have recognized and appreciated that energetic stability of a microhomology may increase proportionately with a length of the microhomology. Accordingly, in some embodiments, the microhomology length k−i may be used for a tuple (k, l), and a default value of 0 may be used if (k, l) does not partake in microhomology.
(201) As another example, the inventors have recognized and appreciated that thermodynamic stability of a microhomology may depend on specific base pairings, and that G-C pairings have three hydrogen bonds and therefore have higher thermodynamic stability than A-T pairings, which have two hydrogen bonds. Accordingly, in some embodiments, a GC fraction, as shown below, may be used as a feature for (k, l), where indicator(boolean) equals 1 if boolean is true, and 0 otherwise. A default value of −1 may be used if (k, l) does not partake in microhomology.
(202) $\frac{\sum_{j=1}^{k-1} \text{indicator}(\text{left}[l][j] = 'G' \text{ or } 'C')}{k-i}$
(203) In some embodiments, a feature for deletion length may be considered, represented as l for the tuple (k, l).
(204) The inventors have also recognized and appreciated (e.g., from experimental data) that 0-genotype and l-genotype repair products may occur despite a lack of microhomology, and may occur through microhomology-free end-joining repair pathways. Accordingly, (k, l) may be featurized with a Boolean for 0-genotype that is equal to 1 if k=0 and (k, l) does not partake in microhomology, and 0 otherwise. A Boolean feature for i-genotypes may also be used where it is equal to 1 if k=l and (k, l) does not partake in microhomology, and 0 otherwise.
(205) FIG. **9** shows another illustrative neural network **900** for computing a frequency distribution over deletion lengths, in accordance with some embodiments.
(206) In some embodiments, the neural network **900** may be parameterized by w[h] and b[h] for each hidden layer h. In some embodiments, these parameters may be initialized randomly, for example, from a spherical Gaussian distribution with some suitable center (e.g., 0) and some suitable variance (e.g., 0.1). These parameters may then be trained using repair genotype data collected from CRISPR/Cas9 experiments.
(207) In some embodiments, the neural network **900** may operate independently for each microhomology, taking as input the length of that microhomology (from the first input node), the GC fraction of that microhomology (from the second input node), Boolean features for 0 and 1-genotypes (from the third and fourth input node, where N-flag corresponds to i-genotypes), and the length of the deletion (from the fifth input node), transforming those five values into 16 values (at the first hidden layer), then transforming those 16 values into 16 other values (at the second hidden layer), and finally outputting a single value (at the output node). In such an embodiment, parameters for the first hidden layer, w[1][i] and b[1][i], are vectors of length 5 for each node i from 1 to 16, whereas parameters for the second hidden layer, w[2][i] and b[2][i], are vectors of length 16 for each node i from 1 to 16, and parameters for the output layer, w[3][1] and b[3][1], are also vectors of length 16.
(208) In some embodiments, the neural network **900** may be applied independently (e.g., as discussed above) to each featurized (k, l) in COLLECTIONS to produce a vector of Q microhomology scores called Z.
(209) In some embodiments, Z may be normalized into a probability distribution over all unique repair genotypes of interest within all deletion lengths of interest (e.g., deletion lengths between 3 and 26). The inventors have recognized and appreciated (e.g., from experimental data) that frequency may decrease exponentially with deletion length. Accordingly, in some embodiments, an exponential linear model may be used to normalize the vector of repair genotype scores. For example, the following formula may be used:
(210) $Y[q] = \frac{\exp(Z[q] - \text{beta} * DL[q])}{\sum_{q'=1}^{Q} \exp(Z[q'] - \text{beta} * DL[q'])}$
where DL [q]=l for each q where COLLECTIONS [q]=(k, l), and beta is a parameter. In some

embodiments, a probability distribution Y over all unique repair genotypes of interest within all deletion lengths of interest may be converted to a probability distribution Y′ over all deletion lengths. The following formula may be used for this:

(211) $$Y'[l] = \frac{\sum_{q=1}^{Q} Y[q] * \text{indicator}(DL[q] = l)}{\sum_{q=1}^{Q} Y[q]}$$

(212) In some embodiments, the parameter beta may be initialized to −1. These parameters may then be trained using repair genotype data collected from CRISPR/Cas9 experiments.

(213) In some embodiments, the parameters w[h] and b[h] for each hidden layer h and the parameters beta may be trained by using a gradient descent method with L2-loss on Y:

$L(\text{predY}, \text{obsY}) = \|\text{predY} - \text{obsY}\|_2^2$,

where predY is a predicted probability distribution on deletion lengths (e.g., as computed by the neural network **900** using current parameter values), and obsY is an observed probability distribution on deletion lengths (e.g., based on repair genotype data collected from CRISPR/Cas9 experiments).

(214) The inventors have recognized and appreciated that one or more of the techniques described herein may be used to identify therapeutic guide RNAs that are expected to produce a therapeutic outcome when used in combination with a genomic editing system without an HDR template. For instance, one or more of the techniques described herein may be used to identify a therapeutic guide RNA that is expected to result in a substantial fraction of genotypic consequences that cause a gain-of-function mutation in DNA in the absence of an HDR template. A therapeutic guide RNA may be used singly, or in combination with other therapeutic guide RNAs. An action of the therapeutic guide RNA may be independent of, or dependent on, one or more genomic consequences of the other therapeutic guide RNAs.

(215) FIG. **10** shows, schematically, an illustrative computer **1000** on which any aspect of the present disclosure may be implemented. In the embodiment shown in FIG. **10**, the computer **1000** includes a processing unit **1001** having one or more processors and a non-transitory computer-readable storage medium **1002** that may include, for example, volatile and/or non-volatile memory. The memory **1002** may store one or more instructions to program the processing unit **1001** to perform any of the functions described herein. The computer **1000** may also include other types of non-transitory computer-readable medium, such as storage **1005** (e.g., one or more disk drives) in addition to the system memory **1002**. The storage **1005** may also store one or more application programs and/or external components used by application programs (e.g., software libraries), which may be loaded into the memory **1002**.

(216) The computer **1000** may have one or more input devices and/or output devices, such as devices **1006** and **1007** illustrated in FIG. **10**. These devices can be used, among other things, to present a user interface. Examples of output devices that can be used to provide a user interface include printers or display screens for visual presentation of output and speakers or other sound generating devices for audible presentation of output. Examples of input devices that can be used for a user interface include keyboards and pointing devices, such as mice, touch pads, and digitizing tablets. As another example, the input devices **1007** may include a microphone for capturing audio signals, and the output devices **1006** may include a display screen for visually rendering, and/or a speaker for audibly rendering, recognized text.

(217) As shown in FIG. **10**, the computer **1000** may also comprise one or more network interfaces (e.g., the network interface **1010**) to enable communication via various networks (e.g., the network **1020**). Examples of networks include a local area network or a wide area network, such as an enterprise network or the Internet. Such networks may be based on any suitable technology and may operate according to any suitable protocol and may include wireless networks, wired networks or fiber optic networks.

(218) Having thus described several aspects of at least one embodiment, it is to be appreciated that

various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be within the spirit and scope of the present disclosure. Accordingly, the foregoing description and drawings are by way of example only.

(219) The above-described embodiments of the present disclosure can be implemented in any of numerous ways. For example, the embodiments may be implemented using hardware, software or a combination thereof. When implemented in software, the software code can be executed on any suitable processor or collection of processors, whether provided in a single computer or distributed among multiple computers.

(220) Also, the various methods or processes outlined herein may be coded as software that is executable on one or more processors that employ any one of a variety of operating systems or platforms. Additionally, such software may be written using any of a number of suitable programming languages and/or programming or scripting tools, and also may be compiled as executable machine language code or intermediate code that is executed on a framework or virtual machine.

(221) In this respect, the concepts disclosed herein may be embodied as a non-transitory computer-readable medium (or multiple computer-readable media) (e.g., a computer memory, one or more floppy discs, compact discs, optical discs, magnetic tapes, flash memories, circuit configurations in Field Programmable Gate Arrays or other semiconductor devices, or other non-transitory, tangible computer storage medium) encoded with one or more programs that, when executed on one or more computers or other processors, perform methods that implement the various embodiments of the present disclosure discussed above. The computer-readable medium or media can be transportable, such that the program or programs stored thereon can be loaded onto one or more different computers or other processors to implement various aspects of the present disclosure as discussed above.

(222) The terms “program” or “software” are used herein to refer to any type of computer code or set of computer-executable instructions that can be employed to program a computer or other processor to implement various aspects of the present disclosure as discussed above. Additionally, it should be appreciated that according to one aspect of this embodiment, one or more computer programs that when executed perform methods of the present disclosure need not reside on a single computer or processor, but may be distributed in a modular fashion amongst a number of different computers or processors to implement various aspects of the present disclosure.

(223) Computer-executable instructions may be in many forms, such as program modules, executed by one or more computers or other devices. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types. Typically the functionality of the program modules may be combined or distributed as desired in various embodiments.

(224) Also, data structures may be stored in computer-readable media in any suitable form. For simplicity of illustration, data structures may be shown to have fields that are related through location in the data structure. Such relationships may likewise be achieved by assigning storage for the fields with locations in a computer-readable medium that conveys relationship between the fields. However, any suitable mechanism may be used to establish a relationship between information in fields of a data structure, including through the use of pointers, tags or other mechanisms that establish relationship between data elements.

(225) Various features and aspects of the present disclosure may be used alone, in any combination of two or more, or in a variety of arrangements not specifically discussed in the embodiments described in the foregoing and is therefore not limited in its application to the details and arrangement of components set forth in the foregoing description or illustrated in the drawings. For example, aspects described in one embodiment may be combined in any manner with aspects described in other embodiments.

(226) In an exemplary embodiment, a computational model described herein is trained with experimental data as outlined in Example 1. The method outlined in Example 1 for training a computational model with experimental data is meant to be non-limiting.
(227) Accordingly, the specification discloses a method for training a computational model described herein, comprising: (i) preparing a library comprising a plurality of nucleic acid molecules each encoding a nucleotide target sequence and a cognate guide RNA, wherein each nucleotide target sequence comprises a cut site; (ii) introducing the library into a plurality of host cells; (iii) contacting the library in the host cells with a Cas-based genome editing system to produce a plurality of genomic repair products; (iv) determining the sequences of the genomic repair products; and (iv) training the computational model with input data that comprises at least the sequences of the nucleotide target sequence and/or the genomic repair products and the cut sites.
(228) In another aspect, the specification discloses a method for training a computational model, comprising: (i) preparing a library comprising a plurality of nucleic acid molecules each encoding a nucleotide target sequence and a cut site; (ii) introducing the library into a plurality of host cells; (iii) contacting the library in the host cells with a DSB-based genome editing system to produce a plurality of genomic repair products; (iv) determining the sequences of the genomic repair products; and (iv) training the computational model with input data that comprises at least the sequences of the nucleotide target sequence and/or the genomic repair products and the cut sites.
(229) Methods for preparing nucleic acid libraries, vectors, host cells, and sequencing methods are well known in the art. The instant description is not meant to be limiting in any way as to the construction and configuration of the libraries described herein for training the computational model.
(230) Accordingly, the specification provides in one aspect a method of introducing a desired genetic change in a nucleotide sequence using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the desired genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site.
(231) A cut site can be at any position in a nucleotide sequence and its position is not particularly limiting.
(232) The nucleotide sequence into which a genetic change is desired is not intended to have any limitations as to sequence, source, or length. The nucleotide sequence may comprise one or more mutations, which can include one or more disease-causing mutations.
(233) In another aspect, the specification provides a method of treating a genetic disease by correcting a disease-causing mutation using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence comprising a disease-causing mutation; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for correcting the disease-causing mutation in the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby correcting the disease-causing mutation and treating the disease.
(234) In yet another aspect, the specification provides a method of altering a genetic trait by introducing a genetic change in a nucleotide sequence using a double-strand brake (DSB)-inducing genome editing system, the method comprising: identifying one or more available cut sites in a nucleotide sequence; analyzing the nucleotide sequence and available cut sites with a computational model to identify the optimal cut site for introducing the genetic change into the nucleotide sequence; and contacting the nucleotide sequence with a DSB-inducing genome editing system, thereby introducing the desired genetic change in the nucleotide sequence at the cut site and consequently altering the associated genetic trait.

(235) In another aspect, the specification provides a method of selecting a guide RNA for use in a Cas-genome editing system capable of introducing a genetic change into a nucleotide sequence of a target genomic location, the method comprising: identifying in a nucleotide sequence of a target genomic location one or more available cut sites for a Cas-based genome editing system; and analyzing the nucleotide sequence and cut site with a computational model to identify a guide RNA capable of introducing the genetic change into the nucleotide sequence of the target genomic location.
(236) In still another aspect, the specification provides a method of introducing a genetic change in the genome of a cell with a Cas-based genome editing system comprising: selecting a guide RNA for use in the Cas-based genome editing system in accordance with the method of the above aspect; and contacting the genome of the cell with the guide RNA and the Cas-based genome editing system, thereby introducing the genetic change.
(237) In various embodiments, the cut sites available in the nucleotide sequence are a function of the particular DSB-inducing genome editing system in use, e.g., a Cas-based genome editing system.
(238) In certain embodiments, the nucleotide sequence is a genome of a cell.
(239) In certain other embodiments, the method for introducing the desired genetic change is done in vivo within a cell or an organism (e.g., a mammal), or ex vivo within a cell isolated or separated from an organism (e.g., an isolated mammalian cancer cell), or in vitro on an isolated nucleotide sequence outside the context of a cell.
(240) In various embodiments, the DSB-inducing genome editing system can be a Cas-based genoe editing system, e.g., a type II Cas-based genome editing system. In other embodiments, the DSB-inducing genome editing system can be a TALENS-based editing system or a Zinc-Finger-based genome editing system. In still other embodiments, the DSB-inducing genome editing system can be any such endonuclease-based system which catalyzes the formation of a double-strand break at a specific one or more cut sites.
(241) In embodiments involving a Cas-based genome editing system, the method can further comprise selecting a cognate guide RNA capable of directing a double-strand break at the optimal cut site by the Cas-based genome editing system.
(242) In certain embodiments, the guide RNA is selected from the group consisting the guide RNA sequences listed in any of Tables 1-6. In various embodiments, the guide RNA can be known or can be newly designed.
(243) In various embodiments, the double-strand brake (DSB)-inducing genome editing system is capable of editing the genome without homology-directed repair.
(244) In other embodiments, the double-strand brake (DSB)-inducing genome editing system comprises a type I Cas RNA-guided endonuclease, or a variant or orthologue thereof.
(245) In still other embodiments, the double-strand brake (DSB)-inducing genome editing system comprises a type II Cas RNA-guided endonuclease, or a functional variant or orthologue thereof.
(246) The double-strand brake (DSB)-inducing genome editing system may comprise a Cas9 RNA-guided endonuclease, or a variant or orthologue thereof in certain embodiments.
(247) In still other embodiments, the double-strand brake (DSB)-inducing genome editing system can comprise a Cpf1 RNA-guided endonuclease, or a variant or orthologue thereof.
(248) In yet further embodiments, the double-strand brake (DSB)-inducing genome editing system can comprise a *Streptococcus pyogenes* Cas9 (SpCas9), *Staphylococcus pyogenes* Cas9 (SpCas9), *Staphyloccocus aureus* Cas (SaCas9), *Francisella novicida* Cas9 (FnCas9), or a functional variant or orthologue thereof.
(249) In various embodiments, the desired genetic change to be introduced into the nucleotide sequence, e.g., a genome, is to a correction to a genetic mutation. In embodiments, the genetic mutation is a single-nucleotide polymorphism, a deletion mutation, an insertion mutation, or a microduplication error.

(250) In still other embodiments, the genetic change can comprises a 2-60-bp deletion or a 1-bp insertion.
(251) The genetic change in other embodiments can comprise a deletion of between 2-20, or 4-40, or 8-80, or 16-160, or 32-320, 64-640, or up to 1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, or 10,000 or more nucleotides. Preferably, the deletion can restore the function of a defective gene, e.g., a gain-of-function frameshift genetic change.
(252) In other embodiments, the desired genetic change is a desired modification to a wildtype gene that confers and/or alters one or more traits, e.g., conferring increased resistance to a pathogen or altering a monogenic trait (e.g., eye color) or polygenic trait (e.g., height or weight).
(253) In embodiments involving correcting a disease-causing mutation, the disease can be a monogenic disease. Such monogenic diseases can include, for example, sickle cell disease, cystic fibrosis, polycystic kidney disease, Tay-Sachs disease, achondroplasia, beta-thalassemia, Hurler syndrome, severe combined immunodeficiency, hemophilia, glycogen storage disease Ia, and Duchenne muscular dystrophy.
(254) In any of the above aspects and embodiments, the step of identifying the available cut sites can involve identifying one or more PAM sequences in the case of a Cas-based genome editing system.
(255) In various embodiments of the above methods, the computational model used to analyze the nucleotide sequence is a deep learning computational model, or a neural network model having one or more hidden layers. In various embodiments, the computational model is trained with experimental data to predict the probability of distribution of indel lengths for any given nucleotide sequence and cut site. In still other embodiments, the computational model is trained with experimental data to predict the probability of distribution of genotype frequencies for any given nucleotide sequence and cut site.
(256) In various embodiments, the computational model comprises one or more training modules for evaluating experimental data.
(257) In various embodiments, the computational model can comprise: a first training module for computing a microhomology score matrix; a second training module for computing a microhomology independent score matrix; and a third training module for computing a probability distribution over 1-bp insertions, wherein once trained with experimental data the computational model computes a probability distribution over indel genotypes and a probability distribution over indel lengths for any given input nucleotide sequence and cut site.
(258) In other embodiments, the computational model predicts genomic repair outcomes for any given input nucleotide sequence and cut site.
(259) In various embodiments, the genomic repair outcomes can comprise microhomology deletions, microhomology-less deletions, and/or 1-bp insertions.
(260) In still other embodiments, the computational model can comprise one or more modules each comprising one more input features selected from the group consisting of: a target site nucleotide sequence; a cut site; a PAM-sequence; microhomology lengths relative at a cut site, % GC content at a cut site; and microhomology deletion lengths at a cut site, and type of DSB-genome editing system.
(261) In various embodiments, the nucleotide sequence analyzed by the computational model is between about 25-100 nucleotides, 50-200 nucleotides, 100-400 nucleotides, 200-800 nucleotides, 400-1600 nucleotides, 800-3200 nucleotides, and 1600-6400 nucleotide, or even up to 7K, 8K, 9K, 10K, 11K, 12K, 13K, 14K, 15K, 16K, 17K, 18K, 19K, 20K nucleotides, or more in length.
(262) In another aspect, the specification relates to guide RNAs which are identified by various methods described herein. In certain embodiments, the guide RNAs can be any of those presented in Tables 1-6, the contents of which form part of this specification.
(263) According to various embodiments, the RNA can be purely ribonucleic acid molecules. However, in other embodiments, the RNA guides can comprise one or more naturally-occurring or

non-naturally occurring modifications. In various embodiments, the modifications can including, but are not limited to, nucleoside analogs, chemically modified bases, intercalated bases, modified sugars, and modified phosphate group linkers. In certain embodiments, the guide RNAs can comprise one or more phosphorothioate and/or 5′-N-phosphporamidite linkages.
(264) In still other aspects, the specification discloses vectors comprising one or more nucleotide sequences disclosed herein, e.g., vectors encoding one or more guide RNAs, one or more target nucleotide sequences which are being edited, or a combination thereof. The vectors may comprise naturally occurring sequences, or non-naturally occurring sequences, or a combination thereof.
(265) In still other aspects, the specification discloses host cells comprising the herein disclosed vectors encoding one more more nucleotide sequences embodied herein, e.g., one or more guide RNAs, one or more target nucleotide sequences which are being edited, or a combination thereof.
(266) In other aspects, the specification discloses a Cas-based genome editing system comprising a Cas protein (or homolog, variant, or orthologue thereof) complexed with at least one guide RNA. In certain embodiments, the guide RNA can be any of those disclosed in Tables 1-6, or a functional variant thereof.
(267) In still other aspects, the specification provides a Cas-based genome editing system comprising an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA, wherein the guide RNA can be identified by the methods disclosed herein for selecting a guide RNA.
(268) In yet another aspect, the specification provides a Cas-based genome editing system comprising an expression vector having at least one expressible nucleotide sequence encoding a Cas protein (or homolog, variant, or orthologue thereof) and at least one other expressible nucleotide sequence encoding a guide RNA, wherein the guide RNA can be identified by the methods disclosed herein for selecting a guide RNA.
(269) In still a further aspect, the specification provides a library for training a computational model for selecting a guide RNA sequence for use with a Cas-based genome editing system capable of introducing a genetic change into a genome without homology-directed repair, wherein the library comprises a plurality of vectors each comprising a first nucleotide sequence of a target genomic location having a cut site and a second nucleotide sequence encoding a cognate guide RNA capable of directing a Cas-based genome editing system to carry out a double-strand break at the cut site of the first nucleotide sequence.
(270) In another aspect, the specification provides a library and its use for training a computational model for selecting an optimized cut site for use with a DSB-based genome editing system (e.g., Cas-based system, TALAN-based system, or a Zinc-Finger-based system) that is capable of introducing a desired genetic change into a nucleotide sequence (e.g., a genome) at the selected cut site without homology-directed repair, wherein the library comprises a plurality of vectors each comprising a nucleotide sequence having a cut site, and optionally a second nucleotide sequence encoding a cognate guide RNA (in embodiments involving a Cas-based genome editing system).
(271) Also, the concepts disclosed herein may be embodied as a method, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

EXAMPLES

(272) In order that the invention described herein may be more fully understood, the following examples are set forth. It should be understood that these examples are for illustrative purposes only and are not to be construed as limiting this invention in any manner.

Example 1: Demonstration of Predictable and Precise Template-Free CRISPR Editing of Pathogtenic Variants

SUMMARY

(273) DNA double-strand break repair following cleavage by Cas9 is generally considered stochastic, heterogeneous, and impractical for applications beyond gene disruption. Here, it is shown that template-free Cas9 nuclease-mediated DNA repair is predictable in human and mouse cells and is capable of precise repair to a predicted genotype in certain sequence contexts, enabling correction of human disease-associated mutations. A genomically integrated library of guide RNAs (gRNAs) was constructed, each paired with its corresponding DNA target sequence, and trained a machine learning model, inDelphi, on the end-joining repair products of 1,095 sequences cleaved by Cas9 nuclease in mammalian cells. The resulting model accurately predicted frequencies of 1- to 60-bp deletions and 1-bp insertions (median r=0.87) with single-base resolution at 194 held-out library sites and −90 held-out endogenous sequence contexts in four human and mouse cell lines. The inDelphi model predicts that 26% of all *Streptococcus pyogenes* Cas9 (SpCas9) gRNAs targeting the human genome result in outcomes in which a single predictable product accounts for ≥30% of all edited products, while 5% of gRNAs are “high-precision guides” that result in repair outcomes in which one product accounts for ≥50% of all edited products. It was experimentally confirmed that 183 human disease-associated microduplication alleles can each be corrected to their wild-type genotypes with ≥50% frequency among edited products following Cas9 cleavage in mammalian cells. Using these insights, genotypic and functional rescue of pathogenic LDLR microduplication alleles was achieved in human and mouse cells, and restored to wild-type an endogenous genomic Hermansky-Pudlak syndrome (HPS1) pathogenic allele in primary patient-derived fibroblasts. This study establishes that template-free Cas9 nuclease activity can be harnessed for precise genome editing applications.

(274) More in particular, this study developed a high-throughput *Streptococcus pyogenes* Cas9 (SpCas9)-mediated repair outcome assay to characterize end-joining repair products at Cas9-induced double-stranded breaks using 1,872 target sites based on sequence characteristics of the human genome. The study used the resulting rich set of repair product data to train the herein disclosed machine-learning algorithm (i.e., inDelphi), which accurately predicts the frequencies of the substantial majority of template-free Cas9-induced insertion and deletion events at single-base resolution (which is further described in M. Shen et al., “Predictable and precise template-free CRISPR editing of pathogenic variants,” Nature, vol. 563, Nov. 29, 2018, pp. 646-651, and including Extended Data). This study finds that in contrast to the notion that end-joining repair is heterogeneous, inDelphi identifies that 5-11% of SpCas9 gRNAs in the human genome induce a single predictable repair genotype in ≥50% of editing products.

(275) Building on this idea of precision gRNAs, this study further uses inDelphi to design 14 gRNAs for high-precision template-free editing yielding predictable 1-bp insertion genotypes in endogenous human disease-relevant loci and experimentally confirmed highly precise editing (median 61% among edited products) in two human cell lines. As described herein, inDelphi was used to reveal human pathogenic alleles that are candidates for efficient and precise template-free gain-of-function genotypic correction and achieved template-free correction of 183 pathogenic human microduplication alleles to the wild-type genotype in ≥50% of all editing products. Finally, these developments were integrated to achieve high-precision correction of five pathogenic low-density lipoprotein receptor (LDLR) microduplication alleles in human and mouse cells, as well as correction of endogenous pathogenic microduplication alleles for Hermansky-Pudlak syndrome (HPS1) and Menkes disease (ATP7A) to the wild-type sequence in primary patient-derived fibroblasts.

(276) Results

(277) Cas9-Mediated DNA Repair Products are Predictable

(278) To capture Cas9-mediated end-joining repair products across a wide variety of target sequences, a genome-integrated gRNA and target library screen was designed in which many unique gRNAs are paired with corresponding 55-bp target sequences containing a single canonical

"NGG" SpCas9 protospacer-adjacent motif (PAM) that directs cleavage to the center of each target sequence (FIG. **11**A). To explore repair products among sequences representative of the human genome, 1,872 target sequences were computationally designed that collectively span the human genome's distributions of % GC, number of nucleotides participating in microhomology, predicted Cas9 on-target cutting efficiency.sup.4, and estimated precision of deletion products.sup.24 (FIGS. **16**A-**16**C). Through a multi-step process (FIGS. **16**A-**16**C), the library (Lib-A —see Table 4) was cloned into a plasmid backbone allowing Tol2 transposon-based integration into the genome.sup.25, gRNA expression, and hygromycin selection for cells with genomically integrated library members.

(279) Lib-A was stably integrated into the genomes of mouse embryonic stem cells (mESCs). Next, these cells were targeted with a Tol2 transposon-based SpCas9 expression plasmid containing a blasticidin expression cassette and selected for cells with stable Cas9 expression. Sufficient numbers of cells were maintained throughout the experiment to ensure >2,000-fold coverage of the library. After one week, genomic DNA was collected from three independent replicate experiments from these cells and performed paired-end high-throughput DNA sequencing (HTS) using primers flanking the gRNA and the target site to reveal the spectrum of repair products at each target site. Using a sequence alignment procedure, the resulting 96,838,690 sequence reads were tabulated into observed frequencies of, on average, 1,262 unique repair genotypes for each target site.

(280) To test the correspondence between library repair products and endogenous repair products, Lib-A included the 55-bp sequences surrounding 90 endogenous genomic loci for which the products of Cas9-mediated repair were previously characterized by HTS.sup.24. Previously reported repair products from this endogenous dataset (VO) in three human cell lines (HCT116, K562, and HEK293) reveal that 94% of endogenous Cas9-mediated deletions are 30 bp or shorter (FIGS. **16**A-**16**C), suggesting that the Lib-A analysis method is capable of assessing the vast majority of Cas9-mediated editing products. It was found that repair outcomes for these Lib-A members corresponding to the VO sites in mESCs are consistent with previously reported endogenous repair products in human cells (median r=0.76, FIGS. **17**A-**17**D). Lib-A repair genotype frequencies are also consistent between experimental replicates (median r=0.89, FIGS. **17**A-**17**D), confirming that Cas9-mediated editing products of the target library reflect previously reported endogenous target locus editing products in human cells.

(281) In Lib-A data from mESCs and in the three VO datasets from endogenous HEK293, K562, and HCT116 cells, end-joining repair of Cas9-mediated double-strand breaks primarily causes deletions (73-87% of all products) and insertions (13-25% of all products) (FIGS. **11**B, **11**C, FIGS. **17**A-**17**D). Rarer Cas9-mediated repair products were also detected such as combination insertion/deletions (0.5-2% of all products) and deletions and insertions distal to the cutsite (3-5% of all products), which occur more often on the PAM-distal side of the double-strand break (FIGS. **17**A-**17**D). The majority of products are deletions containing microhomology consistent with MMEJ (53-58% of all products, and 70-75% of deletions) (FIGS. **11**B, **11**C, FIGS. **17**A-**17**D for a definition of microhomology-containing deletions).

(282) Using the wealth of Cas9 outcome data provided by Lib-A, a novel machine learning model, inDelphi, was trained to predict the spectrum of Cas9-mediated editing products at a given target site. This model consists of three interconnected modules aimed at predicting the three major classes of repair outcomes: microhomology deletions (MH deletions), microhomology-less deletions (MH-less deletions), and single-base insertions (1-bp insertions, FIG. **12**A). These three repair classes are defined as constituting all major editing outcomes and note that they comprise 80-95% of all observed editing products (FIGS. **11**B, **11**C). Motivated by the abundance of MH deletion products in Lib-A and VO data, a deep neural network was designed to predict MH deletions as one module of inDelphi. This module simulates MH deletions using the MMEJ repair mechanism, where 5′-to-3′ end resection at a double-strand break reveals two 3′ ssDNA overhangs that can anneal through sequence microhomology. Extraneous ssDNA overhangs are eliminated,

and DNA synthesis and ligation generates a dsDNA repair product.sup.26 (FIG. **12**B). Through this mechanism, each potential microhomology results in a distinct deletion genotype, allowing a 1:1 mapping between possible microhomologies at a target site and available MH deletion outcome genotypes (FIG. **12**B). inDelphi models MH deletions as a competition between different MH-mediated hybridization possibilities. Using the input features of MH length, MH % GC, and deletion length, inDelphi outputs a score (phi) reflecting the predicted strength of each microhomology (FIG. **12**A). From training data, inDelphi learned that strong microhomologies tend to be long and have high GC content (FIGS. **18**A-**18**H).

(283) To account for all deletions that cannot be simulated through the MMEJ mechanism, inDelphi also contains a second neural network module that predicts the distribution of MH-less deletion lengths using the minimum required resection length as the only input feature (FIG. **12**A). Because there are many MH-less genotypes for each deletion length with frequencies that do not fit a simple pattern, inDelphi predicts the frequencies of deletion lengths but not of genotypic outcomes for MH-less deletions. This module learned from training data that the frequency of MH-less deletions decays rapidly with increasing length (FIGS. **18**A-**18**H). It is hypothesized that MH-less deletions arise primarily from the activity of the classical and alternative NHEJ pathways.sup.27. The two neural networks were jointly trained using observed distributions of deletion genotypes from 1,095 Lib-A target sites (FIG. **12**A).

(284) The inDelphi model contains a third module to predict 1-bp insertions (FIG. **12**A). In VO and Lib-A data, insertions represent a major class of DNA repair at Cas9-mediated double-strand breaks (13-25% of all products, FIGS. **11**B, **11**C, FIGS. **17**A-**17**D). Among insertions, 1-bp insertions are dominant (9-21% of all products, FIGS. **11**B, **11**C, FIGS. **17**A-**17**D). Surprisingly, it was found that the frequency of 1-bp insertions and their resultant genotypes depend strongly on local sequence context. In endogenous and library settings, 1-bp insertions predominantly comprise duplications of the −4 nucleotide (counting the NGG PAM as nucleotides 0-2, FIG. **12**A), with higher precision when the −4 nucleotide is an A or T and with lower precision when it is a C or G (FIG. **12**C). While 1-bp insertions were observed occurring in 9% of products on average in Lib-A, this frequency varies significantly depending on the nucleotide at position −4, falling to less than 4% on average when the −4 nucleotide is G (FIG. **12**D, P<10.sup.−34). While position −4 is most strongly associated with 1-bp insertion frequency, other surrounding bases also contribute to insertion frequency (FIG. **12**E). In addition, it was found that target sites with poor microhomology (low total phi score) and target sites with imprecise deletion product distributions are more likely to contain insertions at the expense of deletions (FIGS. **18**A-**18**H). Based on these empirical observations, inDelphi models insertions and deletions as competitive processes in which the total deletion phi score (overall microhomology strength) and predicted deletion precision influence the relative frequency of 1-bp insertions, and the local sequence context influences the relative frequency and genotypic outcomes of 1-bp insertions (FIG. **12**A). inDelphi integrates these factors into predictions of 1-bp insertion genotype frequencies using a k-nearest neighbor approach. Collectively, from sequence context alone, inDelphi predicts the indel lengths of 80-95% of Cas9-mediated editing products and the single-base resolution genotypes of 65-80% of all products (FIG. **13**A, FIGS. **19**A-**19**D).

(285) Trained on data from 1,095 Lib-A sequence contexts in mESCs, inDelphi demonstrates highly accurate genotypic prediction of 1-bp insertions and 1-60-bp deletions at 87-90 VO target sequences previously characterized experimentally in endogenous K562, HCT116, and HEK293 cells (median r=0.87, FIG. **13**B). It is noted that the Lib-A versions of these target sites were held out of inDelphi training. The inDelphi model also performs well when predicting indel length distributions from 1-bp insertions to 60-bp deletions at the endogenous VO sites in three human cell lines (median r=0.84, FIG. **13**C). Additionally, inDelphi accurately predicts relative frequencies of genotypic outcomes (median r=0.94) and indel length distributions (median r=0.91) of 189 held-out Lib-A targets in mESCs (FIGS. **19**A-**19**D). As a control that the features used in training

inDelphi are crucial for its performance, the MH length feature was deleted from the inDelphi MH deletion module and found that inDelphi's performance predicting genotype frequency was reduced to the performance of a model with random weights. A second control in which the deep neural networks were replaced with linear models showed 10-24% reduced performance on the genotype frequency and indel length prediction tasks. Together, these controls indicate that inDelphi's computational structure is important for its accuracy. An online implementation of inDelphi is provided to predict the spectrum of Cas9-mediated products at any target site (crisprindelphi.design). Taken together, these results establish that in data from human and mouse cells, the relative frequencies of most Cas9 nuclease-mediated editing outcomes are highly predictable.
(286) The ability of Cas9-mediated end-joining repair to induce frameshifts enables efficient gene knockout.sup.28. It was reasoned that inDelphi's accurate prediction of the indel length distribution of 80-95% of template-free Cas9-mediated editing products should also enable accurate prediction of Cas9-induced frameshifts. This task was simulated in 86 endogenous VO target sequences in HEK293 by tabulating the observed frequency of indels resulting in +0, +1, and +2 frameshifts. The observed frequency of indels in each frame predicted by inDelphi (median r=0.81) compare favorably to those generated by Microhomology Predictor (median r=0.37), a previously published method.sup.29 (FIG. **13**D). Thus, it is expected that inDelphi facilitates Cas9-mediated gene knockout approaches by allowing a priori selection of gRNAs that induce high or low knockout frequencies. To this end, an online tool is provided to predict frameshift frequencies for any SpCas9 gRNA targeting the coding human and mouse genome (crisprindelphi.design). It is noted that human exons have a significant tendency (p<10.sup.−100, FIGS. **19**A-**19**D) to favor frame-preserving deletion repair compared to shuffled exon sequences or non-coding human DNA. Taken together, the results show that inDelphi provides accurate single-base resolution predictions for the relative frequencies of most Cas9 nuclease-mediated end-joining repair outcomes, including frameshifts.
(287) Designing High-Precision Template-Free Cas9 Nuclease-Mediated Editing
(288) While end-joining repair is highly efficient at inducing mutations after Cas9 treatment, its tendency to induce a heterogeneous mixture of repair genotypes has limited its application primarily to gene disruption and removal of intervening sequences between two double-stranded breaks30-33. Motivated by inDelphi's ability to predict Cas9-mediated repair outcomes from target sequences alone, it was sought to identify target sites for which the repair profile is highly skewed toward a single outcome. In principle, the ability of inDelphi to identify such sites may enable efficient, template-free, nuclease-mediated precision gene editing.
(289) It was reasoned that a single strong microhomology hybridization possibility with a high phi score would outcompete a background of weaker alternative microhomologies to yield efficient and precise repair to a single deletion genotype. Microduplications, in which a stretch of DNA is repeated in tandem, contain stretches of exact microhomology and thus are predicted by inDelphi to collapse precisely through deletion upon MMEJ repair (FIG. **14**A).
(290) To test this prediction, a second high-throughput Cas9 substrate library (Lib-B—see Table 5) was designed and constructed that contains three families of target sequences with microduplications of each length from 7-25 bp. Cas9-mediated double-strand break repair products were analyzed in Lib-B in mESCs and in human U2OS and HEK293T cells using the same procedure as for Lib-A evaluation. Highly precise repair was consistently observed in which 40-80% of all repair events correspond to a single repair genotype (FIG. **14**B), substantially higher than the 21% median frequency of the most abundant deletion genotype in 90 VO sites that were not pre-selected for microhomology. The fraction of microduplication repair to a single collapsed product as compared to other outcomes increased with microduplication length in mESCs, U20S, and HEK293T cells (r=0.35, p<7×10.sup.−5, FIG. **14**B, FIGS. **20**A-**20**E). It is noted that these sites have significantly higher phi scores and precision scores compared to VO sites, and significantly

fewer 1-bp insertions (FIGS. **20**A-**20**E). Thus, sites with strong MH deletion candidates are enriched in that specific deletion outcome at the expense of MH-less deletion and 1-bp insertion outcomes.

(291) It is also hypothesized that sequence contexts with no strong MHs (low total phi scores) could enable precise 1-bp insertion repair. To test this possibility, three target sequence frameworks with low total phi scores were included in Lib-B (FIGS. **20**A-**20**E) containing randomization at the four positions surrounding the Cas9 cleavage site (positions −5 to −2 with respect to the PAM at positions 0-2; see FIG. **12**A). Cas9 nuclease treatment of 205 such sequences in Lib-B resulted in highly precise (up to 90% of all repair events) and reproducible (r=0.90 between mESC replicates) 1-bp insertions (FIG. **14**C, FIGS. **20**A-**20**E). Strikingly, the efficiency of 1-bp insertions is strongly influenced by the nucleotide identities in positions −5 to −2 (FIG. **14**C). Similar to the findings from Lib-A (FIG. **12**E, FIGS. **20**A-**20**E), −4T and −3G correlate with higher relative frequencies of 1-bp insertion among all products while −4G correlates with lower frequencies of insertion (FIG. **14**D). Among these three fixed sequence contexts in Lib-B with low total MH, 1-bp insertions comprise a median of 29% of all repair products, which is significantly higher than in VO sites (FIGS. **20**A-**20**E). Moreover, a median of 61% of all products are 1-bp insertions at sites with TG at the −4 and −3 positions (FIG. **14**D), revealing that precise 1-bp insertion can be obtained through Cas9-mediated end-joining at specific, predictable sequence contexts.

(292) It is noted that sequences that support higher insertion efficiencies (>50%) have on average 33% lower total efficiencies of Cas9-mediated indels than sequences that yield lower insertion efficiencies (r=−0.35, p=3.3×10.sup.−7, FIGS. **20**A-**20**E), possibly because the lower efficiency of MMEJ at such sites decreases the likelihood of mutagenic repair of the Cas9-induced double-strand break. These observations collectively establish that Cas9-mediated repair of target sites with predictable sequence features can lead to precise editing favoring one particular outcome.

(293) Based on these findings, inDelphi was used to predict gRNAs that lead to such precise outcomes. A metric was defined using information entropy to measure the precision of a repair outcome spectrum as a score ranging from zero (highest entropy, lowest precision) to one (lowest entropy, highest precision) and demonstrated that inDelphi is capable of predicting the precision of Cas9-induced deletions in 86 VO target sequences in HEK293 cells (median r=0.64, FIG. **14**E). inDelphi was then used to discover SpCas9 gRNAs that support precise end-joining repair in the human genome.

(294) It was found that substantial fractions of all genome-targeting SpCas9 gRNAs are predicted to produce relatively precise outcomes (Table 2). Indeed, inDelphi predicts that 26% of SpCas9 gRNAs that target human exons and introns are “precision gRNAs” (FIG. **14**F), which are defined as gRNAs predicted to produce a single genotypic outcome in >30% of all major editing products, with 20% of gRNAs predicted to be produce a single deletion genotype at ≥30% efficiency and 6.2% predicted to produce a single 1-bp insertion genotype at >30% efficiency. Moreover, inDelphi predicts that 4.8% of SpCas9 gRNAs targeting human exons and introns are “high-precision gRNAs,” which are defined as gRNAs that produce a single genotype in ≥50% of all major editing products (FIG. **14**F, 3.8% producing high-precision deletion, 0.94% producing high-precision 1-bp insertion). These findings suggest that Cas9-mediated end-joining outcomes at many target sites are both predictable and precise, and that precision and high-precision gRNAs offer new opportunities for precision deletion and insertion by Cas9 nuclease-mediated editing. An online tool is provided to predict the precision of a given gRNA and to identify precision and high-precision SpCas9 gRNAs targeting the human and mouse genomes (crisprindelphi.design).

(295) Efficient Template-Free Repair of Pathogenic Alleles to Wild-Type Genotypes

(296) Next, inDelphi-classified high-precision gRNAs were used to identify new targets for therapeutic genome editing. Starting with 23,018 insertion, short deletion, and microduplication disease genotypes from the ClinVar and HGMD databases.sup.16,17, inDelphi was tasked with identifying pathogenic alleles that are suitable for template-free Cas9-mediated editing to effect

precise gain-of-function repair of the pathogenic genotype. Two genetic disease allele categories that have not been previously identified as targets for Cas9-mediated repair are predicted by inDelphi to be candidates for high-precision repair. The first category is a selected subset of pathogenic frameshifts in which, because of high-precision repair, inDelphi predicts that 50-90% of Cas9-mediated deletion products will correct the reading frame compared to the average frequency of 34% among all disease-associated frameshift mutations. The second category is pathogenic microduplication alleles in which a short sequence duplication leads to a frameshift or loss-of-function protein sequence changes (FIG. **15**A).

(297) To test the accuracy of inDelphi at predicting repair genotypes of therapeutically relevant alleles, 1,592 pathogenic human loci that inDelphi identified to have the highest predicted rates of frameshift or microduplication repair to the wild-type sequence, were included in Lib-B. Cas9-mediated repair of genome-integrated Lib-B in mESCs and human U2OS and HEK293T cells confirmed highly efficient and precise gain-of-function editing. It was observed that 183 human disease microduplication alleles included in Lib-B were repaired to wild-type in ≥50% of all products FIG. **15**B), and 508 pathogenic human frameshift alleles were restored into proper reading frame in ≥50% of all products in mESCs (FIG. **15**C), in agreement with inDelphi's predictions (r=0.64 for frame restoration, r=0.64 for wildtype repair). Similar results were observed in HEK293T and U2OS cells (FIGS. **21**A-**21**D). While microduplication repair to the wild-type genotype unambiguously restores wild-type protein function, it is noted that frameshift restoration that alters coding sequence requires case-by-case analysis to validate rescue of protein function.

(298) To determine if the efficiency of microduplication repair can be increased by manipulation of DNA repair pathways, Cas9 cleavage of Lib-B was performed in Prkdc.sup.−/−Lig4.sup.−/− mESCs.sup.34, which are deficient for two proteins involved in NHEJ repair.sup.35. As expected, the frequency of MH-less deletion repair in cells with impaired NHEJ is decreased (25% to 16%) (FIGS. **22**A-**22**E). It was also observed that the precision of 1-bp insertions that result in duplication of the −4 nucleotide is increased in Prkdc.sup.−/−Lig4.sup.−/− mESCs (FIGS. **22**A-**22**E). Importantly, the frequency of MH-dependent deletion repair is substantially increased (58% to 72%) in Prkdc.sup.−/−Lig4.sup.−/− mESCs, enabling a subset of pathogenic alleles to be repaired to wild-type with strikingly high precision. In wild-type mESCs, 183 pathogenic alleles are repaired to wild-type in ≥50% of all edited products and 11 pathogenic alleles are repaired to wild-type in ≥70% of all edited products, while in Prkdc.sup.−/−Lig4.sup.−/− mESCs, 286 pathogenic alleles are repaired to wildtype in ≥50% of all edited products and 153 pathogenic alleles are repaired to wild-type in ≥70% of products (FIG. **15**D, Table 6). Thus, impairing NHEJ can further increase the precise repair of pathogenic microduplications to wild-type (p=7.8*10.sup.−12, FIG. **15**D). These data support the model that competing end-joining repair mechanisms determine the relative frequencies of specific editing outcome types and demonstrate that template-free genotypic correction of hundreds of pathogenic microduplication alleles in genes such as PKD1 (corrected in 92% of edited Prkdc.sup.−/−Lig4.sup.−/− mESC alleles), GJB2 (91%), MSH2 (88%), LDLR (87%), and BRCA1 (82%) can be optimized to occur with strikingly high efficiency by manipulation of repair pathways. inDelphi's prediction of highly efficient wild-type repair was further tested on pathogenic LDLR microduplication alleles, which cause dominantly inherited familial hypercholesterolemia.sup.36. Five pathogenic LDLR microduplication alleles were separately introduced within a full-length LDLR coding sequence upstream of a P2A-GFP cassette into the genome of human and mouse cells, such that Cas9-mediated repair to the wild-type LDLR sequence should induce phenotypic gain of LDL uptake and restore the reading frame of GFP. Cas9 and a gRNA that is specific to each pathogenic allele and does not target the wild-type repaired sequence were then deleivered. Robust restoration of LDL uptake was observed as well as restoration of GFP fluorescence in mESCs, U2OS cells, and HCT116 cells in up to 79% of cells following transfection with Cas9 and inDelphi gRNAs (FIGS. **15**E, **15**F, FIGS. **23**A-**23**E). HTS confirms efficient genotypic repair to wild-type of these five LDLR microduplication alleles in

human and mouse cells as well as of three other pathogenic microduplication alleles in the GAA, GLB1, and PORCN genes introduced to cells using the same method (Table 1, Table 3). Importantly, in these experiments, high-frequency LDLR phenotypic correction was observed when cutting with either SpCas9 or *Streptococcus aureus* Cas9 (SaCas9).sup.37 (Table 3), suggesting that microduplication repair is a feature of cellular repair after a Cas9-mediated double-strand break that does not require a specific nuclease.

(299) Finally, precise template-free Cas9-mediated MMEJ was used to repair an endogenous pathogenic 16-bp microduplication in primary fibroblasts from a Hermansky-Pudlak syndrome (HPS1) patient. HPS1 causes blood clotting deficiency and albinism in patients and is particularly common in Puerto Ricans.sup.38. Simultaneous delivery of Cas9 and gRNA specific to the pathogenic microduplication allele induced high-efficiency correction to the wild-type sequence (mean frequency=71% of edited alleles, N=3, Table 1). These findings suggest the potential of template-free, precise Cas9 nuclease-mediated repair of microduplication alleles to achieve efficient repair to the wild-type sequence for therapeutic gain-of-function genome editing.

(300) The following tables are referenced in this specification.

(301) TABLE-US-00003 TABLE 1 Repair of microduplication pathogenic alleles through template-free Cas9- nuclease treatment. Predicted frequency Observed frequency Observed frequency of repair to wild- of repair to wild-type of repair to wild- type Pathogenic type genotype in all genotype in all genotype in all edited microduplication major editing edited products in outcomes in genotype Cell type products (%) Lib-B, mESCs (%) endogenous data (%) HPS1:c.1472_1487 Primary 88 53 71 dup16 patient fibrobtasts LDLR:c.1662_1689 mESCs 85 61 65 dupGCTGGTGA LDLR:c.1662_1669 HCT116 85 61 89 dupGCTGGTGA LDLR:c.1662_1669 USOS 85 51 77 dupGCTGGTGA

(302) TABLE-US-00004 TABLE 2 Frequency of gRNAs in the human genome with denoted Cas9-mediated outcome precision. Fraction of 1,003,524 SpCas9 gRNAs in human exons and introns for which the most-common repair genotype comprises >XX % of all major editing products Precision gRNA Precise product Precise product Any precise frequency is a deletion is a 1-bp insertion product (%) (% of gRNAs) (% of gRNAs) of gRNAs) 10 86 35 96 15 63 23 78 30 43 14 55 35 29 10 39 30 20 6.2 26 35 13 4.1 17 40 8.6 2.6 11 45 5.8 1.3 7.1 50 3.8 0.94 4.8 55 2.4 0.52 3.0 60 1.5 0.39 1.9 65 0.98 0.086 1.1 70 0.58 0.026 0.61 75 0.29 0.024 0.32 80 0.12 0 0.12 85 0.040 0 0.040 90 0.0049 0 0.0049

(303) TABLE-US-00005 TABLE 3 Repair of eight pathogenic microduplication alleles in individual cellular experiments. Pathogenic allele LDLRdup1 LDLRdup2 LDLRdup2 LDLRdup3 LDLRdup4 LDLRdup5 PORCNdup GAAdup GAADup GLB1dup HPS1dup ATP7Adup #AlleleID 245617 245706 245706 245709 245715 246266 25739 354180 354180 98805 ND ND Predicted frequency of 79 98 96 95 86 94 90 76 93 95 ND ND deletions restoring frame Flow cytometric frameshift 57 95 57 90 72 87 ND 79 74 85 ND ND frequency (%) Predicted frequency of 72 90 83 94 85 86 89 74 91 79 88 43 repair to wild-type genotype among all major editing products (%) Flow cytometric 36 69 30 53 33 78 ND ND ND ND ND ND phenotypic repair frequency, mESC (%) Observed frequency of ND 67 39 25 15 65 48 76 59 42 ND ND repair to wild-type genotype among all edited products in HTS, mESC (%) Observed frequency of 100 88 ND ND ND 77 ND ND ND ND ND ND repair to wild-type genotype among all edited products in HTS, U2OS (%) Observed frequency of ND ND ND 24 ND 89 ND ND ND ND ND ND repair to wild-type genotype among all edited products in HTS, HCT116 (%) Observed frequency of ND ND ND ND ND 58 42 ND 63 41 ND ND repair to wild-type genotype among all edited products in Lib-B, mESCs (%) Observed frequency of ND ND ND ND ND ND ND ND ND ND 88 ± 14* 98 repair to wild-type genotype among all edited products in primary patient fibroblasts (%) gRNA

sequence TGCGA GCAAG TCCTC CTGCA TTTCC ACATC CTGTC AGCTG CTGCA TGTGA CAGCA TTTTT (SEQ ID NOs: 11577-11588) AGATG GATAA GTCAG AGGAC TCGTC TACTC CCTGG CAGAA GAAGG ACTAT GGGGA CCATA from left to right GCTCG ATTTG ATTTG AAATC AGATT GCTGG CTTTT GGTGA TGACT GGTGC GGCCC TAAGA GAGGC ACAGG TCCTG TGAGG TGTCG TGAGC ATCCC CTGCA GCAGA ATATA CCAGC TAAGA Cas9 Type KKH KKH KKH KKH KKH SaCas9 SaCas9 KKH SaCas9 SaCas9 SaCas9 SaCas9 SaCas9 SaCas9 SaCas9 SaCas9 SaCas9 SaCas9 ND, not determined. LDLRdup1, LDLR:c.526_533dupGGCTCGGA. LDLRdup2, LDLR:c.668_681dupAGGACAAATCTGAC. (SEQ ID NO: 1) LDLRdup3, LDLR:c.669_680dupGGACAAATCTGA. LDLRdup4, LDLR:c672_683dupCAAATCTGACGA. LDLRdup5, LDLR:c.1662_1669dupGCTGGTGA. (SEQ ID NOs: 2 and 3) PORCNdup, PORC:c.1059_1071dupCCTGGCTTTTATC. GAAdup, GAA:c.2704_2716dupCAGAAGGTGACTG. (SEQ ID NOs: 4 and 5) GLB1dup, GLB1:c.1456_1466dupGGTGCATATAT (SEQ ID NO: 6)
(304) Table 4: Lib-A sequences (presented below between the end of this specification and Table 5).
(305) Table 5: Lib-B sequences (presented below between Table 4 and Table 6).
(306) Table 6: inDelphi predictions and observed results. Table 6 comprises Table 6A: inDelphi predictions and observed results for Lib-B, showing all sequences with replicate-consistent mESC results; Table 6B (continued from 6A); Table 6C (continued from 6B); Table 6D (continued from 6C); and Table 6E (continued from 6D) (presented below between Table 5 and the claims).
(307) TABLE-US-00006 TABLE 7 Frequency of gRNAs in the human genome with denoted Cas9-mediated outcome precision inDelphi trained on Lib-A data InDelphi trained on Lib-A data from mESCs for 1-bp ins. module from U2OS cells for 1-bp ins. module Precision-X Precise product Precise product Total % of Precise product Precise product Total % of threshold is a deletion is a 1-bp insertion gRNAs that is a deletion is a 1-bp insertion gRNAs that (%) (% of gRNAs) (% of gRNAs) are precise-X (% of gRNAs) (% of gRNAs) are precise-X 10 82 38 93 70 78 97 15 61 23 75 44 64 87 20 43 15 55 27 53 72 25 30 10 39 17 44 58 30 21 6.6 28 11 36 46 35 15 4.4 19 6.9 28 34 40 10 2.9 13 4.1 21 25 45 6.5 1.9 6.4 2.4 15 18 50 4.3 1.3 5.6 1.4 10 12 55 2.8 0.8 3.6 0.6 6.7 7.5 60 1.8 0.5 2.3 0.5 4.0 4.4 65 1.1 0.3 1.5 0.2 2.2 2.4 70 0.7 0.2 0.9 0.1 1.1 1.2 75 0.4 0.1 0.5 0.04 0.5 0.5 80 0.2 0.06 0.3 0.01 0.2 0.2 85 0.06 0.04 0.1 0.003 0.07 0.06 90 0.03 0.02 0.05 0.0007 0.03 0.03
(308) TABLE-US-00007 TABLE 8 Endogenous repair of 24 designed high- precision gRNAs in human cell lines Observed frequency among all edited products from deep sequencing at endogenous loci (%) Gene, Most Most exon/chr, frequent frequent outsite Frameshift, genotype, Frameshift genotype, (hg19) U2OS U2OS HEK293T HEK293T VEGFA 91, 67 36, 34* 90, 90 43, 40* exon1: 456 VEGFR2 91, 91 50, 53* 91, 91 50, 24* exon5: 2 PDCD1 90, 90 20, 21* 91, 90 29, 13* exon5: 208 APOB 83, 83 22, 21* 87, 85 35, 18* exon25: 147 VEGFA 85, 89 27, 29* 93, 91 55, 32* exon3: 127 CCR5 82, 81 20, 21* 86, 84 43, 27* exon1: 1941 CD274 85, 86 9, 10* 84, 82 31, 14* exon2: 271 APOB 91, 89 26, 25* 88 37* exon26: 5590 VEGFR2 82, 82 35, 33* 82, 82 40, 24* exon26: 19 CXCR4 86, 86 32, 33* 91 54* exon1: 825 PCSK9 81, 78 28, 25.sup.† 78 27.sup.† exon11: 15 CCR5 84, 85 55, 52.sup.† 67 46.sup.† exon1: 885 CCR5 92, 94 61, 60.sup.† 91, 92 49, 58.sup.† exon1: 1027 APOB 93, 93 75, 74.sup.† 93, 95 69, 81.sup.† exon26: 5573 CCR5 94, 94 37, 25.sup.† 83, 89 29, 38.sup.† exon1: 61 CCR5 61, 61 28, 29.sup.† 80, 83 29, 43.sup.† exon1:1577 APOB 89, 89 25, 27.sup.† 91, 89 23, 38.sup.† exon22: 100 APOBEC3B 83, 84 50, 52.sup.† 75, 88 51, 60.sup.† exon3: 202 MACCHC 97, 95 80, 77.sup.†‡ 97, 98 78, 85.sup.†‡ chr1: 45973892 PROK2 93, 94 44, 41.sup.†‡ 93, 93 45, 53.sup.†‡ chr3: 71821967 IDS 95, 95 72, 74.sup.†‡ 93, 95 64, 80.sup.†‡ chrX: 148564700 ECM1 87, 89 44, 47.sup.†‡ 89, 89 32, 35.sup.†‡ chr1: 150484936 KCNH2 40 25.sup.†‡ 65, 85 35, 14.sup.†‡ chr7: 150644566 LDLR 90, 91 76, 77.sup.†‡ 90, 96 77, 83.sup.†‡ chr19: 11222303
Discussion

(309) The Cas9-mediated end-joining repair products of thousands of target DNA loci integrated into mammalian cells were used to train a machine learning model, inDelphi, that accurately predicts the spectrum of genotypic products resulting from double-strand break repair at a target DNA site of interest. The ability to predict Cas9-mediated products enables new precision genome editing research applications and facilitates existing applications.
(310) The inDelphi model identifies target loci in which a substantial fraction of all repair products consists of a single genotype. The findings suggest that 26% of SpCas9 gRNAs targeting the human genome are precision gRNAs, yielding a single genotypic outcome in ≥30% of all major repair products, and 5% are high-precision gRNAs in which ≥50% of all major repair products are of a single genotype. Such precision and high-precision gRNAs enable uses of Cas9 nuclease in which the major genotypic products can be predicted a priori. Indeed, it was experimentally shown that high-precision, template-free Cas9-mediated editing can mediate efficient gain-of-function repair at hundreds of pathogenic alleles including microduplications (FIGS. **15**B, **15**E, **15**F) in cell lines and in patient-derived primary cells (Table 1).
(311) Moreover, evidence is presented that manipulation of available DNA repair pathways can further increase the precision of template-free repair outcomes. Suppressing NHEJ augments repair of pathogenic microduplication alleles, suggesting that temporary manipulation of DNA repair pathways could be combined with Cas9-mediated editing to favor specific editing genotypes with high precision. Genome editing currently lacks flexible strategies to correct indels in post-mitotic cells because of the limited efficiency of HDR in non-dividing cells.sup.39. As MMEJ is thought to occur throughout the cell cycle.sup.40, inDelphi may provide access to predictable and precise post-mitotic genome editing in a wider range of cell states. It is also anticipated that, given appropriate training data, inDelphi will also be able to accurately predict repair genotypes from other double-strand break creation methods, including other Cas9 homologs, Cpf1, transcription activator-like effector nucleases (TALENs), and zinc-finger nucleases (ZFNs).sup.37,41-43 This work establishes that the prediction and judicious application of template-free Cas9 nuclease-mediated genome editing offers new capabilities for the study and potential treatment of genetic diseases.
(312) Cellular Repair of Double-Stranded DNA Breaks and inDelphi
(313) DNA double-strand breaks are detrimental to genomic stability, and as such the detection and faithful repair of genomic lesions is crucial to cellular integrity. A large number of genes have evolved to respond to and repair DNA double-strand breaks, and these genes can be broadly grouped into a set of DNA repair pathways.sup.26, each of which differs in the biochemical steps it takes to repair DNA double-strand breaks. Accordingly, these pathways tend to produce characteristically distinguishable non-wildtype genotypic outcomes.
(314) The goal of the machine learning algorithm, inDelphi, is to accurately predict the identities and relative frequencies of non-wildtype genotypic outcomes produced following a CRISPR/Cas9-mediated DNA double-strand break. To accomplish this goal, parameters were developed to classify three distinct categories of genotypic outcomes, microhomology deletions, microhomology-less deletions, and insertions, informed by the biochemical mechanisms underlying the DNA repair pathways that typically give rise to them.
(315) Double strand breaks are thought to be repaired via four major pathways: classical non-homologous end-joining (c-NHEJ), alternative-NHEJ (alt-NHEJ), microhomology-mediated end-joining (MMEJ), and homology-directed repair (HDR)1. To create inDelphi, three machine learning modules were developed to model genotypic outcomes assuming characteristic of the c-NHEJ, microhomology mediated alt-NHEJ, and MMEJ pathways. While template-free CRISPR/Cas9 DNA double-strand break may lead to HDR repair via endogenous homology templates that exist in trans.sup.45, HDR-characteristic outcomes are not explicitly modeled using the algorithm.
(316) Before proceeding, it is important to note that while specific DNA repair pathways are

characteristically associated with distinct genotypic outcomes, the proteins involved in the various pathways and the resulting repair products may at times overlap. This fact has several implications. First, conclusive statements cannot be made about the role of specific proteins or pathways in specific genotypic outcomes without perturbation experiments (e.g. the comparison of wildtype and Prkdc.sup.−/−Lig4.sup.−/− mESCs can illuminate the roles of these proteins, specifically). Second, because assigning genotypic outcomes to biochemical mechanisms is likely imperfect, machine learning methods were used to identify trends and patterns in genotype frequencies that refine this crude binning process.
(317) In the first step of the inDelphi method, genotypic outcomes were separated into three classes: microhomology deletions (MH deletions), microhomology-less deletions (MH-less deletions), and single-base insertions (1-bp insertions) (FIG. **12**A). Below the algorithmic definitions of each genotypic outcome class are outlined, the pathways associated with each class, and the DNA sequence parameters included in inDelphi training of each class. For more detailed technical algorithmic definitions of the genotypic outcome classes.
(318) MH Deletions are Predicted from MH Length, MH GC Content, and Deletion Length
(319) The majority of Cas9-mediated double-strand break repair genotypes observed in the datasets are what are classified as MH deletions (53-58% in mESC, K562, HCT116, and HEK293). It is hypothesized that these deletions occur through MMEJ-like processes and use known features of this pathways to inform a machine learning module to predict MH deletion outcomes. Following 5′-end resection as occurs in MMEJ, alt-NHEJ, and HDR.sup.26, microhomologous basepairing of single-stranded DNA (ssDNA) sequences occurs across the border of the double strand breakpoint.sup.46, 47. To restore a contiguous double-strand DNA chain, the 5′-overhangs not participating in the microhomology are removed up until the paired microhomology region, and the unpaired ssDNA sequences are extended by DNA polymerase using the opposing strand as a template (FIG. **12**B, FIGS. **18**A-**18**H).
(320) Assuming these same processes, inDelphi calculates the set of all MH deletions available given a specific sequence context and cleavage site.
(321) As an example workflow, given the following sequence and its cleavage site:
(322) TABLE-US-00008 (SEQ ID NO: 7) ACGTG|CATGA (SEQ ID NO: 11589) TGCAC|GTACT
for every possible deletion length from 1-bp to 60-bp deletions, the 3′-overhang is overlapped downstream of the cut site under the upstream 3′-overhang and it is determined if there is any microhomologous basepairing. As an example, given the 4-bp deletion length:
(323) TABLE-US-00009 ACGTG | || GTACT
it is seen that there are three microhomologous basepairing events.
(324) Then a particular microhomology is chosen (here, the italicized C:G):
(325) TABLE-US-00010 ***AC***GTG | || *G***TACT**
then generate its unique repair genotype by following left-to-right along the top strand and jumping down to the complement of the bottom strand to simulate DNA polymerase fill-in.
Here, this yields:
(326) TABLE-US-00011 ACATGA TGTACT
(327) This can also be displayed as an alignment. It is noted that by “jumping down” after the first base in the top strand, this outcome can also be described using the delta-position 1. (See section on delta-positions). A deletion at delta-position 0 yields the same genotype.
(328) TABLE-US-00012 Deletion b: ACGTG----A Wt: ACGTGCATGA (SEQ ID NO: 7)
(329) Thus, there may be multiple MH deletion outcome genotypes for a given deletion length, and there is always a 1:1 mapping between the microhomologous basepairing used in that MH deletion and the resultant genotypic outcome. The set of MH deletions thus includes all 1-bp to 60-bp deletions that can be derived from the steps above that simulate the MMEJ mechanism.
(330) MMEJ efficiency has been reported to depend on the thermodynamic favorability and

stability of a candidate microhomology.sup.46, 47. To parameterize MH deletions using the biochemical sequence features that influence this form of DNA repair, inDelphi calculates the MH length, MH GC content, and resulting deletion length for each possible MH deletion. These features are input into a machine learning module as the microhomology neural network (MH-NN) to learn the relationship between these features and the frequency of an MH deletion outcome in a training CRISPR/Cas9 genotypic outcome dataset. While it was predicted and empirically found that favored MH deletions have long MH lengths relative to total deletion length and high MH GC-contents, any explicit direction or comparative weighting to these parameters are not provided at the outset. inDelphi then outputs a phi-score for any MH deletion genotype (whether it was in the training data or not) that represents the favorability of that outcome as predicted by MH-NN.
(331) It is important to emphasize that the phi-score of a particular MH deletion does not itself represent the likelihood of that MH deletion occurring in the context of all MH deletions at a given site. Some CRISPR/Cas9 target sites may have many possible favorable MH deletion outcomes while other sites have few, and thus phi-score must be normalized for a given target site to generate the fractional likelihood of that genotypic outcome at that site. Total unnormalized MH deletion phi-score is one factor that is further used to predict the relative frequency of the different repair classes: MH deletions, MH-less deletions, and insertions.
(332) MH-Less Deletions are Predicted from their Length
(333) MH-less deletions are defined as all possible deletions that have not been accounted for by the workflow described above for MH deletions. Mechanistically, the data analysis suggests that MH deletions are associated with repair genotypes produced by c-NHEJ and microhomology-mediated alt-NHEJ pathways.
(334) Following a double-strand break, c-NHEJ-associated proteins rapidly bind the DNA strands flanking the double-strand DNA breakpoint and recruit ligases, exonucleases, and polymerases to process and re-anneal the breakpoint in the absence of 5′-end resection (FIGS. **18**A-**18**H).sup.26, 35. Commonly, c-NHEJ repair is error-free; however, in the context of Cas9-mediated cutting, faithful repair leads to repeated cutting, thereby increasing the eventual likelihood of mutagenic repair. Erroneous c-NHEJ repair products are mainly thought to consist of small insertions or deletions or combinations thereof that most frequently occur in the direct vicinity of the DNA break point.sup.35, 48, 49. The resulting deletions, which are referred to as medial end-joining MH-less deletions, have often lost bases both upstream and downstream of the cleavage site.
(335) Microhomology-mediated alt-NHEJ is a distinct pathway that produces MH-less deletion products. In contrast to c-NHEJ, which is microhomology independent, this form of alt-NHEJ repair occurs following 5′-end resection and is mediated by microhomology in the sequence surrounding the double-strand break-point1. Microhomologous basepairing stabilizes the 3′-ssDNA overhangs following 5′-end resection, similarly to in MMEJ, allowing DNA ligases to join the break across one of the strands of this temporarily configured complex. The opposing un-annealed flap is then removed, and newly synthesized DNA templated off of the remaining strand is annealed to repair the lesion (FIGS. **18**A-**18**H).
(336) While alt-NHEJ uses microhomology, the repair products it produces do not follow the predictable genotypic patterns induced by MMEJ and are thus grouped into MH-less deletion genotypes. MH deletions are a direct merger of both annealed strands, in which the outcome genotype switches from top to bottom strand at the exact end-point of a microhomology. In contrast, while alt-NHEJ employs microhomology in its repair mechanism, the deletion outcomes it generates comprise bases exclusively derived from either the top or bottom strand. Mechanistically, this occurs because ligation of a 3′-overhang to its downstream ligation partner results in removal of the entire opposing ssDNA overhang up until the point of ligation. This process prevents any deletion from occurring in the 3′-overhang strand that is first attached to the DNA backbone, while inducing loss of an indeterminant length of sequence on the opposing strand. The resulting deletion genotypes, which are referred to as unilateral end-joining MH-less deletions, do not retain

information on the exact microhomology causal to their occurrence, and are thus also referred to as MH-less.
(337) Consequently, the various mechanisms that give rise to MH-less deletions are capable of generating a vast number of genotypic outcomes for any given deletion length. Having less information on the biochemical mechanisms that impact the relative frequency of NHEJ deletion products, inDelphi models these deletions without assuming any particular mechanism. inDelphi detects MH-less deletions from training data as the set of all deletions that are not MH deletions and parameterizes them solely by the length of the resulting deletion. This is based on the simple assumption that c-NHEJ and alt-NHEJ processes are most likely to produce short deletions, supported by the empirical observation. As with MH deletions, this assumption is not explicitly coded into the inDelphi MH-less deletion prediction module, instead allowing it to be "learned" by a neural network called MHless-NN.
(338) MHless-NN optimizes a phi-score for a given MH-less deletion length, grounded in the frequency of MH-less deletion outcomes of that length observed in the training data. It was observed that MHless-NN learns a near-exponential decaying phi-score for increasing deletion length, that reflects the sum total frequency of all MH-less deletion genotypes. The total unnormalized MH-less deletion phi-score for a given target and cut site is also employed to inform the relative frequency of different repair classes.
(339) 1-Bp Insertions are Predicted from Sequence Context and Deletion Phi-Scores
(340) Lastly, inDelphi predicts 1-bp insertions from both the broader sequence context and the immediate vicinity of the cleavage site. It was empirically found that 1-bp insertions are far more common than longer insertions, so the focus is on their prediction. It is classically assumed that short sequence insertions are the result of c-NHEJ.sup.48,49, however, little else is known about their biochemical mechanism as it pertains to local sequence context to help inform prediction. Nonetheless, powerful correlations were found between the identities of the bases surrounding the Cas9 cleavage site and the frequency and identity of the inserted base (see main text). Motivated by these empirical observations, inDelphi is fed with training data on 1-bp insertion frequencies and identities at each training site parameterized with the identities of the −3, −4, and −5 bases upstream of the NGG PAM-sequence (when the training set is sufficiently large, and the −4 base alone when training data is limited) as features. Also added as features are the precision score of the deletion length distribution and the total deletion phi-score at that site. These features are combined into a k-nearest neighbor algorithm that predicts the relative frequencies and identities of 1-bp insertion products at any target site.
(341) The Combination of the MH, MH-Less, and Insertion Model Predict Genotype Fractions
(342) Altogether, informed by known paradigms of DNA repair, 2 neural networks and a k-nearest neighbor model were built to predict genotypic outcomes following Cas9 cutting. These models compete and collaborate in inDelphi to generate predictions of the relative frequencies of these products. This competition within inDelphi among repair types reflects empirical evidence from Lib-A and Lib-B that sequence contexts do influence classes of repair outcomes. Sequence contexts with high phi scores (high microhomology) have higher efficiencies of MH deletions among all editing outcomes (FIG. **14**B, FIGS. **20**A-**20**E), and sequence contexts with low phi scores (low microhomology) have higher efficiencies of 1-bp insertions among all editing outcomes (FIGS. **14**C, **14**D, FIGS. **18**A-**18**H,FIGS. **20**A-**20**E). While it is tempting to generalize that the competition and collaboration among outcome classes modeled by inDelphi reflects interactions among components of distinct DNA repair pathways, the classes of outcomes considered by inDelphi do not necessarily arise from distinct DNA repair pathways as they are described above. inDelphi is trained on the repair outcomes only and cannot distinguish between the nature of genotypes when they may occur through MH-mediated and MH-less mechanisms, and it is imaginable that some repair products result through more than one repair pathway.
(343) As an additional note, while NHEJ is generally assumed to dominate double-strand break

repair from environmentally induced damage.sup.35, it was found in the context of Cas9 cutting that MH deletion genotypes are more common than MH-less deletions and insertions. It is possible that error-free c-NHEJ is occurring frequently in response to Cas9 cutting but that its perfect repair allows for recurring Cas9 cutting that goes undetected by the workflow, thus skewing the observed relative frequency profile of mutagenic outcomes toward MMEJ-type repair.

(344) Prkdc.sup.−/−Lig4.sup.−/− Mutants have Distinct and Predictable DNA Repair Product Distributions

(345) While it is generally true that the work cannot establish roles for specific DNA repair pathways in specific types of Cas9-mediated outcomes, an experiment has been performed in which Cas9-mediated genotypic outcomes were measured from mESCs that are lacking Prkdc and Lig4, two proteins known to be key in c-NHEJ5. An increase in relative frequency of MH deletions was found as compared to MH-less deletions in Prkdc.sup.−/−Lig4.sup.−/− mESCs as compared to wild-type mESCs (see main text), which is suggestive of an increase in MMEJ outcomes at the expense of NHEJ outcomes.

(346) Intriguingly, it was also found that Prkdc.sup.−/−Lig4.sup.−/− mESCs are impaired in unilateral deletions, where only bases from one side of the cutsite are removed, but not medial MH-less deletion outcomes that have loss of bases on both sides of the breakpoint. (FIGS. **22**A-**22**E). As discussed earlier, microhomology-mediated alt-NHEJ, which it was hypothesized may give rise to unilateral MH-less deletions, proceeds through a mechanism in which DNA repair intermediates that mimic MMEJ-mediated repair are formed initially (FIGS. **18**A-**18**H), as microhomology basepairing temporarily stabilizes 3′-overhangs following 5′-end resection. Subsequently, ligation joins one 3′ overhang with the sequence on the other side of the DNA double-strand break, giving rise to a unilateral deletion. If the unilateral joining products observed in the experiments indeed arise through similar mechanisms as those described by this form of alt-NHEJ, it is conceivable that the MMEJ pathway may overtake 3′-end ligation at this microhomology-containing intermediate step when ligation is impaired through loss of Lig4. Thus, cross-talk of microhomology-mediated repair pathways could account for loss of unilateral end-joining MH-less outcomes and concomitant increase in MH deletion outcomes. Medial joining outcomes are not hypothesized to originate from intermediates that overlap with microhomology-mediated deletion products (FIGS. **18**A-**18**H). Therefore, the repair genotypes generated via this orthogonal pathway may be afforded more time to be completed by ligases other than Lig4, thus explaining why these outcomes appear unaffected by NHEJ impairment.

(347) While DNA repair products in Prkdc.sup.−/−Lig4.sup.−/− mESCs differ substantially from those in wild-type cells, it was found that these DNA repair products are also highly predictable. In particular, inDelphi performed well on held-out Prkdc.sup.−/−Lig4.sup.−/− data when trained on Prkdc.sup.−/−Lig4.sup.−/− data (indel genotype prediction median Pearson correlation=0.84, indel length frequency prediction Pearson correlation=0.80), showing that the modeling approach is robustly capable of learning accurate predictions for Cas9 editing data in not just wild-type experimental settings but also settings with significant biochemical perturbation. As such, it is suggested here that inDelphi's modeling approach can be useful on additional tasks unexplored here provided that inDelphi is supplied with appropriate training data.

Methods

(348) Library Cloning.

(349) Specified pools of 2000 oligos were synthesized by Twist Bioscience and amplified with NEBNext polymerase (New England Biolabs) using primers OligoLib_Fw and OligoLib_Rv (see below), to extend the sequences with overhangs complementary to the donor template used for circular assembly. To avoid over-amplification in the library cloning process, qPCR was first performed by addition of SybrGreen Dye (Thermo Fisher) to determine the number of cycles required to complete the exponential phase of amplification. The PCR reaction was run for half of the determined number of cycles at this stage. Extension time for all PCR reactions was extended

to 1 minute per cycle to prevent skewing towards GC-rich sequences. The 246-bp fragment was purified using a PCR purification kit (Qiagen).
(350) Separately, the donor template for circular assembly was amplified with NEBNext polymerase (New England Biolabs) for 20 cycles from an SpCas9 sgRNA expression plasmid (Addgene 71485).sup.34 using primers CircDonor_Fw and CircDonor_Rv (see below) to amplify the sgRNA hairpin and terminator, and extended further with a linker region meant to separate the gRNA expression cassette from the target sequence in the final library. The 146-bp amplicon was gel-purified (Qiagen) from a 2.5% agarose gel.
(351) The amplified synthetic library and donor templates were ligated by Gibson Assembly (New England Biolabs) in a 1:3 molar ratio for 1 hour at 50° C., and unligated fragments were digested with Plasmid Safe ATP-Dependent DNase (Lucigen) for 1 hour at 37° C. Assembled circularized sequences were purified using a PCR purification kit (Qiagen), linearized by digestion with SspI for ≥3 hours at 37° C., and the 237-bp product was gel purified (Qiagen) from a 2.5% agarose gel.
(352) The linearized fragment was further amplified with NEBNext polymerase (New England Biolabs) using primers PlasmidIns_Fw and Plasmidlns_Rv (see below) for the addition of overhangs complementary to the 5′- and 3′-regions of a Tol2-transposon containing gRNA expression plasmid (Addgene 71485).sup.34 previously digested with BbsI and XbaI (New England Biolabs), to facilitate gRNA expression and integration of the library into the genome of mammalian cells. To avoid over-amplification, qPCR was performed by addition of SybrGreen Dye (Thermo Fisher) to determine the number of cycles required to complete the exponential phase of amplification, and then ran the PCR reaction for the determined number of cycles. The 375-bp amplicon was gel-purified (Qiagen) from a 2.5% agarose gel.
(353) The 375-bp amplicon and double-digested Tol2-transposon containing gRNA expression plasmid were ligated by Gibson Assembly (New England Biolabs) in a 3:1 ratio for 1 hour at 50° C. Assembled plasmids were purified by isopropanol precipitation with GlycoBlue Coprecipitant (Thermo Fisher) and reconstituted in milliQ water and transformed into NEB10beta (New England Biolabs) electrocompetent cells. Following recovery, a small dilution series was plated to assess transformation efficiency and the remainder was grown in liquid culture in DRM medium overnight at 37° C. A detailed step-by-step library cloning protocol is provided below.
(354) The plasmid library was isolated by Midiprep plasmid purification (Qiagen). Library integrity was verified by restriction digest with SapI (New England Biolabs) for 1 hour at 37° C., and sequence diversity was validated by high-throughput sequencing (HTS) as described below.
(355) Library Cloning Primers
(356) TABLE-US-00013 OligoLib_Fw (SEQ ID NO: 8)
TTTTTGTTTTCTGTGTTCCGTTGTCCGTGCTGTAACGAAAGGATGGGTGC
GACGCGTCAT OligoLib_Rv (SEQ ID NO: 9)
GTTGATAACGGACTAGCCTTATTTAAACTTGCTATGCTGTTTCCAGCATA GCTCTTAAAC
CircDonor_Fw (SEQ ID NO: 10) GTTTAAGAGCTATGCTGGAAACAGC CircDonor_Rv
(SEQ ID NO: 11)
ATGACGCGTCGCACCCATCCTTTCGTTACAGCACGGACAACGGAACACAG
AAAACAAAAAGCACCGACTC PlasmidIns_Fw (SEQ ID NO: 12)
GTAACTTGAAAGTATTTCGATTTCTTGGCTTTATATATCTTGTGGAAAGG ACGAAACACC
PlasmidIns_Rv (SEQ ID NO: 13)
TTGTGGTTTGTCCAAACTCATCAATGTATCTTATCATGTCTGCTCGAAGC
GGCCGTACCTCTAGATTCAGACGTGTGCTCTTCCGATCT
Cloning.
(357) A base plasmid was constructed starting from a Tol2-transposon containing plasmid (Addgene 71485).sup.34. The sequence between Tol2 sites was replaced with a CAGGS promoter, multi-cloning site, P2A peptide sequence followed by eGFP sequence, and Puromycin resistance cassette to produce p2T-CAG-MCS-P2A-GFP-PuroR. The full sequence of this plasmid is

appended in the Sequences section below, and this plasmid has been submitted to Addgene.
(358) Plasmids with this backbone and containing wildtype and micro-duplication mutation versions of LDLR and three other genes, GAA, GLB1, and PORCN, were constructed. Information on cloning these genes is provided below, and the gene sequences are appended below.
(359) LDLR: To generate p2T-CAGGS-LDLRwt-P2A-GFP-PuroR, LDLR (NCBI Gene ID #3949, transcript variant 1 CDS) was PCR amplified from a base plasmid ordered from the Harvard PlasmID resource core and cloned between the BamHI and NheI sites of the base plasmid.
(360) The following mutants were generated through InFusion (Clontech) cloning. Sequences are provided below, and the internal allele nomenclature is in parentheses:
(361) TABLE-US-00014 LDLR:c.526_533dupGGCTCGGA (LDLRdup252) (SEQ ID NO: 14) LDLR:c.668_681dupAGGACAAATCTGAC (LDLRdup254/255) (SEQ ID NO: 15) LDLR:c.669_680dupGGACAAATCTGA (LDLRdup258) (SEQ ID NO: 16) LDLR:c.672_683dupCAAATCTGACGA (LDLRdup261) LDLR:c.1662_1669dupGCTGGTGA (LDLRdup264)
(362) PORCN: NCBI Gene ID #64840, transcript variant C CDS was PCR amplified from HCT116 cDNA and cloned between the BamHI and NheI sites of the base plasmid. PORCN:c.1059_1071dupCCTGGCTTTTATC (SEQ ID NO: 17) (PORCNdup20) was generated through InFusion cloning.
(363) GLB1: NCBI Gene ID #2720, transcript variant 1 CDS was PCR amplified from HCT116 cDNA and cloned between the BamHI and NheI sites of the base plasmid. GLB1:c.1456_1466dupGGTGCATATAT (SEQ ID NO: 18) (GLBldup84) was generated through InFusion cloning.
(364) GAA: NCBI Gene ID #2548, transcript variant 1 CDS was PCR amplified from a base plasmid ordered from the Harvard PlasmID resource core and cloned between the BamHI and NheI sites of the base plasmid. GAA:c.2704_2716dupCAGAAGGTGACTG (SEQ ID NO: 19) (GAAdup327/328) was generated through InFusion cloning.
(365) SpCas9.sup.1 and KKH SaCas9.sup.9 were constructed starting from a Tol2-transposon containing plasmid (Addgene 71485).sup.34. The sequence between Tol2 sites was replaced with a CAGGS promoter, Cas9 sequence, and blasticidin resistance cassette to produce p2T-CAG-SpCas9-BlastR and p2T-CAG-KKHSaCas9-BlastR. These plasmids have been submitted to Addgene.
(366) SpCas9 guide RNAs were cloned as a pool into a Tol2-transposon containing gRNA expression plasmid (Addgene 71485).sup.34 using BbsI plasmid digest and Gibson Assembly (NEB). SaCas9 guide RNAs were cloned into a similar Tol2-transposon containing SaCas9 gRNA expression plasmid (p2T-U6-sgsaCas2xBbsI-HygR) which has been submitted to Addgene using BbsI plasmid digest and Gibson Assembly. Protospacer sequences used are listed below, using the internal nomenclature which matches the duplication alleles.
(367) TABLE-US-00015 LDLR gRNAs sgsaLDLRdup252: (SEQ ID NO: 20) GCTGCGAAGATGGCTCGGAGGC sgsaLDLRdup254: (SEQ ID NO: 21) GTGCAAGGACAAATCTGACAGG sgsaLDLRdup255: (SEQ ID NO: 22) GTTCCTCGTCAGATTTGTCCTG sgsaLDLRdup258: (SEQ ID NO: 23) GACTGCAAGGACAAATCTGAGG sgsaLDLRdup261: (SEQ ID NO: 24) GTTTTCCTCGTCAGATTTGTCG sgspLDLRdup264: (SEQ ID NO: 25) GACATCTACTCGCTGGTGAGC PORCN gRNAs sgspPORCNdup20: (SEQ ID NO: 26) GCTGTCCCTGGCTTTTATCCC GLB1 gRNAs sgspGLB1dup84: (SEQ ID NO: 27) GTGTGAACTATGGTGCATATA GAA gRNAs sgsaGAAdup327: (SEQ ID NO: 28) GCAGCTGCAGAAGGTGACTGCA sgspGAAdup328: (SEQ ID NO: 29) GCTGCAGAAGGTGACTGCAGA

Cell Culture.

(368) Mouse embryonic stem cell lines used have been described previously and were cultured as

described previously.sup.44. HEK293T, HCT116, and U2OS cells were purchased from ATCC and cultured as recommended by ATCC. For stable Tol2 transposon plasmid integration, cells were transfected using Lipofectamine 3000 (Thermo Fisher) using standard protocols with equimolar amounts of Tol2 transposase plasmid.sup.25 (a gift from Koichi Kawakami) and transposon-containing plasmid. For library applications, 15-cm plates with >10.sup.7 initial cells were used, and for single gRNA targeting, 6-well plates with >10.sup.6 initial cells were used. To generate lines with stable Tol2-mediated genomic integration, selection with the appropriate selection agent at an empirically defined concentration (blasticidin, hygromycin, or puromycin) was performed starting 24 hours after transection and continuing for >1 week. In cases where sequential plasmid integration was performed such as integrating gRNA/target library and then Cas9 or micro-duplication plasmid and then Cas9 plus gRNA, the same Lipofectamine 3000 transfection protocol with Tol2 transposase plasmid was performed each time, and >1 week of appropriate drug selection was performed after each transfection.

(369) Deep Sequencing.

(370) Genomic DNA was collected from cells after >1 week of selection. For library samples, 16 μg gDNA was used for each sample; for individual locus samples, 2 μg gDNA was used; for plasmid library verification, 0.5 μg purified plasmid DNA was used.

(371) For individual locus samples, the locus surrounding CRISPR/Cas9 mutation was PCR amplified in two steps using primers >50-bp from the Cas9 target site. PCR1 was performed using the primers specified below. PCR2 was performed to add full-length Illumina sequencing adapters using the NEBNext Index Primer Sets 1 and 2 (NEB) or internally ordered primers with equivalent sequences. All PCRs were performed using NEBNext polymerase (New England Bioscience). Extension time for all PCR reactions was extended to 1 min per cycle to prevent skewing towards GC-rich sequences. The pooled samples were sequenced using NextSeq (Illumina) at the Harvard Medical School Biopolymers Facility, the MIT BioMicro Center, or the Broad Institute Sequencing Facility.

(372) Library Prep Primers:

(373) TABLE-US-00016 For LDLRDup252, 254, 255, 258, 261: 120417_LDLRDup254_r1_seq_A (SEQ ID NO: 30) CTTTCCCTACACGACGCTCTTCCGATCT NNN ACTCCAGCTGGCGCTGTGAT 120417_LDLR254_r2seq_A (SEQ ID NO: 31) GGAGTTCAGACGTGTGCTCTTCCGATCT CAACTTCATCGCTCATGTCCTTG For LDLRDup264: 120817_LDLR264_r1seq_B (SEQ ID NO: 32) CTTTCCCTACACGACGCTCTTCCGATCT NNNAACTCCCGCCAAGATCAAGAAAG 120817_LDLR264_r2seq_B (SEQ ID NO: 33) GGAGTTCAGACGTGTGCTCTTCCGATCT CAGCCTCTTTTCATCCTCCAAGA For PORCDup20: 120517_PORCN20_r1_seq (SEQ ID NO: 34) CTTTCCCTACACGACGCTCTTCCGATCT NNN CCTCCTACATGGCTTCAGTTTCC 120517_PORCN20_r2seq (SEQ ID NO: 35) GGAGTTCAGACGTGTGCTCTTCCGATCT CCAGAGCTCCAAAGAGCAAGTTT For GLB1Dup84: 120517_GLB_184_r1seq (SEQ ID NO: 36) CTTTCCCTACACGACGCTCTTCCGATCT NNN AGCCACTCTGGACCTTCTGGTA 120517_GLB_184_r2seq (SEQ ID NO: 37) GGAGTTCAGACGTGTGCTCTTCCGATCT CCAGTCCGTGAGGATATTGGAAC For GAADup327/328: 120517_GAA327_r1seq (SEQ ID NO: 38) CTTTCCCTACACGACGCTCTTCCGATCT NNN GATCGTGAATGAGCTGGTACGTG 120517_GAA327_r2seq (SEQ ID NO: 39) GGAGTTCAGACGTGTGCTCTTCCGATCT AACAGCGAGACACAGATGTCCAG

Data Availability.

(374) High-throughput sequencing data have been deposited in the NCBI Sequence Read Archive database under accession codes SRP141261 and SRP141144.

(375) Code Availability.
(376) All data processing, analysis, and modeling code is available atgithub.com/maxwshen/inDelphi-dataprocessinganalysis. The inDelphi model is available online at the URLcrisprindelphi.design.
(377) Library Cloning Protocol
(378) Synthesized Oligo Library Sequence
(379) TABLE-US-00017 (SEQ ID NO: 40) embedded image embedded image PROTOSPACER depending on whether it naturally starts with a G] GTTTAAGAGCTATGCTGGAAACAGC Linker Region/Oligo Library Amplification Primer Anneal Region Read 2 Sequencing Primer Stub
SspI Restriction Site U6-Promoter Stub sgRNA-Hairpin Stub
1. Oligo library QPCR to determine number of amplification cycles for Oligo Library PCR Notes: Amplification of oligos with relatively low GC-content is less efficient than GC-rich sequences. It was found that NEBNext polymerase was the least biased in amplification of the library. Increasing the elongation time to 1 min per cycle for all cloning and sequencing library prep PCRs eliminates GC-skewing of library sequences and reduces the rate of PCR-recombination. Set up the following reaction:
(380) TABLE-US-00018 0.4 ng Synthesized Oligo Library 10 ul NEBNext 2x Master Mix 0.5 ul 20 uM OligoLib_Fw 0.5 ul 20 uM OligoLib_Rv 0.2 ul SybrGreen Dye (100x) to 20 ul H.sub.2O
67° C. annealing temperature Check 246 bp amplicon size on 2.5% agarose gel. Determine the point that signal amplification has plateaued.
2. Oligo Library PCR Amplification Set up the following reaction:
(381) TABLE-US-00019 4 ng Synthesized Oligo Library 50 ul NEBNext 2x Master Mix 2.5 ul 20 uM OligoLib_Fw 2.5 ul 20 uM OligoLib_Rv to 100 ul H.sub.2O
67° C. annealing temperature, 1 minute extension time.
Cycle number is half the number of cycles needed to reach signal amplification plateau in the QPCR in step 1, reduced by 1 cycle to scale for DNA input. PCR purify amplified sequence. 3. Donor template amplification Set up the following reaction:
(382) TABLE-US-00020 5 ng spCas9 sgRNA plasmid (71485) 50 ul NEBNext 2x Master Mix 2.5 ul 20 uM CircDonor_Fw 2.5 ul 20 uM CircDonor_Rv to 100 ul H.sub.2O
62° C. annealing temperature
20 cycles
Gel purify 167 bp band from 2.5% agarose gel.
4. Circular assembly and restriction digest linearization
Note: A molar ratio of donor template to amplified oligo library of 3:1 was used. An increase in amplified oligo library compounds cross-over within library members resulting in mismatch of protospacer and target sequences. Set up the following reaction:
(383) TABLE-US-00021 429 ng Donor template 239 ul Amplified Oligo Library 30 ul Gibson Assembly 2x Master Mix to 60 ul H.sub.2O
50° C. incubation for 1 hour. Exonuclease treatment
(384) TABLE-US-00022 60 ul Circular assembly reaction 9 ul ATP (25 mM) 9 ul 10x Plasmid Safe Buffer 3 ul Plasmid Safe Nuclease 9 ul H.sub.2O
37° C. incubation for 1 hour. PCR purify and elute in 50 ul. Digest to linearize library
(385) TABLE-US-00023 50 ul Purified assemblies 10 ul 10x CutSmart Buffer 3 ul Sspl-HF 37 ul H.sub.2O
37° C. incubation for ≥3 hours. Gel purify 273 bp band from 2.5% agarose gel.
Note: Band is sometimes fuzzy and poorly visible. If not clearly discernible, proceed with gel isolation between 200-300 bp.
5. Linearized Library QPCR to Determine Number of Amplification Cycles for PCR Amplification Set up the following reaction:

(386) TABLE-US-00024 0.5% Purified linearized library 10 ul NEBNext 2x Master Mix 0.5 ul 20 uM PlasmidIns_Fw 0.5 ul 20 uM PlasmidIns_Rv 0.2 ul SybrGreen Dye (100x) to 20 ul H.sub.2O
65° C. annealing temperature Determine the point that signal amplification has plateaued.
6. Linearized Library PCR amplification Set up the following reaction:
(387) TABLE-US-00025 50% Purified linearized library 50 ul NEBNext 2x Master Mix 2.5 ul 20 uM PlasmidIns_Fw 2.5 ul 20 uM PlasmidIns_Rv to 100 ul H.sub.2O
65° C. annealing temperature, 1 minute extension time.
Cycle number is number of cycles needed to reach signal amplification plateau in the QPCR in step 5, reduced by 4 cycles to scale for increased DNA input. Gel purify 375 bp band from 2.5% agarose gel.
7. Vector backbone digest Set up the following reaction:
(388) TABLE-US-00026 2 ug spCas9 sgRNA plasmid (71485) 10 10x Buffer 2.1 3 BbsI 2 XbaI to 100 ul H.sub.2O
37° C. incubation for ≥3 hours. Gel purify 5.9 kb band from 1% agarose gel.
8. Vector assembly and cleanup
Note: Include a ligation with water for insert as a control. Set up the following reaction:
(389) TABLE-US-00027 300 ng Digested vector backbone 42 ng Amplified Oligo Library 30 ul Gibson Assembly 2x Master Mix to 60 ul H.sub.2O
50° C. incubation for 1 hour. Isopropanol precipitation
(390) TABLE-US-00028 40 ul Vector assembly reaction 0.4 ul GlycoBlue Coprecipitant 0.8 ul 50 mM NaCl 38.8 ul Isopropanol Vortex and incubate at room temperature for 15 minutes. Spin down at ≥15.000 g for 15 minutes, and carefully remove supernatant. Wash pellet with 300 ul 80% EtOH and repeat spin at ≥15.000 g for 5 minutes. Carefully remove all liquid without disturbing pellet, and let air dry for 1-3 minutes.—Dissolve dried pellet in 10 ul H2O at 55° C. for 10 minutes.
9. Transformation
Note: Electroporation competent cells give a higher transformation efficiency than chemically competent cells. NEB10beta electro-competent cells were used, however these can be substituted for other lines and transformed according to the manufacturer's instructions.
Note: DRM was used as recovery and culture medium to enhance yield. If substituting for a less rich medium such as LB, it is recommended scaling up the culture volume to obtain similar plasmid DNA quantities.
Note: Antibiotic-free recovery time should be limited to 15 minutes to prevent shedding of transformed plasmids from replicating bacteria.
Note: Also transform water ligation as control. Pre-warm 3.5 mL recovery medium per electroporation reaction, at 37° C. for 1 hour. Pre-warm LB-agar plates containing appropriate antibiotic. Per reaction, add 1 ul purified vector assembly to 25 ul competent cells on ice. Perform 8 replicate reactions. Electroporate according to the manufacturer's instructions. Immediately add 100 ul pre-warmed recovery media per cuvette and pool all replicates into culture flask. Add 1 mL recovery media per replicate reaction to culture flask and shake at 200 rpm 37° C. for 10-15 minutes. Plate a dilution series from 1:104-1:106 on LB-agar plates containing antibiotic and grow overnight at 37° C. Add 2 mL media per replicate reaction and admix appropriate antibiotic. Grow overnight in shaking incubator at 200 rpm 37° C. Assess transformation efficiency from serial dilution LB-agar plates. Expect ˜10.sup.6 clones.
The development of this cloning protocol was guided by work described in Videgal et al. 2015.
Sequence Alignment and Data Processing
(391) For library data, each sequenced pair of gRNA fragment and target was associated with a set of designed sequence contexts G by finding the designed sequence contexts for all gRNAs whose beginning section perfectly matches the gRNA fragment (read 1 in general does not fully sequence the gRNA), and by using locality sensitive hashing (LSH) with 7-mers on the sequenced target to search for similar designed targets. An LSH score on 7-mers between a reference and a sequenced

context reflects the number of shared 7-mers between the two. If the best reference candidate scored, through LSH, greater than 5 higher than the best LSH score of the reference candidates obtained from the gRNA-fragment, the LSH candidate is also added to G. LSH was used due to extensive (−33% rate) PCR recombination between read1 and read2 which in sequenced data appears as mismatched read1 and read2 pairs. The sequenced target was aligned to each candidate in G and the alignment with the highest number of matches is kept. Sequence alignment was performed using the Needleman-Wunsch algorithm using the parameters: +1 match, −1 mismatch, −5 gap open, −0 gap extend. For library data, starting gaps cost 0. For all other data, starting and ending gaps cost 0. For VO data, sequence alignments were derived from SAM files from SRA.
(392) Alignments with low-accuracy or short matching sections flanked by long (10 bp+) insertions and deletions were filtered out as PCR recombination products (observed frequency of −5%). These PCR recombination products are different than that occurring between read1 and read2; these occur strictly in read2. Alignments with low matching rates were removed. Deletions and insertions were shifted towards the expected cleavage site while preserving total alignment score. CRISPR-associated DNA repair events were defined as any alignment with deletions or insertions occurring within a 4 bp window centered at the expected cut site and any alignment with both deletions and insertions (combination indel) occurring with a 10 bp window centered at the expected cut site. All CRISPR-associated DNA repair events observed in control data had their frequencies subtracted from treatment data to a minimum of 0.
(393) Replicate experiments were carried out for library data in each cell type. For each cell-type, each sequence context not fulfilling the following data quality criteria was filtered: data at this sequence context in the two replicates with the highest read-counts must have at least 1000 reads of CRISPR editing outcomes in both replicates, and a Pearson correlation of at least 0.85 in the frequency of microhomology-based deletion events. The class of microhomology-based deletion events was used for this criterion since it is a major repair class with the highest average replicability across experiments. For disease library data in U2OS and HEK, a less stringent read count threshold of 500 was used instead.
(394) Details on Alignment Processing
(395) All alignments with gaps were shifted as much as possible towards the cleavage site while preserving the overall alignment score. Then, the following criteria were used to categorize the alignments into noise, not-noise but not CRISPR-associated (for example, wildtype); as well as primary and secondary CRISPR activity. All data used in modeling and analysis derive solely from outcomes binned into primary CRISPR activity.
(396) The following criteria was used to filter library alignments into “noise” categories. Homopolymer: Entire read is homopolymer of a single nucleotide. Not considered a CRISPR repair product.
(397) Has N: Read contains at least one N. Discarded as noise, not considered a CRISPR repair product.
(398) PCR Recombination: Contains recombination alignment signature: (1) if a long indel (10 bp+) followed by chance overlap followed by long indel (10 bp+) of the opposite type, e.g., insertion-randommatch-deletion and deletion-randommatch-insertion. OR, if one of these two indels is 30 bp+, the other can be arbitrarily short. If either criteria is true, and if the chance overlap is length 5 or less, or any length with less than 80% match rate, then it satisfies the recombination signature. In addition, if both indels are 30 bp+, regardless of the middle match region, it satisfies the recombination signature. Finally, if randommatch is length 0, then indel is allowed to be any length. Not considered a CRISPR repair product.
(399) Poor-Matches: 55 bp designed sequence context has less than 5 bp representation (could occur from 50 bp+ deletions or severe recombination) or less than 80% match rate. Not considered a CRISPR repair product.
(400) Cutsite-Not-Sequenced: The read does not contain the expected cleavage site.

(401) Other: An alignment with multiple indels where at least one non-gap region has lower than an 80% match rate. Or generally, any alignment not matching any defined category above or below. In practice, can include near-homopolymers. Not considered a CRISPR repair product.
(402) The following criteria was used to filter library alignments into "main" categories.
(403) Wildtype: No indels in all of alignment. Not considered a CRISPR repair product.
(404) Deletion: An alignment with only a single deletion event. Subdivided into: Deletion—Not CRISPR: Single deletion occurs outside of 2 bp window around cleavage site. Not considered a CRISPR repair product.
(405) Deletion—Not at cut: Single deletion occurring within 2 bp window around cleavage site, but not immediately at cleavage site. Considered a CRISPR repair product.
(406) Deletion: Single deletion occurring immediately at cleavage site. Considered a CRISPR repair product.
(407) Insertion: An alignment with only a single insertion event. Subdivided into:
(408) Insertion—Not CRISPR: Single insertion occurs outside of 10 bp window around cleavage site. Not considered a CRISPR repair product.
(409) Insertion—Not at cut: Single insertion occurring within 2 bp window around cleavage site, but not immediately at cleavage site. Considered a CRISPR repair product.
(410) Insertion: Single insertion occurring immediately at cleavage site. Considered a CRISPR repair product.
(411) Combination indel: An alignment with multiple indels where all non-gap regions have at least 80% match rate. Subdivided into:
(412) Combination Indel: All indels are within a 10 bp window around the cleavage site. Considered a primary CRISPR repair product.
(413) Forgiven Combination Indel: At least two indels, but not all, are within a 10 bp window around the cleavage site. Considered a rarer secondary CRISPR repair product, ignored.
(414) Forgiven Single Indel: Exactly one indel is within a 10 bp window around the cleavage site. Considered a rarer secondary CRISPR repair product, ignored.
(415) Combination Indel—Not CRISPR: No indels are within a 10 bp window around the cleavage site. Not considered a CRISPR repair product.
(416) It is noted that deletion and insertion events, even those spanning many bases, are defined to occur at a single location between bases. As such, events occurring up to 5 bp away from the cleavage site are defined as events where there are five or fewer matched/mismatched alignment positions between the event and the cleavage site, irrespective of the number of gap dashes in the alignment.
(417) Selection of Variants from Disease Databases
(418) Disease variants were selected from the NCBI ClinVar database (downloaded Sep. 9, 2017).sup.16 and the Human Gene Mutation Database (publicly available variant data from before 2014.3).sup.17 for computational screening and subsequent experimental correction.
(419) A total of 4,935 unique variants were selected from Clinvar submissions where the functional consequence is described as complete insertions, deletions, or duplications where the reference or alternate allele is of length less than or equal to 30 nucleotides. Variants were included where at least one submitting lab designated the clinical significance as 'pathogenic' or 'likely pathogenic' and no submitting lab had designated the variant as 'benign' or 'likely benign', including variants will all disease associations. More complex indels and somatic variants were included. A total of 18,083 unique insertion variants were selected from HGMD which were between 2 to 30 nucleotides in length. Variants were included with any disease association with the HGMD classification of 'DM' or disease-causing mutation.
(420) SpCas9 gRNAs and their cleavage sites were enumerated for each disease allele. Using a previous version of inDelphi, genotype frequency and indel length distributions were predicted for each tuple of disease variant and unique cleavage site. Among each unique disease, the single best

gRNA was identified as the gRNA inducing the highest predicted frequency of repair to wildtype genotype, and if this was impossible (due to, for example, a disease allele with 2+ bp deletion), then the single best gRNA was identified as the gRNA inducing the highest predicted frameshift repair rate. 1327 sequence contexts were designed in this manner for Lib-B. An additional 265 sequence contexts were designed by taking the 265 sequence contexts in any disease in decreasing order of predicted wildtype repair rate, starting with Clinvar, stopping at 45% wildtype repair rate, then continuing with HGMD. This yielded 1592 total sequences derived from Clinvar and HGMD.
(421) Definition of Delta-Positions
(422) Using the MMEJ mechanism, deletion events can be predicted at single-base resolution. For computational convenience, the tuple (deletion length, delta-position) was used to construct a unique identifier for deletion genotypes. A delta-position associated with a deletion length N is an integer between 0 and N inclusive (FIGS. **19**A-**19**D). In a sequence alignment, a delta-position describes the starting position of the deletion gap in the read w.r.t. the reference sequence relative to the cleavage site. For a deletion length N and a cleavage site at position C such that seq[:C] and seq[C:] yield the expected DSB products where the vector slicing operation vector[index1:index2] is inclusive on the first index and exclusive on the second index (python style), a delta-position of 0 corresponds to a deletion gap at seq[C−N+0: C+0], and generally with a delta-position of D, the deletion gap occurs at seq[C−N+D: C+D]. Microhomologies can be described with multiple delta-positions. To uniquely identify microhomology-based deletion genotypes, the single maximum delta-position in the redundant set is used. Microhomology-less deletion genotypes are associated with only a single delta position and deletion length tuple; this was used as its unique identifier.
(423) Another way to define delta-positions can be motivated by the example workflow shown above on MH deletions describing how each microhomology is associated with a deletion genotype. In that workflow, the delta-position is the number of bases included on the top strand before “jumping down” to the bottom strand.
(424) MH-less medial end-joining products correspond to all MH-less genotypes with delta-position between 1 and N−1 where N is the deletion length. MH-less unilateral end-joining products correspond to MH-less genotypes with delta-position 0 or N. It is noted that a deletion genotype with delta position N does not immediately imply that it is a microhomology-less unilateral end-joining product since it may contain microhomology (it's possible that delta-positions N−j, N−j+1, . . . , N all correspond to the same MH deletion.)
(425) Definition of Precision Score
(426) For a distribution X, where |XI indicates its cardinality (or length when represented as a vector):
(427) 0PrecisionScore$(X) = 1 - \frac{-\sum_{i=1}^{n} P(x_i)\log(P(x_i))}{\log(|X|)}$
This precision score ranges between zero (minimally precise, or highest entropy) to one (maximally precise, or lowest entropy).
inDelphi Deletion Modeling: Neural Network Input and Output
(428) inDelphi receives as input a sequence context and a cleavage site location, and outputs two objects: a frequency distribution on deletion genotypes, and a frequency distribution on deletion lengths.
(429) To model deletions, inDelphi trains two neural networks: MH-NN and MHless-NN.
(430) MH-NN receives as input a microhomology that is described by two features: microhomology length and GC fraction in the microhomology. Using these features, MH-NN outputs a number (psi). MHless-NN receives as input the deletion length. Using this feature, MHless-NN outputs a number (psi).
(431) A phi score is obtained from a psi score using: phi_i=exp(psi_i−0.25*deletion_length), where 0.25 is a “redundant” hyperparameter that serves to reduce training speed by helpful scaling. This relationship between psi and phi is differentiable and encodes the assumption that the frequency of

an event exponentially increases with neural network output psi (which empirically appears to reflect MH strength) and exponentially decreases with its minimum necessary resection length (deletion length).

(432) The architecture of the MH-NN and MHless-NN networks are input-dimension .fwdarw.16 .fwdarw.16 .fwdarw.1 for a total of two hidden layers where all nodes are fully connected. Sigmoidal activations are used in all layers except the output layer. All neural network parameters are initialized with Gaussian noise centered around 0.

(433) inDelphi Deletion Modeling: Making Predictions

(434) Given a sequence context and cleavage site, inDelphi enumerates all unique deletion genotypes as a tuple of its deletion length and its delta-position for deletion lengths from 1 bp to 60 bp. For each microhomology enumerated, an MH-phi score is obtained using MH-NN. In addition, for each deletion length from 1 bp to 60 bp, an MHindep-phi score is obtained using MHless-NN.

(435) inDelphi combines all MH-phi and MHindep-phi scores for a particular sequence context into two objects—a frequency distribution on deletion genotypes, and a frequency distribution on deletion lengths—which are both compared to observations for training. The model is designed to output two separate objects because both are of biological interest, and separate but intertwined modeling approaches are useful for generating both. By learning to generate both objects, inDelphi jointly learns about microhomology-based deletion repair and microhomology-less deletion repair.

(436) To generate a frequency distribution on deletion genotypes, inDelphi assigns a score for each microhomology. Score assignment considers the concept of “full” microhomology and treats full and not full MHs differently.

(437) A microhomology is “full” if the length of the microhomology is equal to its deletion length. The biological significance of full microhomologies is that there is only a single deletion genotype possible for the entire deletion length, while in general, a single deletion length is consistent with multiple genotypes. In addition, this single genotype can be generated through not just the MH-dependent MMEJ mechanism but also through MH-less end-joining, for example as mediated by Lig4. Therefore, full microhomologies were modelled as receiving contributions from both MH-containing and MH-less mechanisms by scoring full microhomologies as MH-phi[i]+MHindep-phi[j] for deletion length j and microhomology index i. Microhomologies that are not “full” are assigned a score of MH-phi[i] for MH index i.

(438) Scores for all deletion genotypes assigned this way are normalized to sum to 1 to produce a predicted frequency distribution on deletion genotypes.

(439) To generate a frequency distribution on deletion lengths, inDelphi assigns a score for each deletion length. Score assignment integrates contributions from both MH-dependent and MH-independent mechanisms via the following procedure: For each deletion length j, its score is assigned as MHindep-phi[j] plus the sum of MH-phi for each microhomology with that deletion length. Scores for all deletion lengths are normalized to sum to 1 to produce a frequency distribution.

(440) inDelphi trains its parameters using a single sequence context by producing both a predicted frequency distribution on deletion genotypes and deletion lengths and minimizing the negative of the sum of squared Pearson correlations for both objects to their observed versions. In practice, deletion genotype frequency distributions are formed from observations for deletion lengths 1-60, and deletion length frequency distributions are formed from observations for deletion lengths 1-28. Both neural networks are trained simultaneously on both tasks. inDelphi is trained with stochastic gradient descent with batched training sets. inDelphi is implemented in Python using the autograd library. A batch size of 200, an initial weight scaling factor of 0.10, an initial step size of 0.10, and an exponential decaying factor for the step size of 0.999 per step were used.

(441) inDelphi Deletion Modeling: Summary and Revisiting Assumptions

(442) In summary, inDelphi trains MH-NN, which uses as input (microhomology length, microhomology GC content) to output a psi score which is translated into a phi score using deletion

length. This phi score represents the “strength” of the microhomology corresponding to a particular MH deletion genotype. It also trains MHless-NN which uses as input (deletion length) to directly output a phi score representing the “total strength” of all MH-independent activity for a particular deletion length.

(443) While the model assumes that microhomology and microhomology-less repair can overlap in contributions to a single repair genotype, this assumption is made conservatively by assuming that their contributions overlap only when there is no alternative. Specifically, in the context of a deletion length with full microhomology, the model assumes that they must overlap, while in the context of a deletion length without full microhomology, inDelphi allows MHindep-phi to represent all MH-less repair genotypes and none of the MH-dependent repair genotypes which are represented solely using their MH-phi scores. This can be seen by noting that at a deletion length j without full microhomology, MH genotypes are scored using their MH-phi scores, while the length j is scored by MHindep-phi[j] plus the sum of MH-phi for each microhomology. Therefore, the subset of MH-less genotypes at this deletion length have a score MHindep-phi[j].

(444) When the subset of MH-less genotypes includes only one MH-less genotype, this single genotype's score is equal to MHindep-phi[j]. In general, multiple MH-less genotypes are possible, in which case the total score of all of the MH-less genotypes is equal to MHindep-phi[j].

(445) The relative frequency of MH deletions and MH-less deletions is learned implicitly by the balancing between the sum of all MH-phi and MHindep-phi. Since MHindep-phi does not vary by sequence context while MH-phi does, the model assumes that variation in the fraction of deletions that use MH can at least partially be explained by varying sequence microhomology as represented by MH-NN.

(446) inDelphi Insertion Modeling

(447) Once inDelphi is trained on both deletion tasks, it predicts insertions from a sequence context and cleavage site by using the precision score of the predicted deletion length distribution and total deletion phi (from all MH-phi and MHindep-phi). inDelphi also uses one-hot-encoded binary vectors encoding nucleotides −4 and −3. In a training set, these features are collected and normalized to zero mean and unit variance, and the fraction of 1-bp insertions over the sum counts of 1-bp insertions and all deletions are tabulated as the prediction goal. A k-nearest neighbor model is built using the training data. inDelphi uses the default parameter k=5.

(448) On test data, the above procedure is used to predict the frequency of 1-bp insertions out of 1-bp insertions and all deletions for a particular sequence context. Once this frequency is predicted, it is used to make frequency predictions for each of the 4 possible insertion genotypes, which are predicted by deriving from the training set the average insertion frequency for each base given its local sequence context. When the training set is small, only the −4 nucleotide is used. When the training set is relatively large, nucleotides −5, −4, and −3 are used.

(449) To produce a frequency distribution on 1-bp insertions and 1-60 bp deletion genotypes, scores for all deletion genotypes and all 1-bp insertions are normalized to sum to 1. To produce a frequency distribution on indel lengths (+1 to −60), scores for all deletion lengths and 1-bp insertions are normalized to sum to 1.

(450) inDelphi: Repair Classes Predicted at Varying Resolution

(451) inDelphi predicts MH-deletions and 1-bp insertions at single base resolution. Measuring performance on the task of genotype frequency prediction considers this subset of repair outcomes only (about 60-70% of all outcomes). inDelphi predicts MH-less deletions to the resolution of deletion length. That is, inDelphi predicts a single frequency corresponding to the sum total frequency of all unique MH-less deletion genotypes possible for a particular deletion length. This modeling choice was made because genotype frequency replicability among MH-less deletions is substantially lower than among MH deletions.

(452) Measuring performance on the task of indel length frequency considers MH deletions, MH-less deletions, and 1-bp insertions (90% of all outcomes).

(453) In practice, if end-users desire, they can extend inDelphi predictions to frequency predictions for specific MH-less deletion genotypes by noting that MH-less deletions are distributed uniformly between 0 delta-position genotypes, medial genotypes, and N delta-position genotypes.
(454) Comparison with a Linear Baseline Model
(455) inDelphi was compared to a baseline model with the same model structure but replacing the deep neural networks with linear models. The comparison was done using Lib-A mESC data. While inDelphi achieves a mean held-out Pearson correlation of 0.851 on deletion genotype frequency prediction and 0.837 on deletion length frequency prediction, the linear baseline model achieves a mean held-out Pearson correlation of 0.816 on deletion genotype frequency prediction and 0.796 on deletion length frequency prediction. When including the third model component for 1-bp insertion modeling and testing on genotype frequency prediction for 1-bp insertions and all deletions, inDelphi achieves a median held-out Pearson correlation of 0.937 and 0.910 on the task of indel length frequency prediction. The linear baseline model achieves a median held-out Pearson correlation of 0.919 and 0.900 on the two tasks respectively.
(456) From these results, it is shown that much of the model's power is derived from its designed structure which is independent of the choice of linear or non-linear modeling. While the baseline does not significantly cripple the model, the use of deep nonlinear neural networks offers a substantial performance improvement (10-24%) above linear modeling. In addition, the strong performance of the linear baseline model highlights that the prediction task, given the model structure, is relatively straightforward. This suggests that the model should be able to generalize well to unseen data.
(457) The deep neural network version of MH-NN learns that microhomology length is more important than % GC (FIGS. **18**A-**18**H). The linear version learns the same concept, with a weight of 1.1585 for MH length and 0.332 for % GC.
(458) Comparison with a Baseline Model Lacking Microhomology Length as a Feature
(459) Microhomology length is an important feature for MH-NN (FIGS. **18**A-**18**H). A model was trained that uses only % GC as input to MH-NN while keeping the rest of the model structure identical. On held-out data, this baseline model at convergence achieves to a mean Pearson correlation of 0.59 on the task of predicting deletion genotype frequencies, and a mean Pearson correlation of 0.58 on the task of predicting deletion length frequencies. Notably, a model at iteration 0 with randomly initialized weights achieves mean Pearson correlations of 0.55 and 0.54 on the two respective tasks on held-out data. This basal Pearson correlation is relatively high due to the model structure, in particular, the exponential penalty on deletion length. In sum, removing MH length as a feature severely impacts model performance, restricting it to predictive performance not appreciably better than random chance.
(460) inDelphi Training and Testing on Data from Varying Cell-Types
(461) For predicting genotype and indel length frequencies in any particular cell-type C where data D is available, inDelphi's deletion component was first trained on a subset of Lib-A mESC data. Then, k-fold cross-validation was applied on D where D is iteratively split into training and test datasets. For each cross-validation iteration, the training set is used to train the insertion frequency model (k-nearest neighbors) and insertion genotype model (matrix of observed probabilities of each inserted base given local sequence context, which is just the −4 nucleotide when the training dataset is small, and −5, −4 and −3 nucleotides when the training dataset is large). For each cross-validation iteration, predictions are made at each sequence context in the test set which are compared to observations for each sequence context to yield a Pearson correlation. For any particular sequence context, the median test-time Pearson correlation across all cross-validation iterations is used as a single number summary of the overall performance of inDelphi. For all reported results, 100-fold cross-validation was used with 80%/20% training and testing splits. Empirically, small variance in test-time Pearson correlation was observed, highlighting the stability of inDelphi's modeling approach.

(462) inDelphi Testing on Endogenous VO Data
(463) On this task, the deletion component of inDelphi was trained on a subset of the Lib-A mESC data. For each cell type in HCT116, K562, and HEK293T, all VO sequence contexts (about 100) were randomly split into training and test datasets 100 times. During each split, the training set was used for k-nearest neighbor modeling of 1-bp insertion frequencies. Feature normalization to zero mean and unit variance was not performed. The average frequency of each 1-bp insertion genotype was derived from the training set as well. For each of the ˜100 sequence contexts, the median test-time Pearson correlation was used for plotting in FIGS. **13**A-**13**D. Due to the small size of the training set, only the −4 nucleotide was used for modeling both the insertion frequency and insertion genotype frequencies.
(464) inDelphi Testing on Library Data
(465) On this task, the deletion component of inDelphi was trained on a subset of the Lib-A mESC data. The remaining test set was used for measuring test-time prediction performance on Lib-A. Nucleotides −5, −4, and −3 were used for the insertion genotype model. For testing on Lib-B, Lib-B was split into training and test datasets in the same manner as with VO data. Nucleotide −4 was used for the insertion genotype model. The median test-time Pearson correlation is used as a single number summary of the overall performance of inDelphi on any particular sequence context. For reporting predictive results in FIGS. **15**A-**15**F, sequence contexts with low replicability (less than 0.85 Pearson correlation) in observed editing outcome frequencies were first removed.
(466) inDelphi Training and Testing on Prkdc.sup.−/−Lig4.sup.−/− Data
(467) inDelphi was trained on data from 946 Lib-A sequence contexts and tested on 168 held-out Lib-A sequence contexts. Nucleotide −4 was used for insertion rate modeling, all other modeling choices were standard as described above. On held-out data, this version of inDelphi achieved a median Pearson correlation of 0.84 on predicting indel genotype frequencies, and 0.80 on predicting indel length frequencies.
(468) Training the Online Public Version of inDelphi and its Expected Properties
(469) For general-use on arbitrary cell types, a version of inDelphi was trained using additional data from diverse types of cells. Deletion modeling was trained using data from 2,464 sequence contexts from high-replicability Lib-A and Lib-B data (including clinical variants and microduplications, fourbp, and longdup) in mES and data from VO sequence contexts in HEK293 and K562. Insertion frequency modeling is implemented as above. Insertion genotype modeling uses nucleotides −5, −4, and −3. The insertion frequency model and insertion genotype model are trained on VO endogenous data in K562 and HEK293T, Lib-A data in mESC, and Lib-B data (including clinical variants and microduplications, fourbp, and longdup) in mESC and U2OS.
(470) Though MHless-NN, as trained on library data, never receives information on deletion lengths beyond 28, it was allowed to generalize its learned function and make predictions on deletion lengths up to 60 bp to match the supported range of MH-NN.
(471) inDelphi makes predictions on 1-bp insertions and 1−60-bp deletions, which were empirically shown to consist of higher than 90% of all Cas9 editing outcomes in data from multiple human and mouse cell lines. Nevertheless, there is a subset of repair (about 8% on average) that inDelphi does not attempt to predict. It is suggested that end-users, depending on what predictive quantities are of interest, take this into account when using inDelphi. For example, if inDelphi predicts that 60% of 1-bp insertions and 1−60-bp deletions at a disease allele correspond to repair to wildtype genotype, a quantity of interest may be the rate of wildtype repair in all Cas9 editing outcomes (including the 8% not predicted by inDelphi). In such a situation, this quantity can be calculated as (92%*60%)=55.2%.
(472) By the design of 1872 sequence contexts in Lib-A, the training dataset has rich and uniform representation across all quintiles of several major axes of variation including GC content, precision, and number of bases participating in microhomology as measured empirically in the human genome. This design strategy enables inDelphi to generalize well to arbitrary sequence

contexts from the human genome.
(473) These training data further include data in the outlier range of statistics of interest, including extremely high and low precision repair distributions, and extremely weak and strong microhomology (minimal microhomology and extensive microduplication microhomology sequences). The availability of such sequences in the training data enables inDelphi to generalize well to sequence contexts of clinical interest and sequence contexts supporting unusually high frequencies of precision repair. In particular, by training on more than 1000 examples of repair at clinical microduplications, inDelphi has received strong preparation for accurate prediction on other clinical microduplications.
(474) By training on data from many cell-types, inDelphi was enabled to make predictions that are generally applicable to many human cell-types. It is noted that the HCT116 human colon cancer cell line experiences a markedly higher frequency of single base insertions compared to all other cell lines that were studied, possibly due to the MLH1 deficiency of this cell line leading to impaired DNA mismatch repair. For this reason, HCT116 data was excluded from the training dataset. For best results, it is suggested that end-users keep in mind that repair class frequencies can be cell type-dependent, and this issue has not been well-characterized thus far.
(475) It is noted that inDelphi's main error tendency is on the side of overestimating rather than underestimating the precision of repair (FIGS. **14**A-**14**F, FIGS. **15**A-**15**F). In general, this tendency can be explained by noting that inDelphi only considers sequence microhomology as a factor, while it's plausible and likely in biological experimental settings that even sequence contexts with very strong sequence microhomology may not yield precise results due to noise factors that are not considered by inDelphi. For best results, it is recommended that end-users take this tendency into account when using inDelphi predictions for further experiments. In particular, if gRNAs are designed by using a minimum precision threshold, end-users should recognize that observed repair outcomes may have empirical precision under this threshold. However, conversely, it is unlikely that a gRNA will have precision higher than what inDelphi predicts.
(476) Lib-A Design (See Table 4)
(477) All designed sequence contexts were 55 bp in length with cutting between the 27.sup.th and.sub.28th base.
(478) 1872 sequence contexts were designed by empirically determining the distribution of four statistics in sequence contexts from the human genome. These four statistics are GC content, total sum of bases participating in microhomology for 3−27-bp deletions, Azimuth predicted on-target efficiency score, and the statistical entropy of the predicted 3−27-bp deletion length distribution from a previous version of inDelphi. For each of these statistics, empirical quintiles were derived by calculating these statistics in a large number of sequence contexts from the human genome. For the library, sequence contexts were designed by randomly generated DNA that categorized into each combination of quintiles across each of the four statistics. For example, a sequence context falling into the 1.sup.st quintile in GC, 2.sup.nd quintile of total MH, 1.sup.st quintile of Azimuth score, and 5.sup.th quintile of entropy, was found by random search. With four statistics and five bins each (due to quintiles), there are 5.sup.4=625 possible combinations. For each combination, it was attempted to design three sequence contexts for a total of 1875; 3 sequences could not be designed (for a total of 1872) though each bin was filled. 90 sequence contexts were designed from VO sequence contexts. Other sequence contexts were also designed for a total of 2000 sequence contexts in Lib-A. Lib-A sequence names, gRNAs, and sequence contexts are listed in Table 4 (appended, forming part of the instant specification).
(479) Lib-B Design (See Table 5)
(480) All designed sequence contexts were 55 bp in length with cutting between the 27.sup.th and 28.sup.th base.
(481) 1592 sequence contexts were designed from Clinvar and HGMD (see section on Selection of variants from disease databases). Some disease sequence contexts were designed that such that the

corrected wildtype or frameshift allele supports further cutting by the original gRNA; data from such sequence contexts were ignored during analysis. 57 “longdup” sequence contexts were designed by repeating the following procedure three times: for N=7 to 25, an N-mer was randomly generated, then duplicating and surrounded by randomly generated sequences, while ensuring that SpCas9 NGG was included and appropriately positioned for cutting between positions 27 and 28. 90 sequence contexts were designed from VO sequence contexts. 228 “fourbp” sequence contexts were designed at 3 contexts with random sequences (with total phi score on average lower than VO sequence contexts) while varying positions −5 to −2; for each of the 3 “low-microhomology” contexts, 76 four bases were randomly designed while ensuring representation from all possible 2 bp microhomology patterns including no microhomology, one base of microhomology at either position, and full two bases of microhomology. Other sequence contexts were also designed for a total of 2000 sequence contexts in Lib-B. Lib-B sequence names, gRNAs, and sequence contexts are listed in Table 5.

(482) Generating a DNA Motif for 1-Bp Insertion Frequencies

(483) Nucleotides from positions −7 to 0 were one-hot-encoded and used in ridge regression to predict the observed frequency of 1-bp insertions out of all Cas9 editing events in 1996 sequence contexts from Lib-A mESC data. The data were split into training and testing sets (80/20 split) 10,000 times to calculate a bootstrapped estimate of linear regression weights and test-set predictive Pearson correlation. The median test-set Pearson correlation was found to be 0.62. To generate a DNA motif, any features that included 0 within the bootstrapped weight range were excluded (probability that the weight is zero >1e-4). The average bootstrapped weight estimate was used as the “logo height” for all remaining features. Each feature is independent; vertical stacking of features follows the published tradition of DNA motifs.

(484) TABLE-US-00029 Plasmid and insert sequences P2T-CAG-MCS-P2A-GFP-PuroR complete plasmid sequence (SEQ ID NO: 41)

CCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTTAAATTTTTGTTAAAT
CAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCCTTATAAATCAAAAGA
ATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACAAGAGTCCACTATTAA
AGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTATCAGGGCGATGGCCC
ACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGGTGCCGTAAAGCAC
TAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACGGGGAAAGCCGGC
GAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGGCGCTAGGGCGC
TGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCGCGCTTAATGC
GCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTTGGGAAGG
GCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGATGTGC
TGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGTAAAA
CGACGGCCAGTGAGCGCGCGTAATACGACTCACTATAGGGCGAATTGGGTACCG
GCATATGGTTCTTGACAGAGGTGTAAAAAGTACTCAAAAATTTTACTCAAGTGAAAG
TACAAGTACTTAGGGAAAATTTTACTCAATTAAAAGTAAAAGTATCTGGCTAGAATC
TTACTTGAGTAAAAGTAAAAAAGTACTCCATTAAAATTGTACTTGAGTATTAAGGAA
GTAAAAGTAAAAGCAAGAAAGATCGATCTCGAAGGATCTGGAGGCCACCATGGTG
TCGATAACTTCGTATAGCATACATTATACGAAGTTATCGTGCTCGACATTGATTATT
GACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGA
GTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGAC
CCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGAC
TTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTAC
ATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGG
CCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTA
CATCTACGTATTAGTCATCGCTATTACCATGGTCGAGGTGAGCCCCACGTTCTGCT
TCACTCTCCCCATCTCCCCCCCCTCCCCACCCCCAATTTTGTATTTATTTATTTTTTA

ATTATTTTGTGCAGCGATGGGGGCGGGGGGGGGGGGGGCGCGCGCCAGGCG
GGGCGGGGCGGGGCGAGGGGCGGGGCGGGGCGAGGCGGAGAGGTGCGGCGG
CAGCCAATCAGAGCGGCGCGCTCCGAAAGTTTCCTTTTATGGCGAGGCGGCGGC
GGCGGCGGCCCTATAAAAAGCGAAGCGCGCGGCGGGCGGGGAGTCGCTGCGAC
GCTGCCTTCGCCCCGTGCCCCGCTCCGCCGCCGCCTCGCGCCGCCCGCCCCGG
CTCTGACTGACCGCGTTACTCCCACAGGTGAGCGGGCGGGACGGCCCTTCTCCTC
CGGGCTGTAATTAGCGCTTGGTTTAATGACGGCTTGTTTCTTTTCTGTGGCTGCGT
GAAAGCCTTGAGGGGCTCCGGGAGGGCCCTTTGTGCGGGGGGAGCGGCTCGGG
GGGTGCGTGCGTGTGTGTGTGCGTGGGGAGCGCCGCGTGCGGCTCCGCGCTGC
CCGGCGGCTGTGAGCGCTGCGGGCGCGGCGCGGGGCTTTGTGCGCTCCGCAGT
GTGCGCGAGGGGAGCGCGGCCGGGGGCGGTGCCCCGCGGTGCGGGGGGGCT
GCGAGGGGAACAAAGGCTGCGTGCGGGGTGTGTGCGTGGGGGGGTGAGCAGGG
GGTGTGGGCGCGTCGGTCGGGCTGCAACCCCCCCTGCACCCCCCTCCCCGAGTT
GCTGAGCACGGCCCGGCTTCGGGTGCGGGGCTCCGTACGGGGCGTGGCGCGGG
GCTCGCCGTGCCGGGCGGGGGGTGGCGGCAGGTGGGGGTGCCGGGCGGGGCG
GGGCCGCCTCGGGCCGGGGAGGGCTCGGGGGAGGGGCGCGGCGGCCCCCGGA
GCGCCGGCGGCTGTCGAGGCGCGGCGAGCCGCAGCCATTGCCTTTTATGGTAAT
CGTGCGAGAGGGCGCAGGGACTTCCTTTGTCCCAAATCTGTGCGGAGCCGAAATC
TGGGAGGCGCCGCCGCACCCCCTCTAGCGGGCGCGGGGCGAAGCGGTGCGGCG
CCGGCAGGAAGGAAATGGGCGGGGAGGGCCTTCGTGCGTCGCCGCGCCGCCGT
CCCCTTCTCCCTCTCCAGCCTCGGGGCTGTCCGCGGGGGGACGGCTGCCTTCGG
GGGGGACGGGGCAGGGCGGGGTTCGGCTTCTGGCGTGTGACCGGCGGCTCTAG
AGCCTCTGCTAACCATGTTCATGCCTTCTTCTTTTTCCTACAGCTCCTGGGCAACGT
GCTGGTTATTGTGCTGTCTCATCATTTTGGCAAAGAATTCCTCGAGCGGCCGCCAG
TGTGATGGATATCGGATCCGCTAGCGCTACTAACTTCAGCCTGCTGAAGCAGGCT
GGAGACGTGGAGGAGAACCCTGGACCTGGACCGGTCGCCACCATGGTGAGCAAG
GGCGAGGAGCTGTTCACCGGGGTGGTGCCCATCCTGGTCGAGCTGGACGGCGAC
GTAAACGGCCACAAGTTCAGCGTGTCCGGCGAGGGCGAGGGCGATGCCACCTAC
GGCAAGCTGACCCTGAAGTTCATCTGCACCACCGGCAAGCTGCCCGTGCCCTGG
CCCACCCTCGTGACCACCCTGACCTACGGCGTGCAGTGCTTCAGCCGCTACCCCG
ACCACATGAAGCAGCACGACTTCTTCAAGTCCGCCATGCCCGAAGGCTACGTCCA
GGAGCGCACCATCTTCTTCAAGGACGACGGCAACTACAAGACCCGCGCCGAGGT
GAAGTTCGAGGGCGACACCCTGGTGAACCGCATCGAGCTGAAGGGCATCGACTT
CAAGGAGGACGGCAACATCCTGGGGCACAAGCTGGAGTACAACTACAACAGCCAC
AACGTCTATATCATGGCCGACAAGCAGAAGAACGGCATCAAGGTGAACTTCAAGAT
CCGCCACAACATCGAGGACGGCAGCGTGCAGCTCGCCGACCACTACCAGCAGAA
CACCCCCATCGGCGACGGCCCCGTGCTGCTGCCCGACAACCACTACCTGAGCAC
CCAGTCCGCCCTGAGCAAAGACCCCAACGAGAAGCGCGATCACATGGTCCTGCTG
GAGTTCGTGACCGCCGCCGGGATCACTCTCGGCATGGACGAGCTGTACAAGTAAA
GCGGCCGCCACCGCGGTGGAGCTCGAATTAATTCATCGATGATGATCCAGACATG
ATAAGATACATTGATGAGTTTGGACAAACCACAACTAGAATGCAGTGAAAAAAATG
CTTTATTTGTGAAATTTGTGATGCTATTGCTTTATTTGTAACCATTATAAGCTGCAAT
AAACAAGTTAACAACAACAATTGCATTCATTTTATGTTTCAGGTTCAGGGGGAGGTG
TGGGAGGTTTTTTAAAGCAAGTAAAACCTCTACAAATGTGGTATGGCTGATTATGAT
CCTCTAGAGTCGGTGGGCCTCGGGGGCGGGTGCGGGGTCGGCGGGGCCGCCCC
GGGTGGCTTCGGTCGGAGCCATGGGGTCGTGCGCTCCTTTCGGTCGGGCGCTGC
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GCCCAAAGGACCCTTGATCAATTCTGAATTCTATACTGGCTGGCTAGATCACTGGG
GCCAACCTCACTCCACAATCAAGACCGAAGCAGTGGCTTCCTCCCTCTATGATATA
CTTGCCCGTGGGGCGAGTGTGAACTTGTACATGTTTATAGGTGGGACCAATTTTGC
CTATTGGAATGGGGCCAACTCACCCTATGCAGCACAGCCCACCAGCTACGACTAT
GATGCCCCACTGAGTGAGGCTGGGGACCTCACTGAGAAGTATTTTGCTCTGCGAA
ACATCATCCAGAAGTTTGAAAAAGTACCAGAAGGTCCTATCCCTCCATCTACACCA
AAGTTTGCATATGGAAAGGTCACTTTGGAAAAGTTAAAGACAGTGGGAGCAGCTCT
GGACATTCTGTGTCCCTCTGGGCCCATCAAAAGCCTTTATCCCTTGACATTTATCCA
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GCAACCCAGCACCTCTCTCTTCACCCCTCAATGGAGTCCACGATCGAGCATATGTT
GCTGTGGATGGGATCCCCCAGGGAGTCCTTGAGCGAAACAATGTGATCACTCTGA
ACATAACAGGGAAAGCTGGAGCCACTCTGGACCTTCTGGTAGAGAACATGGGACG
TGTGAACTATGGTGCATATATGGTGCATATATCAACGATTTTAAGGGTTTGGTTTCT
AACCTGACTCTCAGTTCCAATATCCTCACGGACTGGACGATCTTTCCACTGGACAC
TGAGGATGCAGTGTGCAGCCACCTGGGGGGCTGGGGACACCGTGACAGTGGCCA
CCATGATGAAGCCTGGGCCCACAACTCATCCAACTACACGCTCCCGGCCTTTTATA
TGGGGAACTTCTCCATTCCCAGTGGGATCCCAGACTTGCCCCAGGACACCTTTATC
CAGTTTCCTGGATGGACCAAGGGCCAGGTCTGGATTAATGGCTTTAACCTTGGCC
GCTATTGGCCAGCCCGGGGCCCTCAGTTGACCTTGTTTGTGCCCCAGCACATCCT
GATGACCTCGGCCCCAAACACCATCACCGTGCTGGAACTGGAGTGGGCACCCTG
CAGCAGTGATGATCCAGAACTATGTGCTGTGACGTTCGTGGACAGGCCAGTTATT
GGCTCATCTGTGACCTACGATCATCCCTCCAAACCTGTTGAAAAAAGACTCATGCC
CCCACCCCCGCAAAAAAACAAAGATTCATGGCTGGACCATGTA PORCNwt (SEQ ID
NO: 52)
ATGGCCACCTTTAGCCGCCAGGAATTTTTCCAGCAGCTACTGCAAGGCTGTCTCCT
GCCTACTGCCCAGCAGGGCCTTGACCAGATCTGGCTGCTCCTTGCCATCTGCCTC
GCCTGCCGCCTCCTCTGGAGGCTCGGGTTGCCATCCTACCTGAAGCATGCAAGCA
CCGTGGCAGGCGGGTTCTTCAGCCTCTACCACTTCTTCCAGCTGCACATGGTTTG
GGTCGTGCTGCTCAGCCTCCTGTGCTACCTCGTGCTGTTCCTCTGCCGACATTCCT
CCCATCGAGGCGTCTTCCTATCCGTCACCATCCTCATCTACCTACTCATGGGTGAG
ATGCACATGGTAGACACCGTGACATGGCACAAGATGCGAGGGGCACAGATGATTG
TGGCCATGAAGGCAGTGTCTCTGGGCTTCGACCTGGACCGGGGCGAGGTGGGTA
CGGTGCCCTCGCCAGTGGAGTTCATGGGCTACCTCTACTTCGTGGGCACCATCGT
CTTCGGGCCCTGGATATCCTTCCACAGCTACCTACAAGCTGTCCAAGGCCGCCCA
CTGAGCTGCCGGTGGCTGCAGAAGGTGGCCCGGAGCCTGGCACTGGCCCTGCTG
TGCCTTGTGCTGTCCACTTGCGTGGGCCCCTACCTCTTCCCGTACTTCATCCCCCT
CAACGGTGACCGCCTCCTTCGCAAGGGCACCATGGTAAGGTGGCTGCGAGCCTA
CGAGAGTGCTGTCTCCTTCCACTTCAGCAACTATTTTGTGGGCTTTCTTTCCGAGG
CCACGGCCACGTTGGCGGGGGCTGGCTTTACCGAGGAGAAGGATCACCTGGAAT
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AAGTTGTCACAAGCTGGAACCTGCCCATGTCTTATTGGCTAAATAACTATGTTTTCA
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CGCCCTCCTACATGGCTTCAGTTTCCACCTGGCTGCGGTCCTGCTGTCCCTGGCT
TTTATCACTTACGTGGAGCATGTCCTCCGGAAGCGCCTGGCTCGGATCCTCAGTG
CCTGTGTCTTGTCAAAGCGGTGCCCGCCAGACTGTTCGCACCAGCATCGCTTGGG
CCTGGGGGTGCGAGCCTTAAACTTGCTCTTTGGAGCTCTGGCCATCTTCCACCTG
GCCTACCTGGGCTCCCTGTTTGATGTCGATGTGGATGACACCACAGAGGAGCAGG

GCTACGGCATGGCATACACTGTCCACAAGTGGTCAGAGCTCAGCTGGGCCAGTCA CTGGGTCACTTTTGGATGCTGGATCTTCTACCGTCTCATAGGC PORCNDup20 (SEQ ID NO: 53)
ATGGCCACCTTTAGCCGCCAGGAATTTTTCCAGCAGCTACTGCAAGGCTGTCTCCT GCCTACTGCCCAGCAGGGCCTTGACCAGATCTGGCTGCTCCTTGCCATCTGCCTC GCCTGCCGCCTCCTCTGGAGGCTCGGGTTGCCATCCTACCTGAAGCATGCAAGCA CCGTGGCAGGCGGGTTCTTCAGCCTCTACCACTTCTTCCAGCTGCACATGGTTTG GGTCGTGCTGCTCAGCCTCCTGTGCTACCTCGTGCTGTTCCTCTGCCGACATTCCT CCCATCGAGGCGTCTTCCTATCCGTCACCATCCTCATCTACCTACTCATGGGTGAG ATGCACATGGTAGACACCGTGACATGGCACAAGATGCGAGGGGCACAGATGATTG TGGCCATGAAGGCAGTGTCTCTGGGCTTCGACCTGGACCGGGGCGAGGTGGGTA CGGTGCCCTCGCCAGTGGAGTTCATGGGCTACCTCTACTTCGTGGGCACCATCGT CTTCGGGCCCTGGATATCCTTCCACAGCTACCTACAAGCTGTCCAAGGCCGCCCA CTGAGCTGCCGGTGGCTGCAGAAGGTGGCCCGGAGCCTGGCACTGGCCCTGCTG TGCCTTGTGCTGTCCACTTGCGTGGGCCCCTACCTCTTCCCGTACTTCATCCCCCT CAACGGTGACCGCCTCCTTCGCAAGGGCACCATGGTAAGGTGGCTGCGAGCCTA CGAGAGTGCTGTCTCCTTCCACTTCAGCAACTATTTTGTGGGCTTTCTTTCCGAGG CCACGGCCACGTTGGCGGGGGCTGGCTTTACCGAGGAGAAGGATCACCTGGAAT GGGACCTGACGGTGTCCAAGCCACTGAATGTGGAGCTGCCTCGGTCAATGGTGG AAGTTGTCACAAGCTGGAACCTGCCCATGTCTTATTGGCTAAATAACTATGTTTTCA AGAATGCTCTCCGCCTGGGGACCTTCTCGGCTGTGCTGGTCACCTATGCAGCCAG CGCCCTCCTACATGGCTTCAGTTTCCACCTGGCTGCGGTCCTGCTGTCCCTGGCT TTTATCCCTGGCTTTTATCACTTACGTGGAGCATGTCCTCCGGAAGCGCCTGGCTC GGATCCTCAGTGCCTGTGTCTTGTCAAAGCGGTGCCCGCCAGACTGTTCGCACCA GCATCGCTTGGGCCTGGGGGTGCGAGCCTTAAACTTGCTCTTTGGAGCTCTGGCC ATCTTCCACCTGGCCTACCTGGGCTCCCTGTTTGATGTCGATGTGGATGACACCAC AGAGGAGCAGGGCTACGGCATGGCATACACTGTCCACAAGTGGTCAGAGCTCAG CTGGGCCAGTCACTGGGTCACTTTTGGATGCTGGATCTTCTACCGTCTCATAGGC

EXECUTIVE SUMMARY

(485) It was found that template-free DNA repair of Cas9-cleaved and Cpf1-cleaved DNA produces a predictable set of repair genotypes that can result in the gain-of-function repair of human disease mutations. Contrary to the assumption that end-joining following double-strand breaks is random and difficult to harness for applications beyond gene disruption, here it is shown that template-free end-joining repair of DNA cleaved by CRISPR-associated nucleases produces a predictable set of repair genotypes. A library of 2000 guide RNAs paired with target DNA sites was constructed, and they were integrated into mouse and human genomes, applied Cas9, and performed high-throughput sequencing of repair genotypes. Data from this assay are consistent with results from 98 endogenous loci. Building upon prior work, it is shown that the majority of repair genotypes in cells with saturated exposure to both CRISPR-Cas9 and Cpf1 are deletions associated with sequence microhomology. Using 1,588 sequence contexts from the data, CRISPR-Texture, a machine learning method that accurately predicted the frequencies of template-free Cas9-mediated microhomology-associated deletions as well as 1 bp insertions, was trained. On 282 held-out sequence contexts, CRISPR-Texture predicted frameshift rates more accurately than published methods and accurately predicted the statistical entropy of repair product distributions. Applied to the human genome, CRISPR-Texture identified an appreciable fraction of Cas9 target sites supporting high-precision repair distributions that are dominated by a single genotype. Further, it was found that a class of human disease-associated micro-duplication mutations can be repaired to wildtype at high frequency by template-free Cas9 nuclease editing and used the assay to validate hundreds of such alleles. Template-free Cas9 nuclease-mediated rescue of pathogenic LDLR alleles to wildtype phenotype in cellular models was also validated. This work establishes a

strategy for predicting the outcomes of template-free end-joining and demonstrates that CRISPR editing can also mediate efficient gain-of-function editing at certain disease alleles without homology-directed repair.

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Example 2: Workflow Description of Using inDelphi to Design Cas9 gRNAs for Efficient Genome Editing to Induce Exon Skipping

(487) One application of CRISPR-Cas9 for therapeutic purposes is to alter the genome to cause RNA editing to skip a pathogenic exon by changing the splicing regulatory sites controlling the inclusion of the pathogenic exon. In some cases, a single exon in a gene may contain a pathogenic variant, and the entire exon can be skipped to produce an alternative isoform with normal function. For example, certain cases of Duchenne muscular dystrophy (DMD) are caused by a deleterious mutation in exon 23 of the Dmd gene that results in a premature stop codon that produces a dysfunctional Dmd protein. To restore function the deleterious exon 23 can skipped by editing the DNA proximal to the diseased exon with a single gRNA (Long, 2016).

(488) This Example tests a new method for selecting gRNAs for CRISPR editing to disrupt splice site sequences to cause the skipping of pathogenic exons to restore wild-type cellular function. Splice site acceptor DNA motifs occur at the boundary between introns and exons, depicted as 5′-intron-AG-exon-3′, where the AG is a highly-conserved element of the splice site acceptor motif and is considered to reside in the intron. The splice site acceptor DNA motif is considered to be as

long as 23 bp in length including the AG. Algorithms such as MaxEntScan (Yeo, 2004) receive as input a DNA sequence and output a numerical score representing how strongly the splice site acceptor motif is present at the DNA sequence.

(489) CRISPR-Cas9 induces a DNA double-strand break at a specific location specified by the gRNA. DNA repair fixes the double-strand break, inducing insertions and deletions through non-homologous end-joining (NHEJ) and microhomology-mediated end-joining (MMEJ) when a homology template is not present for repair through homology-directed repair (HDR). InDelphi, as disclosed herein, predicts the frequency distribution of NHEJ/MMEJ-mediated repair genotypes following Cas9 cutting. In this Example, inDelphi's ability to predict the spectrum of repair genotypes is utilized to identify gRNAs that ablate the splice site acceptor motif at a high frequency out of all non-wild-type repair outcomes.

(490) Other computational methods have focused on predicting on-target efficiency of CRISPR-Cas9 cutting. This Example aims to identify gRNAs that will efficiently cut and induce non-wild-type repair outcomes (otherwise described as high on-target activity). A relevant published method (Doench et al., 2016, Nature, aka “Azimuth”) uses the DNA sequence surrounding the gRNA, as well as the position of the cutsite in the protein, to predict the gRNA's ability to knock out the protein as observed in gRNA enrichment in screens of cell-essential genes, where a higher score indicates higher gRNA cutting efficacy. However, Doench et al. does not directly predict the frequency of non-wild-type repair frequency. It is reasoned here that the frequency of protein knockdown depends on the rate of non-wild-type repair, and the rate of frameshift repair out of all non-wild-type repair outcomes. Despite this concern, gRNAs are filtered based on a minimum threshold Azimuth score in order to maximize the chances that selected gRNAs have high on-target activity (increase true positives), at the risk of filtering away some gRNAs that also have high on-target activity (increase false negatives). In alternative embodiments, this Azimuth filtering step may be skipped to decrease the rate of false negatives at the risk of decreasing the rate of true positives.

(491) To more directly address the question of on-target efficiency, this Example developed an algorithm referred to as the Basic On-Target Model (BOTM) which directly predicts the frequency of non-wild-type observations from DNA sequence features. BOTM uses DNA sequence as input and outputs a predicted frequency of non-wild-type repair, where non-wild-type repair is defined as the sum frequency of CRISPR-associated deletions and insertions (defined as reads aligning to the reference with exactly one gap which resides within 1-bp of the cutsite, using alignment scores +1 match, −1 mismatch, −5 gap open, −0 gap extend), over the denominator of sum frequency of non-noise outcomes consisting of CRISPR-associated indels, wildtype repair, and reads with multiple indels with at least one occurring near the cutsite, and reads with exactly one indel occurring anywhere outside the cutsite). BOTM is implemented as an ensemble of 100 gradient boosted regression trees, each with maximum depth 3, that are fitted in consecutive stages on the negative gradient of the least squares loss function. BOTM uses the following input features: one-hot encoded nucleotides at positions −7 to 0 (such that “NGG” occupies positions 0 to 2), the GC fraction of the 40-bp window around the cutsite, and the following features from inDelphi: log phi score (microhomology score), precision score (ranging from 0 to 1, with 1 being more precise), expected value of the indel length distribution, the frequency of 1-bp insertions, microhomology deletions, and microhomology-less deletions, the highest frequency of any single 1-bp insertion outcome, the highest frequency of any single deletion outcome, and the highest frequency of any single outcome. Trained on deep sequencing data at 3,600 target sites from our genome-integrated library construct in mES also used to train inDelphi, BOTM achieves a Pearson correlation of 0.42 at predicting the observed frequency of non-wild-type repair on 400 held-out target sites from our genome-integrated library construct in mES. On held-out data, it was manually determined a BOTM predicted frequency of 0.65 or greater for gRNAs that have a high frequency of non-wildtype repair.

(492) One computational workflow for identifying Cas9 gRNAs with clinical relevance for the correction of genetic diseases by inducing exon skipping consists of four steps: identify relevant exons, select gRNAs for these exons with effective targeting, determine the genotypic products of each gRNA using inDelphi, and select gRNAs with genotypic products that are predicted to disrupt the relevant splicing motif. Using this approach, we have identified 4000 gRNAs that target splice sites to correct genetic diseases (Appendix attached).
(493) First, 6805 exons with the following characteristics were determined: the exon length is evenly divisible by 3 so that skipping them preserves frame; the exon contains at least one HGMD pathogenic indel, which are likely to disrupt normal protein function (basal frameshift rate ˜66%, column "hgmd_indel_count" in Appendix spreadsheet); the exon is not constitutive, measured by <100% presence in Ensembl transcripts (Ensembl); and the exon does not contain an annotated protein domain in Pfam (Pfam). The last two criteria are used to identify exons that may not be essential for wild-type protein function. The resulting 6,805 exons were candidates for disease correction by exon skipping.
(494) Then, SpCas9 gRNAs (NGG PAM) with cutsites in a 6 bp window surrounding and including the AG motif were selected, resulting in an average of 2.2 SpCas9 gRNAs per exon. We then ensured high predicted on-target editing efficiency by removing all gRNAs with Azimuth score below 0.20. (threshold set manually) or BOTM score below 0.65 which is chosen to separate gRNAs with high versus low frequencies of non-wild-type repair).
(495) Each gRNA and exon target site for splice site motif disruption were scored. We obtained this prediction by first using inDelphi to predict the frequency distribution of 1-bp insertion and deletion (1−60 bp) genotypes resulting from template-free DNA repair of a CRISPR gRNA induced cut at the target exon site.
(496) Finally, for each genotype predicted by inDelphi, we classified a genotype as "motif disrupting" when its MaxEntScan score is <0.9 of its unedited MaxEntScan score; otherwise we classified the genotype as "no effect". This classification ruleset was provided by (Tang, 2016) and validated on experimental splicing data to achieve a sensitivity of 83.6% and specificity of 79.2% (Tang, 2016). The total frequency of all motif-disruption repair genotypes was used to predict the splice site motif disruption frequency out of all inDelphi predicted genotypes.
(497) The top 4000 gRNA and target site pairs were selected based on this predicted frequency of splice site disruptions. Long, 2016 identified several SpCas9 gRNAs that, in a mouse model of muscular dystrophy, restored some degree of dystrophin protein expression and improved skeletal muscle function by inducing exon skipping of exon 23 (containing a non-sense mutation) via NHEJ-mediated DNA repair of a Cas9-induced cut. Without considering the results of their experiments and focusing solely on the DNA sequence context and background biological knowledge, our computational workflow recognizes that exon 23 of DMD is a good candidate for disease correction via exon skipping: the exon has a length evenly divisible by 3, is associated with a pathogenic non-sense variant that destroys normal protein function, and is not constitutive or required for normal protein function. Long 2016 reports results for only one SpCas9 gRNA targeting the 5′ end of exon 23 called sgRNA-L8 ATAATTTCTATTATATTACA (SEQ ID NO: 11590) with PAM GGG. In their experiments with sgRNA-L8, they observe 9/18 pups with exon 23 skipping. This gRNA targets mm10 chrX: 83,803,134−83,803,156 (minus strand), while the exon 23 boundary is 149-bp downstream at mm10 chrX: 83,803,305. This Example's computational workflow for now only identifies gRNAs cutting within a 6-bp window of the AG motif at the exon 5′ boundary, so as described our workflow does not identify Long's sgRNA-L8. Other methods of selecting exons for splice site acceptor removal include selecting exons with mutant splice regulatory sites that result in the inappropriate inclusion of exons in RNA transcripts (Sterne-Weiler 2014). Alternatively, subsequent expressed exons can be skipped to restore reading frame. In this case, reading frame can be restored by skipping a subsequent expressed exon where the length of the subsequent skipped exon and the length of the indel sum to 0 mod 3. In addition,

constitutive exons and/or exons known to contain annotated protein domains in Pfam can be selected for exon skipping as an alternative method for knocking out a gene.

(498) Correcting Genetic Disorders Using Predictable CRISPR/Cas9-Induced Exon Skipping

(499) Exon skipping has emerged as a powerful method to restore gene function in a number of genetic disorders. These therapies force the splicing machinery to bypass exons that contain deleterious point mutations or frameshifts. The FDA has recently approved an antisense oligonucleotide therapy that induces exon skipping in Duchenne muscular dystrophy to restore dystrophin function, and several other related strategies have shown pre-clinical promise. Yet, oligonucleotide therapies are transient treatments that require frequent dosing.

(500) CRISPR/Cas9 instead promises to alleviate genetic disease permanently, through genome alteration. Using a high-throughput experimental-computational pipeline, as described herein, the inventors have developed an algorithm capable of highly accurate prediction of CRISPR/Cas9 genotypic alterations. At a predictable subset of genomic target sites, CRISPR/Cas9 induces precise sequence deletions. These modifications are highly specific and have excellent potential for therapeutic genome editing through controlled deletion of splice-acceptor sites.

(501) The inventors will systematically evaluate this new approach to treat genetic disorders using CRISPR/Cas9 deletions. At intron-exon junctions, we will induce small deletions to bypass exons containing deleterious variants that affect protein function or alter the reading frame. While not every splice site can be successfully deleted through CRISPR/Cas9 modification, and not every exon can be skipped without compromising gene function, the inventors expect that this approach will succeed in enough genes to have broad therapeutic implications.

(502) To measure the applicability of this approach to treat disease throughout the genome, the inventors will establish a principled computational approach to identify exons known to harbor disease-causing mutations where omission is unlikely to impact gene function. The inventors will then apply a novel, high-throughput CRISPR/Cas9 assay that quantifies the impact of high-precision genome editing on splicing at thousands of these intron-exon boundaries. After determining a set of candidate exons that can be skipped efficiently, the inventors will measure the impact of CRISPR/Cas9-mediated exon skipping on transcript structure and gene function for dozens of human disease exons. This exhaustive approach promises to chart a systematic path toward classifying disease genes that would be most amenable for future pre-clinical evaluation of permanent therapeutic exon skipping.

Induce Exon Skipping Using CRISPR/Cas9 at Thousands of Exons that Harbor Disease Variants.

(503) By mapping coding variants known to be associated with genetic disorders, the inventors will develop a set of exons whose skipping could feasibly provide clinical benefit. The inventors will use measures of selective constraint from large-scale population data and alternative splicing data to prioritize exons whose skipping is least likely to compromise protein function. Using the herein described algorithm which predicts the genotypic consequence of targeting with CRISPR/Cas9, the inventors will refine a list of up to 10,000 intron-exon boundaries where modification is predicted to induce exon skipping at high rates. To test these predictions in high-throughput, the inventors will adapt our CRISPR/Cas9 cutting assay to read out context-specific splicing outcomes in human cells in vitro. This novel assay will allow paired evaluation of CRISPR/Cas9 cutting genotype and splicing phenotype for hundreds of distinct replicates in each of 10,000 human exons. The inventors will perform this assay in several human cell lines and will computationally identify exons that can be skipped at high frequency for further study. The inventors will also explore Cas9 base-editing in the same high-throughput system to determine if splicing can be altered by single base alterations. Using these data, the inventors will derive computational rules for which sequence alterations do and do not lead to exon skipping.

(504) Evaluate the Consequences of CRISPR/Cas9 Exon Skipping on Transcript Structure and Function.

(505) Using results from the high-throughput assay, the inventors will select up to 100 exon-

skipping guide RNAs to pursue in greater depth. The inventors will prioritize exons that are natively excluded from at least one experimentally validated splice isoform, that lack characterized protein domains, and that are under relaxed selective constraint. For these exons, the inventors will edit native genomes in an appropriate cell line given the gene function and disease process. By performing transcript-specific RNA deep sequencing, the inventors will determine the rate of exon skipping and the transcript structure after the exon is skipped, monitoring for the appearance of aberrant splice acceptors. The inventors will also assay the function of these genes with skipped exons, using appropriate cellular and biochemical assays for each gene. This analysis will identify a set of disease genes that are promising candidates for further study in mutated cell lines and animal models. Overall, this systematic study will elucidate which disease genes are compelling candidates for pre-clinical evaluation of CRISPR/Cas9-mediated exon skipping therapy.

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OTHER EMBODIMENTS

(507) The foregoing has been a description of certain non-limiting embodiments of the invention. Those of ordinary skill in the art will appreciate that various changes and modifications to this description may be made without departing from the spirit or scope of the present invention, as defined in the following claims.

EQUIVALENTS AND SCOPE

(508) In the claims articles such as “a,” “an,” and “the” may mean one or more than one unless indicated to the contrary or otherwise evident from the context. Claims or descriptions that include “or” between one or more members of a group are considered satisfied if one, more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process unless indicated to the contrary or otherwise evident from the context. The invention includes embodiments in which exactly one member of the group is present in, employed in, or otherwise relevant to a given product or process. The invention includes embodiments in which more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process.

(509) Furthermore, the invention encompasses all variations, combinations, and permutations in which one or more limitations, elements, clauses, and descriptive terms from one or more of the listed claims is introduced into another claim. For example, any claim that is dependent on another claim can be modified to include one or more limitations found in any other claim that is dependent on the same base claim. Where elements are presented as lists, e.g., in Markush group format, each subgroup of the elements is also disclosed, and any element(s) can be removed from the group. It should it be understood that, in general, where the invention, or aspects of the invention, is/are referred to as comprising particular elements and/or features, certain embodiments of the invention or aspects of the invention consist, or consist essentially of, such elements and/or features. For purposes of simplicity, those embodiments have not been specifically set forth in haec verba herein.

It is also noted that the terms "comprising" and "containing" are intended to be open and permits the inclusion of additional elements or steps. Where ranges are given, endpoints are included. Furthermore, unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or sub-range within the stated ranges in different embodiments of the invention, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.
(510) This application refers to various issued patents, published patent applications, journal articles, and other publications, all of which are incorporated herein by reference. If there is a conflict between any of the incorporated references and the instant specification, the specification shall control. In addition, any particular embodiment of the present invention that falls within the prior art may be explicitly excluded from any one or more of the claims. Because such embodiments are deemed to be known to one of ordinary skill in the art, they may be excluded even if the exclusion is not set forth explicitly herein. Any particular embodiment of the invention can be excluded from any claim, for any reason, whether or not related to the existence of prior art.
(511) Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation many equivalents to the specific embodiments described herein. The scope of the present embodiments described herein is not intended to be limited to the above Description, but rather is as set forth in the appended claims. Those of ordinary skill in the art will appreciate that various changes and modifications to this description may be made without departing from the spirit or scope of the present invention, as defined in the following claims.
(512) TABLE-US-00030 Lengthy table referenced here US12406749-20250902-T00001 Please refer to the end of the specification for access instructions.
(513) TABLE-US-00031 Lengthy table referenced here US12406749-20250902-T00002 Please refer to the end of the specification for access instructions.
(514) TABLE-US-00032 Lengthy table referenced here US12406749-20250902-T00003 Please refer to the end of the specification for access instructions.
(515) TABLE-US-00033 Lengthy table referenced here US12406749-20250902-T00004 Please refer to the end of the specification for access instructions.
(516) TABLE-US-00034 Lengthy table referenced here US12406749-20250902-T00005 Please refer to the end of the specification for access instructions.
(517) TABLE-US-00035 Lengthy table referenced here US12406749-20250902-T00006 Please refer to the end of the specification for access instructions.
(518) TABLE-US-00036 Lengthy table referenced here US12406749-20250902-T00007 Please refer to the end of the specification for access instructions.
(519) TABLE-US-LTS-00001 LENGTHY TABLES The patent contains a lengthy table section. A copy of the table is available in electronic form from the USPTO web site (). An electronic copy of the table will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

Claims

1. A method of introducing a genetic change into a target genomic location encoding a pathogenic allele to modify the pathogenic allele to become a non-pathogenic allele using a Cas-based double strand break genome editing system, the method comprising: using a computer hardware processor to perform: selecting a guide RNA for use in introducing the genetic change into the target genomic location by analyzing inputs indicating a nucleotide sequence of the target genomic location and one or more available cut sites for the Cas-based double strand break genome editing system, the selecting comprising: (a) determining a microhomology score matrix using a first neural network, the determining comprising: determining a plurality of pairs of overhang sequences using the inputs; determining a microhomology length vector and/or a microhomology GC fraction vector

using the inputs; and applying the first neural network to the plurality of pairs of overhang sequences and the microhomology length vector and/or the microhomology GC fraction vector to obtain the microhomology score matrix; (b) determining a microhomology-independent score matrix using a second neural network, the determining comprising: determining a deletion length vector using the inputs and the plurality of pairs of overhang sequences; and applying the second neural network to the deletion length vector to obtain the microhomology-independent score matrix; (c) determining a probability distribution over 1-bp insertions; (d) determining, using the microhomology score matrix, the microhomology-independent score matrix and the probability distribution over 1-bp insertion, a probability distribution over indel genotypes and a probability distribution over indel lengths for the nucleotide sequence of the target genomic location and the one or more available cut sites; (e) determining, using the probability distribution over indel genotypes and the probability distribution over indel lengths, for each guide RNA of a plurality of guide RNAs, a predicted frequency of introducing the genetic change into the target genomic location using the Cas-based double strand break genome editing system and the guide RNA; (f) selecting, using the predicted frequencies of (e), a guide RNA of the plurality of guide RNAs for use in introducing the genetic change into the target genomic location using the Cas-based double strand break genome editing system; and introducing the genetic change into the target genomic location using the guide RNA selected at (f) and the Cas-based double strand break genome editing system, wherein the genetic change is a 1 base pair insertion or a 1−60 base pair deletion that modifies the pathogenic allele to become the non-pathogenic allele.
2. The method of claim 1, wherein analyzing inputs comprises identifying, for a cut site of the one or more available cut sites, a plurality of pairs of overhang sequences, each pair of the plurality of pairs of overhang sequences comprising: (i) a 3′-overhang sequence; and (ii) a 5′-overhang sequence.
3. The method of claim 1, wherein determining the probability distribution over 1-bp insertion comprises: determining, using the inputs, an overall deletion score, a precision score, and/or cut site nucleotides; and applying a third neural network to the overall deletion score, the precision score, and/or the cut site nucleotides to obtain the probability distribution over 1-bp insertion.
4. The method of claim 1, comprising: contacting a plurality of cells with the guide RNA and the Cas-based double strand break genome editing system; and introducing, using the selected guide RNA and the Cas-based double strand break genome editing system, the genetic change into the target genomic location of at least 30% of cells of the plurality of cells.
5. The method of claim 1, comprising: contacting a plurality of cells with the guide RNA and the Cas-based double strand break genome editing system; and introducing, using the selected guide RNA and the Cas-based double strand break genome editing system, the genetic change into the target genomic location of at least 50% of cells of the plurality of cells.
6. The method of claim 1, wherein the Cas-based double strand break genome editing system comprises a type II Cas RNA-guided endonuclease, or a functional variant or ortholog thereof.
7. The method of claim 1, wherein the Cas-based double strand break genome editing system comprises a Cas9 RNA-guided endonuclease, or a variant or orthologue thereof.
8. The method of claim 7, wherein the Cas9 RNA-guided endonuclease is *Streptococcus pyogenes* Cas9 (SpCas9), *Staphylococcus pyogenes* Cas9 (SpCas9), *Staphyloccocus aureus* Cas (SaCas9), *Francisella novicida* Cas9 (FnCas9), or a functional variant or orthologue thereof.
9. The method of claim 1, further comprising: treating a genetic disease in a subject caused by a pathogenic allele in a genome of a cell of the subject by contacting the genome of the cell of the subject with the selected guide RNA and the Cas-based double strand break genome editing system, in order to correct the pathogenic allele into a non-pathogenic allele in the genome of the cell.
10. The method of claim 1, wherein selecting the guide RNA for use in introducing the genetic change into the target genomic location using the Cas-based double strand break genome editing

system comprises selecting a guide RNA that corresponds to a cut site of the target genomic location.

11. The method of claim 1, wherein selecting the guide RNA comprises selecting a guide RNA that has a predicted frequency of introducing the genetic change into the target genomic location of at least 30%.

12. The method of claim 1, wherein selecting the guide RNA comprises selecting a guide RNA that has a predicted frequency of introducing the genetic change into the target genomic location of at least 50%.

13. The method of claim 1, wherein the genetic change is a 1 base pair insertion or a 1-60 base pair deletion that modifies the pathogenic allele to become a wildtype allele.
